WORKING PAPER Nº IDB-WP-1690

Corruption and Political Accountability in Good and Bad Economic Times

Andrés Barinas-Forero
Carlos Scartascini

Inter-American Development Bank
Department of Research and Chief Economist

July 2025

Corruption and Political Accountability in Good and Bad Economic Times

Andrés Barinas-Forero*
Carlos Scartascini*

* Inter-American Development Bank

Inter-American Development Bank
Department of Research and Chief Economist

July 2025

Cataloging-in-Publication data provided by the
Inter-American Development Bank
Felipe Herrera Library

Barinas-Forero, Andres.
Corruption and political accountability in good and bad economic times / Andres Barinas-Forero, Carlos Scartascini.
p. cm. — (IDB Working Paper Series ; 1690)
Includes bibliographical references.
1. Political corruption-Mathematical models. 2. Public goods-Corrupt practices-Mathematical models. I. Scartascini, Carlos G., 19 71- II. Inter-American Development Bank. Department of Research and Chief Economist. III. Title. IV. Series.
IDB-WP-1690

http://www.iadb.org

Copyright © 2025 Inter-American Development Bank ("IDB"). This work is subject to a Creative Commons license CC BY 3.0 IGO (https://creativecommons.org/licenses/by/3.0/igo/legalcode). The terms and conditions indicated in the URL link must be met and the respective recognition must be granted to the IDB.

Further to section 8 of the above license, any mediation relating to disputes arising under such license shall be conducted in accordance with the WIPO Mediation Rules. Any dispute related to the use of the works of the IDB that cannot be settled amicably shall be submitted to arbitration pursuant to the United Nations Commission on International Trade Law (UNCITRAL) rules. The use of the IDB's name for any purpose other than for attribution, and the use of IDB's logo shall be subject to a separate written license agreement between the IDB and the user and is not authorized as part of this license.

Note that the URL link includes terms and conditions that are an integral part of this license.

The opinions expressed in this work are those of the authors and do not necessarily reflect the views of the Inter-American Development Bank, its Board of Directors, or the countries they represent.



Abstract

While the literature extensively explores the structural enablers of corruption and its adverse effects on economic performance, less is known about how the state of the economy influences corruption and political accountability. To address this gap, we develop a theoretical model in which politicians may divert resources from public goods and citizens can respond by punishing corruption. In our model, periods of positive economic conditions increase corruption while weakening accountability. We validate these predictions through a laboratory experiment, finding that corruption rates significantly rise when economic conditions are good. However, citizens' willingness to punish corrupt politicians remains stable across the business cycle. Punishment decisions are driven by observed public good allocations; low allocations prompt significantly higher punishment rates than high allocations, even resulting in the punishment of honest politicians during bad economic times. Additionally, we assess the role of corruption expectations in shaping responses: citizens with prior beliefs that politicians are corrupt are less likely to punish than those who believe politicians are honest when public good provision is low. Accountability becomes more challenging when citizens struggle to clearly identify corruption, and citizens are more forgiving of corruption during good economic times and if they already mistrust politicians. These findings underscore the importance of robust transparency and accountability mechanisms in upholding governance standards, particularly in the face of economic fluctuations and public mistrust.

JEL classifications: D72, D73, H41, C91
Keywords: Rent seeking, Economic Booms, Corruption, Punishment, Laboratory experiment, Downturns

The authors thank participants at REPAL 2024, LACEA-LAMES 2024, MPSA 2025, the University of Virginia seminar, Martín Ardanaz, Hernán Bejarano, Mariana Blanco, Ernesto Calvo, Dan Gingerich, Phil Keefer, Andrea Lopez-Luzuriaga, Julia Seither, and an anonymous reviewer for their comments and suggestions. The research described in this article has Universidad del Rosario IRB Approval CS474 (March 15, 2024) and was preregistered at the OSF Registries (www.osf.io/rmv92, May 8, 2024). This research was financed by the Inter-American Development Bank (IDB). IDB management had no involvement in the study design, analysis, or interpretation of the data, in the writing of the report, or in the decision to submit the article for publication. The opinions expressed here are those of the authors and do not necessarily reflect the views of the Inter-American Development Bank, its Board of Directors, or the countries they represent. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

1 Introduction

Political corruption disrupts growth (Mauro, 1995; Bardhan, 1997), deepens poverty (Gupta et al., 1998), exacerbates inequality (Olken, 2006), erodes trust between citizens and the state (Keefer and Scartascini, 2022; Khlif and Amara, 2019), and leads to significant economic inefficiencies (Dal Bó and Rossi, 2007; Fisman and Svensson, 2007). Economic and political cycles can amplify or mitigate these harmful effects (Saha and Sen, 2023) and may also influence the likelihood of corruption itself.

During an economic bonanza, increased resources may provide more opportunities for corrupt practices (Galbraith, 1997; Leite and Weidmann, 1999). Conversely, during times of economic hardship, corruption risks can intensify as the value of rents grows, and public officials may exploit the population's increased vulnerability (Ivlevs and Hinks, 2015; Torsello, 2010). Economic cycles also shape perceptions of corruption, as citizens may attribute downturns to political mismanagement or become more attuned to the issue during periods of hardship (Márquez Romo and Romero-Vidal, 2023; Zechmeister and Zizumbo-Colunga, 2013).

For instance, data from a nationally representative survey conducted across 26 Latin American and Caribbean countries (LAPOP, 2024), from 2016 to 2023, reveal a notable trend: respondents' perceptions of political corruption are inversely related to their views on economic conditions over the past year (Figure 1.)[1] This correlation suggests that the state of the economy could affect suspicions of corruption or actual corrupt activities. If corruption beliefs are influenced by the cycle, there may be increased incentives for corruption, as political accountability decreases.

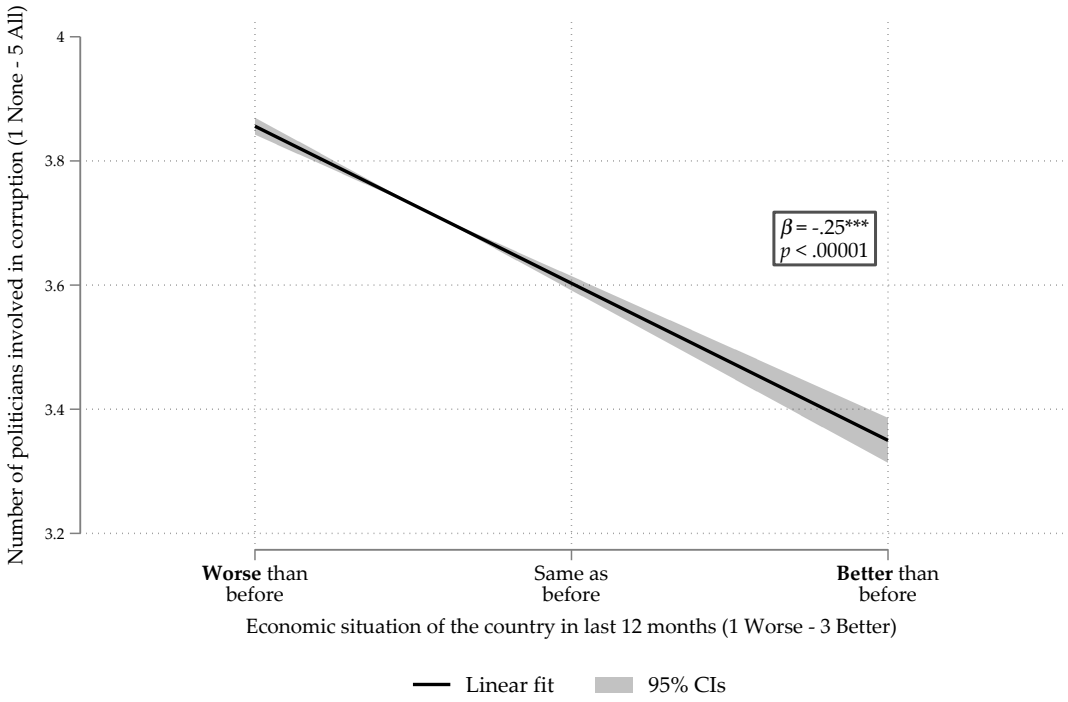
In this paper, we examine how economic good and bad economic times impact the behavior of politicians when it comes to local corruption and citizens when it comes to political accountability.[2] To analyze these dynamics, we develop a theoretical model of political corruption and political accountability, where citizens pay taxes and can punish politicians, and politicians provide public goods (or pocket part of the revenues). We operationalize the economic cycle through an exogenous economic shock that modifies the cost of public

[1] This pattern is consistent across all countries in the region and across support for the current president, as detailed in the Online Appendix (see Figures D1, D2 and D3).

[2] In our model, politicians who act as "rent-seekers" rather than "office-seekers" are the ones who frame local corruption (see Persson and Tabellini, 2002). Our concept of accountability diverges from the traditional understanding found in seminal political accountability literature (see Barro, 1973; Przeworski et al., 1999). Instead, we characterize accountability as citizen's willingness to punish politicians while they are in office (Klašnja et al., 2021).

goods construction, affecting politicians' incentives, the quantity of public goods provided, and citizens' subsequent choices to hold politicians accountable.

Figure 1: Perception of the Number of Politicians Involved in Corruption across National Economic Situation Perception



Notes: The figure depicts the scatterplot binned between the perception of the number of politicians involved in corruption and the perception of the country's economic situation over the past 12 months. Scatter binned estimation for all respondents to the LAPOP question in 26 countries between 2016 and 2023. The dark line indicates the linear fit from a regression model with $n = 79,787$. The gray area represents the 95% clustered confidence intervals at the country level. The analysis includes country and year fixed effects. The regression analysis is presented in the Online Appendix D1. Statistical significance levels: $^{***}p < 0.01$,** $p < 0.05$,* $p < 0.1$, n.s $p > 0.1$.

We model economic conditions as changes in the cost of public goods for two main reasons. First, it allows us to introduce random shocks to citizens' material well-being and observe how these fluctuations shape both corruption and political accountability. Since citizens derive utility from public goods (Persson and Tabellini, 2002), increases in cost reduce provision and consequently lower citizens' welfare, prompting them to potentially punish incumbents even in the absence of corruption (Benton, 2005; Kayser and Peress, 2012; Lewis-Beck and Stegmaier, 2000).[3] Second, in many developing countries, public finances usually employed in the construction of public goods are closely tied to volatile commodity revenues (Martínez, 2023). As such, random fluctuations in production costs provide an externally valid analogue for real-world economic shocks that are independent

[3]We deviate from other studies that employ increments in productivity as the main source of the positive economic shock (see Bai et al., 2017).

of incumbents' decisions and actions.[4]

In our model, a local politician, a national authority, and a citizen interact strategically. In the first stage of the game, an economic shock occurs (the construction cost of the public goods is defined by nature), which is common knowledge for both the citizen and the politician. The representative citizen pays income taxes that are transferred to the politician. The politician receives the transfer and decides how much to invest in public goods.[5] After the politician's allocation decision, the national authority may provide additional funds for public goods construction (an exogenous action). In the final stage, the citizen observes the actual allocation of public goods, considering the economic conditions, and decides whether to punish (or disapprove) the politician——a choice that incurs a cost.

In this model, citizens and politicians can exhibit different intrinsic motivations. Politicians may vary in their moral commitment or integrity (Martinelli, 2022; Tanner et al., 2022), and citizens may differ in their motivation to hold politicians accountable (Adserà et al., 2003; Leight et al., 2020; Nannicini et al., 2013).

The presence of a national authority in our model allows us to incorporate three common dimensions: *i)* in many countries, public goods provision is a shared responsibility between local and national authorities;[6] *ii)* corruption is often facilitated by imperfect information about politicians' actions.[7]; *iii)* it enables us to simulate an environment in which corruption is difficult to detect, thereby introducing real incentives for participants to engage in corrupt behavior.

In our theoretical framework, we identify a signaling equilibrium shaped by two primary dynamics. First, politicians who derive greater utility from honesty are less inclined to

[4]For instance, royalties from natural resource extraction serve as a major income source for municipalities in Colombia and one of the main sources of financing of public goods (Martínez, 2023). Thus, the shift in construction costs we assume in the model for simplicity mimics an economic shock similar to those driven by fluctuations in commodity prices. Moreover, the relevance of economic shocks to commodity prices extends beyond public goods provision, as they have also been shown to influence criminal behavior (Dube and Vargas, 2013) and even the ethical conduct of politicians (Asher and Novosad, 2023).

[5]By allowing the politician to decide how much to allocate, we concentrate on "assignment" problems, which are common approaches when thinking about corruption (see Banerjee et al., 2012).

[6]For example, in Colombia, since the 1990s, local and national authorities have jointly managed essential public services, including drinking water, sanitation, health, and education (Alesina et al., 2005). In developing countries, it is also the case that international agencies and organizations provide local public goods that complement local provision. Our case could then be used to think about the role of foreign aid and corruption.

[7]Details on who funds specific services and the amounts allocated are rarely common knowledge (Burman and Slemrod, 2020; López-Laborda et al., 2023), creating uncertainty that weakens accountability; corruption typically occurs under conditions of secrecy (Hu and Oak, 2023; Shleifer and Vishny, 1993)

engage in corruption, while citizens who place a lower value on political accountability are less likely to incur the cost of disapproving of corrupt politicians. Second, following a positive economic shock that lowers the cost of public goods, politicians may be more prone to corruption, anticipating reduced citizen vigilance in holding them accountable. This expectation of lower disapproval increases the expected benefits of corrupt actions. Thus, our model suggests that economic booms should not only elevate local corruption but also weaken political accountability.[8]

Building on these theoretical insights, we designed and implemented a laboratory experiment to empirically test our hypotheses. In the experiment, participants were assigned to one of two roles: citizens or politicians.[9] Participants were matched using a perfect stranger matching protocol, informed of the game structure with emphasis on the importance of public goods, and repeated the same activity across multiple rounds. The design incorporated within-subject variation in public goods cost, set to either **high cost** or **low cost**, simulating exogenous variation in the economic times. This structure allows us to investigate the effect of economic booms on political behavior, providing empirical support for the theoretical framework outlined earlier. Detailed experimental procedures are presented in Section 3.

Our findings indicate that corruption rates in high-cost environments (economic downturns) are approximately 51%. In contrast, in low-cost environments, such as those following good economic times, corruption rises to 58%, marking a 7-percentage point increase. Regarding political accountability, we observe that citizens' willingness to disapprove of politicians is 16% in high-cost environments and slightly lower, at 15%, during economic booms. These results carry significant economic implications: good economic times are associated with roughly a 14% rise in local political corruption, underscoring the difficulty of maintaining political accountability during periods of economic expansion.

Given the structure of the game, the final allocation of public goods also serves as a quality signal of the politician's actions (Mani and Mukand, 2007). Consequently, we examine how the observed public goods allocation affects the citizen's willingness to disapprove of the politician (Ferejohn, 1986). Our results indicate that citizens are willing to incur personal costs to punish corrupt politicians, with a disapproval rate of 36% for those

[8]This idea is consistent with the theoretical results developed by Bhattacharyya and Hodler (2010), and with empirical results linking economic conditions and perceptions of corruption in the population (Márquez Romo and Romero-Vidal, 2023; Zechmeister and Zizumbo-Colunga, 2013).

[9]The experiment used loaded framing, with participants explicitly informed of their role titles. Prior research indicates that framing effects are typically minimal in corruption games (Abbink and Hennig-Schmidt, 2006; Barr and Serra, 2009; Alekseev, Charness and Gneezy, 2017; Banerjee, 2016).

identified as corrupt. In stark contrast, honest politicians face a disapproval rate of only 3%, illustrating a strategic approach by citizens: they actively seek to "remove" corrupt leaders from office while supporting those who govern with integrity. These findings underscore the critical role of accountability mechanisms, suggesting that when citizens have access to information and the means to act, they are not only discerning but also decisive in promoting good governance.

Citizens' actions are not independent of their priors about the politician's type. Citizens who assume that politicians are generally corrupt are less inclined to punish them for allocating low public goods, unlike those who believe politicians are honest.[10] These findings highlight that, in environments of high mistrust and disillusionment, merely providing information may have limited impact——a result consistent with findings by Chong et al. (2015), Jha (2023), Keefer and Scartascini (2022), and Rivera et al. (2024).

In the literature, a substantial body of empirical research examines the complex relationship between economic development and corruption.[11] However, the nuanced dynamics between economic conditions and corruption remain less explored despite their critical importance. Some studies indicate that positive income shocks can exacerbate corruption (Gokcekus and Suzuki, 2011; Voors et al., 2011), while others suggest these shocks may reduce it (Altunbaş and Thornton, 2012; Kéïta and Laurila, 2021; Tavares, 2003). Moreover, perceptions of corruption are influenced by the state of the economy (Márquez Romo and Romero-Vidal, 2023; Zechmeister and Zizumbo-Colunga, 2013). Resolving these contrasting findings is essential, as understanding the influence of economic fluctuations on corruption can guide more effective policy interventions across varying economic conditions. This paper seeks to address this gap.

Our paper makes a significant contribution to the literature on political accountability and economic cycles by examining how citizens' responses to political corruption fluctuate with economic conditions. Empirical evidence consistently demonstrates that citizens tend to punish corrupt politicians, often reducing their re-election prospects (Ferraz and Finan, 2011; Weitz-Shapiro and Winters, 2013, 2016, 2017). However, the inclination to penalize corrupt officials during prosperous times reveals a more nuanced scenario. On the one hand, improved economic conditions may decrease the relative cost of punitive actions, thereby promoting more frequent punishment. On the other hand, the higher opportunity cost of engaging in such actions during times of prosperity (Lynge and Martinez i Coma,

[10] This result is consistent with findings from Corbacho et al. (2016); Pavão (2018); Persson et al. (2013).
[11] See Ades and Di Tella (1999); Bai et al. (2017); Cieślik and Goczek (2018); Gründler and Potrafke (2019); Huang (2016); Méon and Sekkat (2005); Mocan (2008); Serra (2006); Treisman (2000).

2022), combined with enhanced incentives for citizens in exchange for their approval (Leight et al., 2020), could reduce the likelihood of punitive behavior.

Moreover, economic prosperity often correlates with greater overall happiness and tolerance, potentially diminishing the public's motivation to sanction corruption (Ristei and Gugiu, 2016). Bad times, on the contrary, tend to generate rejection of incumbents despite the origin of the cycle (Lewis-Beck and Stegmaier, 2000). Our findings, however, challenge the assumption that economic prosperity universally tempers the demand for accountability. Even in the context of a low- to middle-income country, where tolerance for corruption is typically higher (Cameron et al., 2009), we find that citizens are willing to incur personal costs to disapprove of corrupt politicians, irrespective of the economic climate. These results underscore the complex relationship between economic conditions and political accountability, highlighting the resilience of civic engagement in countering corruption.

Lastly, our paper contributes to the literature on measuring corruption and political accountability in experimental settings.[12] Measuring corruption is inherently challenging (Armand et al., 2023), but we introduce a novel experimental design that captures corruption dynamics while maintaining realistic incentives for politicians to avoid detection. While similar experimental frameworks have been explored,[13] our approach is unique in incorporating conditions where politicians are either shielded from detection, introducing uncertainty, or explicitly identified as corrupt or honest. This design enables us to assess citizens' willingness to punish under different economic conditions. Moreover, our experiment isolates the causal effect of good economic times on political corruption, controlling for common information asymmetries associated with corrupt behavior (Persson et al., 1997).

Building on the existing literature on the willingness to punish corruption in lab settings, our study distinguishes itself by enabling citizens to be fully aware of economic conditions and providing a clear measure of corruption.[14] This innovation allows us to measure the impact of good economic conditions on political accountability precisely. Furthermore, by incorporating both citizens' and politicians' beliefs regarding corruption and accountability,

[12] Laboratory experiments have shown that corruption observed in controlled environments correlates with real-world corruption in participants' home countries, underscoring the validity of these metrics (Abbink and Serra, 2012).

[13] See Abbink et al. (2002); Armantier and Boly (2011); Berninghaus et al. (2013); Cagala et al. (2019); Cubel et al. (2024); Klašnja and Tucker (2013); Leight et al. (2020); Martinelli and Esandi (2025); Schulze and Frank (2003).

[14] See Sjursen (2023); Martin (2014); Solaz et al. (2019)

our study offers a comprehensive and nuanced understanding of these mechanisms, providing valuable insights into the interplay between good and bad economic times and political behavior.

2 Theoretical Model

Consider a model of local corruption involving three agents: a **citizen**, a local **politician**, and a **national authority**. In this setup, all three agents interact within a local municipality. The citizen has an exogenous income $y \in \Re^+$, and the local politician receives a wage $w \in \Re^+$ for serving as a public official. An exogenous tax rate $\tau \in [0, 1]$ is imposed on the citizen's income, generating tax revenue for the municipality defined as τy. The politician's responsibility is to decide how much tax revenue to allocate toward the construction of public goods. The politician can choose to be *honest*, allocating the entire budget (τy), or *corrupt*, allocating $(\tau y - r)$, where $r \in (0, \tau y)$ represents the amount of rents extracted while in office.

We define the politician's choice as $h \in \{0, 1\}$ with $h = 1$ meaning that she is *honest* and $h = 0$ meaning she is *corrupt*. Note that if the politician decides to extract rents, she receives r, and the rest goes to the construction of public goods satisfying the following condition:

$$\tau y = \boldsymbol{\Theta}(\pi) + r,$$

where $\boldsymbol{\Theta}$ denotes the amount of public goods provided in the municipality, and π denotes the construction cost per unit of public goods. Specifically, $\pi \in \{\overline{\pi}, \underline{\pi}\} \subset \Re^+$ which means that the municipality may face a *high* or *low* cost, with $\overline{\pi} > \underline{\pi}$. Since costs are exogenous, we say that *higher* costs represent **bad** economic times (e.g., exchange devaluation, inflation, declining commodities revenues, etc.), and *lower* costs represent **good** economic times (e.g., exchange revaluation, lower material prices, etc.). Moreover, we say that $\frac{\partial \boldsymbol{\Theta}}{\partial \pi} < 0$, which means that higher costs result in fewer units of public goods.[15] The construction cost (π) is publicly announced to both the citizen and the politician.

Once the local politician chooses, the **national authority** enters the game by randomly deciding whether to allocate additional resources for public goods in the municipality.[16]

[15] This is consistent with an inverse demand model with $p = a - bQ$ assuming that public goods behave like *normal* goods.

[16] This setting of a local and a national authority could also be represented as one of a national authority and a supranational one, such as an international aid agency that steps in to provide public goods. While the way we represent the game directs the reader to think about federal organizations of government, it can

We denote the national authority's decision by $\gamma = \{0, 1\}$, where $\gamma = 1$ indicates the allocation of additional resources. Specifically, when the national authority allocates resources, the amount allocated matches the rents the politician can extract, meaning the national authority can perfectly offset the politician's decision.[17] Since the national authority's decision is random, we say it allocates resources with probability $\mu \in (0, 1)$, and with probability $(1 - \mu)$, it allocates zero additional resources.[18] Thus, the national authority acts as a neutral agent in our model, having no specific payoff or strategic choices.

The total amount allocated to construct public goods equals the sum of resources assigned by both the politician and the national authority. Additionally, neither the politician's nor the national authority's decisions are directly observable to the citizens, who only observe the final allocation of public goods. Thus, we define the final provision of public goods in the municipality as:

$$\boldsymbol{\Theta}(\pi, h, \gamma) = \left(\frac{1}{\pi}\right)(\tau y - (1 - h)r + \gamma r), \tag{1}$$

Notice that higher or lower levels of public goods provision convey information to the citizen. For instance, consider a scenario with an honest politician and a national authority that provides resources. In that case, the citizen would observe a **high-allocation** signal $(\boldsymbol{\Theta}^h(\pi)) = \frac{1}{\pi}(\tau y + r)$. Likewise, if the politician is corrupt and the national authority does not provide resources, the citizen would observe a **low-allocation** signal $(\boldsymbol{\Theta}^l(\pi)) = \frac{1}{\pi}(\tau y - r)$. For completeness, if the politician is corrupt but covered by the national authority, or if she is honest but not supported with additional resources, the citizen would observe an **ambiguous-allocation** signal $(\boldsymbol{\Theta}^a(\pi)) = \frac{1}{\pi}(\tau y)$.

Finally, once the **politician** and the **national authority** have made their choices, the citizen observes the public goods allocation $(\boldsymbol{\Theta})$, and then decides whether to approve or disapprove the politician. We denote the citizen's choice as $\delta \in \{0, 1\}$, where $\delta = 1$ indicates disapproval of the politician. If the citizen approves of the politician, the politician receives her full salary (w), while if the citizen disapproves, then the politician receives a lower

also be used to think about the impact of foreign aid on corruption and accountability.

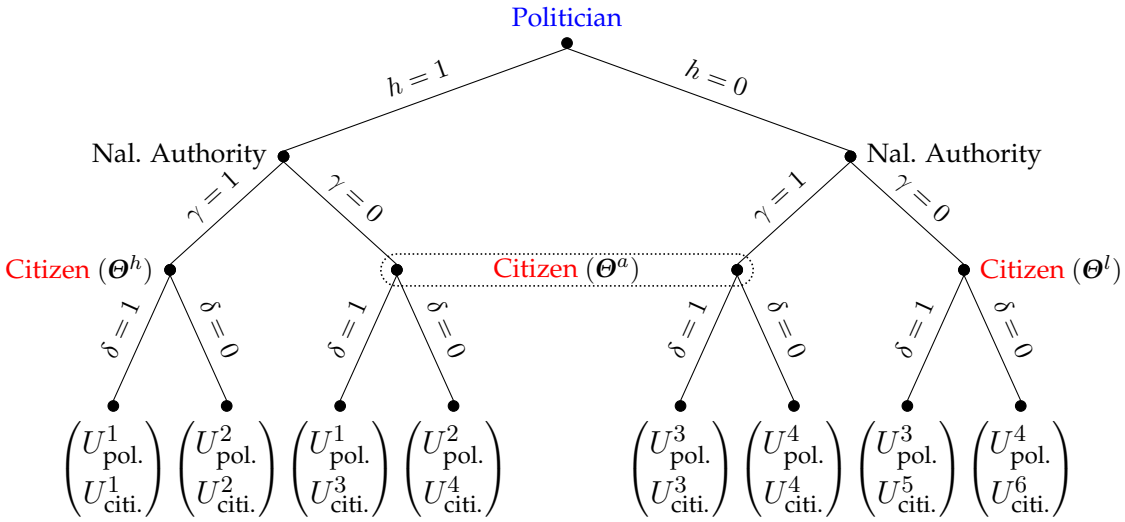
[17] Although this may seem unrealistic, consider an alternative scenario in which the national authority allocates $M \in \Re^+$ with $M \neq r$, indicating different resource amounts are distributed. However, a significant challenge remains: citizens often lack precise knowledge of the specific allocations made by each agent. As a result, distinguishing between these allocations becomes inherently difficult due to limited access to information. This informational deficit can be represented as $M = \epsilon + r$, where $\epsilon \in [-\epsilon, \epsilon]$ represents a parameter that distorts beliefs regarding resources. This distortion effectively makes the model equivalent to our proposed one, highlighting the practical challenges in resource allocation.

[18] It is essential to mention that $\mu \perp\!\!\!\perp c$, indicating that the national authority's choice is independent of the local politician's decision.

salary that is equal to $(w - \eta w)$ where $\eta \in (0, 1)$ represents the severity of the penalty in the municipality. Finally, for the citizen to disapprove of the incumbent, she must pay a fixed cost $\kappa \in \Re^+$, representing the private cost of protesting or participating in social unrest.

Given that the national authority acts as a neutral agent, we model rent-seeking behavior arising from national transfers characterized by a low level of presidential discretionality but with a high level of municipal discretionality (Bonvecchi and Lodola, 2011).[19] This framework accentuates the importance of local politicians' decisions over those of the national authority. Furthermore, our model realistically simulates non-discretionary capital expenditures, which are typically allocated for the construction of public goods, as opposed to current expenditures. This is relevant as capital expenditures exhibit a pro-cyclical pattern in adverse economic times and become acyclical during periods of economic stability (Ardanaz and Izquierdo, 2022; Izquierdo et al., 2018). Therefore, this behavior supports our model's assumption that national transfers do not change during good times and, at least, do not increase during bad times, allowing us to capture a lower-bound estimate of corruption in our setting. Figure 2 presents the extensive representation of the model.

Figure 2: Extensive Representation of the Political Accountability Game



Notes: We assume that the cost of the public goods was announced at the beginning of the game, and was set to be either $\pi \in \{\overline{\pi}, \underline{\pi}\}$. In the citizen's nodes and parentheses, we present the observed allocation of public goods (Θ) given the construction cost. Online Appendix A presents details on the payoff structure.

[19] This model mirrors significant systems of national transfers such as the *Coparticipación Federal de Impuestos* (Bonvecchi and Lodola, 2011) in Argentina and the *Sistema General de Participaciones* (Martínez, 2023) in Colombia.

2.1 Agents Types

2.1.1 Politician's Type

In reality, politicians exhibit a spectrum of behaviors and intentions. Some are genuinely well-intentioned individuals who prioritize public welfare, act with integrity, and derive intrinsic satisfaction from fulfilling their civic duties, while others gain greater benefits from the rents extracted while being in office. To account for this, we define the **politician's type** as ϕ_p where $\phi_p \sim U[0, \widehat{\phi}]$ with $\widehat{\phi} > r$. Specifically, the politician's type represents the intrinsic reward for doing the right thing (Martinelli, 2022) or acting with integrity, so the politician privately observes this "warm-glow" benefit. Finally, the politician receives this moral reward only when she is *honest* ($h = 1$).

2.1.2 Citizen's Type

Similarly, citizens vary in their propensity to engage in political activism (Chenoweth et al., 2022) and to hold their elected officials accountable. Some are inclined to protest and scrutinize politicians relentlessly, irrespective of the circumstances, as they derive a high intrinsic reward from participating in political accountability mechanisms. Conversely, others remain skeptical regarding the efficacy of their actions in influencing political outcomes. To account for this heterogeneity, we define the **citizen's type** as α_c where $\alpha_c \sim f[0, \overline{\alpha}]$ with $\overline{\alpha} > \max\{\boldsymbol{\Theta}(\underline{\pi}, h, \gamma)\}$. This parameter represents the inherent reward for political accountability or for making disapproval of politicians more salient. Once more, the citizen observes this benefit privately and receives the reward only when she disapproves of the politician's mandate.

2.2 Payoffs

In our model, the politician derives utility from her public official salary[20] (w), her own "warm-glow" reward (ϕ_p), and the material payoff from extracting rents while in office (r). Therefore, we can express the politician's payoff as:

$$U_p = w + (1 - h)r - \delta\eta w + h\phi_p \tag{2}$$

[20]While good economic times may influence salaries, we assume that wages remain constant across different settings. Economic shocks might reduce politicians' endowments, potentially decreasing the trust citizens place in them (Barinas-Forero, 2024), or they might increase politicians' salaries, which could lead to a reduction in corruption (Van Rijckeghem and Weder, 2001).

with $\delta\eta w$ being the monetary punishment received by the disapproval from the citizen.

On the other hand, the citizen derives utility from the public good provision ($\boldsymbol{\Theta}$) and from her non-taxed income $y(1-\tau)$. For simplicity, the valuation of each unit of the public good is fixed at $\kappa \in \Re^{+}$. Additionally, note that the marginal benefit for each unit is equal to the fixed cost of disapproval, meaning that choosing to disapprove of the politician represents a loss of one public good unit.[21] We characterize the citizen's payoff as:

$$U_c = y(1-\tau) + \kappa\boldsymbol{\Theta} + \delta\left((\alpha_c - \boldsymbol{\Theta})\lambda - \kappa\right), \tag{3}$$

where $\lambda \in [0,1]$ denotes the citizen's posterior belief about the politician's type once she observes the signal provided by the public goods allocation ($\boldsymbol{\Theta}$). It is worth mentioning that the citizen updates her beliefs in a Bayesian manner,[22] and they are consistent with her prior beliefs of corruption ($\beta \in (0,1)$).

In general, citizens may perceive higher returns from protest when public good provision is scarce (Almer et al., 2017; Grossman, 1991). Therefore, higher allocations of public goods act as a reserve utility for citizens. In our model, this reserve utility is represented by $(\alpha_c - \boldsymbol{\Theta})$, where lower allocations of public goods increase the marginal benefit of protest. Additionally, citizens behave strategically. Therefore, the marginal benefits of non-approval are mediated by the posterior beliefs of corruption (λ).

While the model suggests that politicians possess deeper insights into the public goods financing scheme, it raises the question of why this type of information asymmetry is deemed more critical than the asymmetry concerning economic factors that influence construction costs. Notably, the costs of public goods tend to align closely with those of private goods, and given that citizens regularly purchase these private goods, it is plausible to assume they can reasonably estimate public costs. In contrast, the intricacies of budgetary processes remain largely inaccessible to the general public, underscoring the rationale for focusing on budgetary rather than economic informational asymmetries.

Figure 3 summarizes the model's timeline. The participants' types (α_c, ϕ_p), as well as the

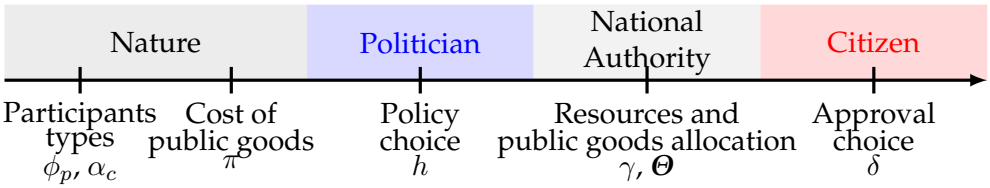
[21] According to Ortiz et al. (2022), social discontent can lead to the devastation of infrastructure and wealth. Images of burning cars or buildings frequently appear in media coverage about protests in many nations. Consequently, it makes sense to express the cost of protests as the loss of one unit of public goods.

[22] For a specific allocation of public goods, the beliefs updating process follows:

$$\lambda \equiv \Pr\left(\sigma = 0 | \boldsymbol{\Theta}(\pi) = \boldsymbol{\Theta}^m(\pi)\right) = \frac{\Pr\left(\boldsymbol{\Theta}(\pi) = \boldsymbol{\Theta}^m(\pi) | \sigma = 0\right) \times \Pr\left(\sigma = 0\right)}{\Pr\left(\boldsymbol{\Theta}(\pi) = \boldsymbol{\Theta}^m(\pi)\right)} \quad \forall\, m = \{h, a, l\}$$

costs for public goods $\pi = \{\overline{\pi}, \underline{\pi}\}$ are exogenously determined at the beginning of the game. Then, the politician makes the policy decision, the national authority randomly decides whether to provide resources to the municipality, and the public goods allocation is shown to the citizen. Finally, once the citizen observes the final allocation of public goods, she decides whether to approve or disapprove the politician, and payoffs are realized.

Figure 3: Timeline for the Political Corruption and Accountability Model



Notes: Participant types are drawn from their corresponding distributions ($\phi_p \sim U[0, \widehat{\phi}]$), ($\alpha_c \sim f[0, \overline{\alpha}]$. The cost of public goods (π) follows a Bernoulli distribution, and $\pi \in \{\overline{\pi}, \underline{\pi}\}$. Finally, the national authority decides to allocate additional resources ($\gamma = 1$) with a probability of $\mu \in (0, 1)$.

2.3 Signaling Equilibrium

Now, we define strategies, beliefs, and equilibrium for the extensive game of incomplete information played by the citizen and the politician. The information set available to the politician before making the policy choice is characterized by their own type and the public good cost. Thus, we define a *strategy for politician* as a mapping σ,

$$\sigma : \mathcal{I}_p \rightarrow [0, 1] \quad \textit{where} \quad \mathcal{I}_p \equiv (\phi_p, \pi) \in [0, 1] \times \{\overline{\pi}, \underline{\pi}\},$$

which specifies the probability of adopting the policy decision h as a function of the information received by the politician at the beginning of the game.

The citizen knows the public good cost, observes the allocation of public goods given that cost, and makes a decision given the intrinsic motivation for disapproval α_c. Thus, we define a *strategy for the citizen* as a mapping ν,

$$\nu : \{\boldsymbol{\Theta}^h(\pi), \boldsymbol{\Theta}^a(\pi), \boldsymbol{\Theta}^l(\pi)\} \times [0, \overline{\alpha}] \rightarrow [0, 1],$$

that goes from the set of possible public goods allocations given a specific cost and the distribution of possible disapproval rates $[0, \overline{\alpha}]$ to a range between $[0, 1]$ that denotes the probability of enforcing the punishment against the politician.

Finally, we define a *belief system* for the citizen as a mapping λ,

$$\lambda : \{\boldsymbol{\Theta}^h, \boldsymbol{\Theta}^a, \boldsymbol{\Theta}^l\} \times \{0, 1\} \to [0, 1]$$

that goes from the observed allocation of public goods $\boldsymbol{\Theta}$ and prior beliefs $\beta \in (0, 1)$ of corruption to a range of updated beliefs of corruption in $[0, 1]$.

Definition 1. *(Signaling Equilibrium) A signaling Bayesian equilibrium is given by*

(i) a strategy for the politician (σ)
(ii) a strategy for the citizen (ν)
(iii) a belief system for the citizen (λ) *such that: the citizen strategy* (ν) *is optimal given her beliefs, the citizen beliefs are consistent with the politician's strategy—meaning that* λ *is derived from prior beliefs* (β) *and the politician's strategy using Bayes' rule for every signal, and a politician strategy* (σ) *that is optimal given the citizen strategy.*

2.4 Equilibrium Analysis

Given the structure and timing of the game, in the final stage of the interaction, the citizen observes the final allocation of public goods and updates her posterior beliefs about corruption in a Bayesian manner. As illustrated in Figure 3, the sequential nature of the game allows the politician to anticipate the strategic behavior of the citizen in the later stage, enabling her to choose the optimal strategy in the early stage of the game. Therefore, to characterize the equilibrium, we first determine the citizen's optimal strategy ν and then the politician's optimal strategy σ.

Lemma 1. *In every signal equilibrium, regardless of the cost* (π)*, if the citizen observes a* ***high-allocation*** *signal* $\boldsymbol{\Theta}^h$*, the citizen plays a strategy of no punishment.*

$$\nu(\alpha_c, \boldsymbol{\Theta}^h) = 0$$

Proof. See Online Appendix B.1. □

From Lemma 1, we can establish that regardless of the cost, a high allocation of public goods consistently signals an honest politician, and consequently, even citizens who prioritize political accountability have no incentive to disapprove of the incumbent. This result aligns with the idea that citizens reward good politicians to keep them in office and punish bad politicians to remove them from office.

Definition 2. *(Cutoff strategy). A strategy* ν *is a cutoff strategy if there is some* $\alpha^*(\boldsymbol{\Theta}^m(\pi)) \in$

$(0, \overline{\alpha}]$, *with* $m = \{a, l\}$ *such that:*

$$\nu(\alpha_c, \boldsymbol{\Theta}^m(\pi)) = \begin{cases} 1 & \text{if } \alpha_c \geq \alpha^*(\boldsymbol{\Theta}^m(\pi)) \\ 0 & \text{if } \alpha_c < \alpha^*(\boldsymbol{\Theta}^m(\pi)) \end{cases}$$

Lemma 2. *In every signaling equilibrium, if the citizen observes an* ***ambiguous*** $(\boldsymbol{\Theta}^a(\pi))$ *or* ***low-allocation*** *signal* $(\boldsymbol{\Theta}^l(\pi))$*, the citizen plays a cutoff strategy following Definition 2.*

Proof. See Online Appendix B.2. □

From Lemma 2, we conclude that for each possible allocation $\boldsymbol{\Theta}^m(\pi)$ there exists an $\alpha^*(\boldsymbol{\Theta}^m(\pi))$ cutoff that defines the citizen's strategy ν. This means that the citizen observes her private intrinsic reward for political accountability, the construction cost, and the final allocation of public goods, and if her intrinsic reward for disapproval is higher than $\alpha^*(\boldsymbol{\Theta}^m(\pi))$, she would disapprove of the politician. Moreover, as citizens receive different information about the public goods allocation, the cutoff may shift according to the construction cost (π) and the observed allocation.

Lemma 3. *In every signaling equilibrium,* $\pi \in \{\overline{\pi}, \underline{\pi}\}$, $\alpha^*(\boldsymbol{\Theta}^l(\pi)) \leq \alpha^*(\boldsymbol{\Theta}^a(\pi))$

Proof. See Online Appendix B.3. □

From Lemma 3, the deciding cutoff under an ambiguous allocation is always greater than or equal to the deciding cutoff under a low allocation. Thus, citizens who are willing to punish under ambiguous allocations must have a higher intrinsic reward for disapproval than citizens who punish under low allocations. Furthermore, a citizen who punishes a politician under an ambiguous allocation——where there is uncertainty of the politician's corruption——will also punish under a low allocation, where there is certainty that the politician was corrupt.

Given that we have already characterized the citizen's optimal strategy (ν), which follows a consistent belief system (λ), it remains to characterize the politician's optimal strategy consistent with the citizen's strategy to define a signaling equilibrium, as presented in Definition 3.

Definition 3. *(Threshold strategy). A strategy* σ *is a threshold strategy if there is some* $\phi^*(\pi) \leq \widehat{\phi}$,

and for every $\pi \in \{\overline{\pi}, \underline{\pi}\}$ *such that:*

$$\sigma(\phi_p) = \begin{cases} 1 & \text{if } \phi_p \geq \phi^*(\pi) \\ 0 & \text{if } \phi_p < \phi^*(\pi) \end{cases}$$

Lemma 4. *In every signaling equilibrium, the politician enforces a threshold strategy following Definition 3.*

Proof. See Online Appendix B.4. □

Lemma 4 establishes that there must exist a threshold $\phi^*(\pi)$ which defines the politician's strategy σ. This means that the politician observes her private intrinsic reward for doing the "right thing" (ϕ_p), the construction cost, and then she compares the expected payoff of being corrupt versus being honest. If her intrinsic reward for acting with integrity exceeds $\phi^*(\pi)$, she will choose to be honest and allocate the entire budget to public goods.

2.5 Impacts of Economic Times on Political Corruption and Accountability

Regarding the impacts of good economic times on political corruption and political accountability, we first define **good economic** times as a shift in construction costs from a high cost $(\overline{\pi})$ to a low cost $(\underline{\pi})$. Moreover, these cost changes are randomly determined and announced to both citizens and politicians at the beginning of the game. With this in mind,

Theorem 1. *In a signaling equilibrium, and for all* $m = \{a, l\}$*, if* $\overline{\pi} \geq \underline{\pi}$*, then*

$$\alpha^*(\boldsymbol{\Theta}^m(\overline{\pi})) \geq \alpha^*(\boldsymbol{\Theta}^m(\underline{\pi}))$$

Proof. See Online Appendix B.5. □

The main result of our theoretical model is that good economic times, captured by a decline in the cost of public goods provision from a high level $(\overline{\pi})$ to a low level $(\underline{\pi})$, reduce citizens' willingness to punish politicians by raising the optimal punishment threshold. This shift is driven by a higher expected material payoff, as the expected public goods allocation is higher during good economic times. Importantly, the increase in the cutoff applies regardless of the observed allocation, reflecting a general tendency toward more lenient behavior in both low and ambiguous allocation scenarios.

This leads to the following corollary:

Corollary 1. *In a signaling equilibrium, and for all* $\boldsymbol{\Theta}^m$ *with* $m = \{a, l\}$*, if* $\overline{\pi} \geq \underline{\pi}$*, then*

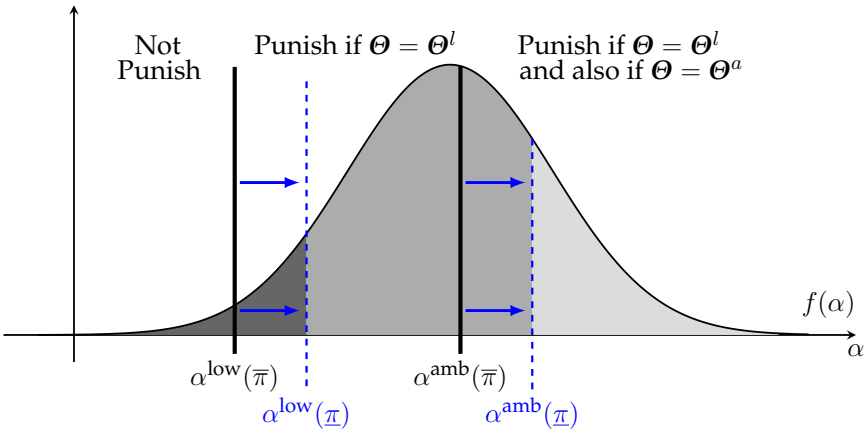
$$\int_0^{\alpha^*(\boldsymbol{\Theta}^m(\overline{\pi}))} f(\alpha)\, d\alpha \geq \int_0^{\alpha^*(\boldsymbol{\Theta}^m(\underline{\pi}))} f(\alpha)\, d\alpha$$

Proof. See Online Appendix B.6. □

As a direct implication of Theorem 1, Corollary 1 demonstrates that, given an increase in the optimal cutoff points and a given signal $\alpha^*(\boldsymbol{\Theta}^m)$, a politician is more likely to be punished in high-cost environments than in low-cost settings,

Figure 4 visually illustrates the changes in the probability of punishment. Specifically, it compares the initial set of cutoffs, $\alpha^*(\overline{\pi}) = \{\alpha^*(\boldsymbol{\Theta}^l(\overline{\pi})), \alpha^*(\boldsymbol{\Theta}^a(\overline{\pi}))\}$ with the new pair of cutoffs $\alpha^*(\underline{\pi}) = \{\alpha^*(\boldsymbol{\Theta}^l(\underline{\pi})), \alpha^*(\boldsymbol{\Theta}^a(\underline{\pi}))\}$, following the cost reduction.

Figure 4: Citizen Distribution of Types (α)



Notes: The figure depicts the distribution of the types $(\alpha_c \sim f[0, \overline{\alpha}]$ for the citizen. In dark gray, the mass of citizens that are *not willing to punish*. In medium gray, the mass of citizens who are *willing to punish* under **low-allocations** of public goods. In light gray, the mass of citizens that are *willing to punish* under *low-allocations* and **ambiguous-allocations** of public goods. In black (solid, $\phi^*(\overline{\pi})$), we present the optimal cutoff under a high-cost setting. In blue (dashed, $\phi^*(\underline{\pi})$), we present the optimal cutoff under a low-cost setting.

Finally, we examine how favorable economic conditions influence a politician's marginal propensity to engage in corruption.

Theorem 2. *In a signaling equilibrium, if* $\overline{\pi} \geq \underline{\pi}$

$$\phi^*(\overline{\pi}) \leq \phi^*(\underline{\pi})$$

Proof. See Online Appendix B.7. □

As established in Corollary 1, an economic boom reduces the likelihood that citizens will punish corrupt politicians. This decline in perceived accountability lowers the expected cost of corrupt behavior, thereby increasing its net expected benefit. Theorem 2 formalizes this result, showing that the equilibrium propensity to engage in corruption rises as the cost of public goods provision falls.

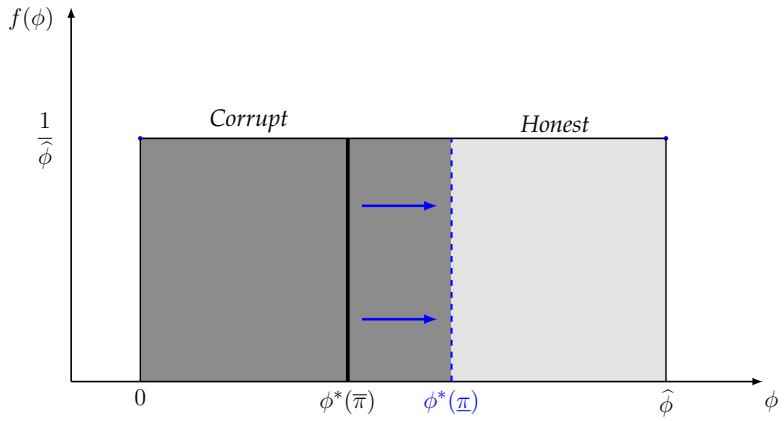
Corollary 2. *In a signaling equilibrium with* $\phi^*(\underline{\pi}) \in \mathfrak{R}^+$, *if* $\overline{\pi} \geq \underline{\pi}$

$$\int_0^{\phi^*(\overline{\pi})} U(\phi)\, d\phi \leq \int_0^{\phi^*(\underline{\pi})} U(\phi)\, d\phi$$

Proof. This result follows directly from the procedure described in Online Appendix B.6, by changing $\alpha^*(\boldsymbol{\Theta}^m(\pi))$ to $\phi^*(\pi)$. □

Finally, Corollary 2 presents a direct implication of Theorem 2, showing that the probability of a politician engaging in corrupt activities is greater when construction costs are low compared to when they are high. These theoretical results are further illustrated in Figure 5. Taken together, our findings highlight the substantial influence of prevailing economic conditions on the incidence of corruption and the mechanisms of political accountability.

Figure 5: Politician Distribution of Types (ϕ) after the Economic Shock



Notes: The figure depicts the distribution of the types $(\phi_p \sim U[0, \widehat{\phi}])$ for the politician. In medium gray, the mass of politicians who are *corrupt*. In light gray, the mass of politicians who are *honest*. In black (solid, $\phi^*(\overline{\pi})$), we present the optimal cutoff under a high-cost setting. In blue (dashed, $\phi^*(\underline{\pi})$), we present the optimal cutoff under a low-cost setting.

3 Experimental Design

Building on these theoretical results, we conducted a series of laboratory experiments using oTree (Chen et al., 2016). The experiments involved nearly 800 college students recruited

through the Rosario Experimental and Behavioral Economics Lab (REβEL, Greiner (2015)) at Universidad del Rosario in Bogota, Colombia. We collected data from 24 independent sessions, each lasting an average of one and a half hours.

3.1 Political Corruption Game (PCG)

The main activity of the experiment was a political corruption game (PCG) designed to replicate our theoretical model. Participants were randomly assigned to one of two roles: **politician** (P) or **citizen** (C), and this role remained fixed for the entire session. Each citizen received an endowment of 100 experimental points (EP),[23] while each politician had a potential salary of 120 EP. Participants were paired with a member of the opposite role under a perfect stranger-matching design.

Once paired, the citizen paid 40 EP in taxes, which were transferred to the politician. The politician's main task was to decide how much of this budget to allocate toward constructing public goods: either the entire amount (40 EP) or half (20 EP), keeping the remaining half as personal income. In this setup, public goods represent an additional source of income for citizens. At the beginning of the experiment, both participants were informed that each unit of public goods constructed would provide the citizen with an additional 30 EP, highlighting the social benefits of higher public goods allocations and reinforcing the politician's duty to act in the citizen's interest.

In this setup, each unit of public goods has a cost. Participants are aware that the construction cost is either 20 EP (high cost) or 10 EP (low cost), with each option having an equal probability. Both participants were informed about the public goods cost (π_r) at the beginning of each round, ensuring no information asymmetry about the state of the economy.

After observing the public goods cost (π_r), and once the politician made her allocation decision, the computer simulated a national authority by randomly deciding whether to allocate additional funds. If the computer allocated resources, 20 EP were added to the construction of public goods; otherwise, only the politician's contribution was used. Importantly, the computer's decision remained private, meaning citizens observed only the final public goods allocation without knowing the individual contributions from the politician or the higher-level government (in this case, the computer's random allocation).

[23] Each EP was valued at USD 0.08. Participants received a USD 3 show-up fee. Citizens (Politicians) earned an average of USD 13.1 (USD 12.9). Given that the minimum wage in Colombia is USD 1.42/hour, the earnings obtained are about nine times the hourly minimum wage.

Public goods provision follows equation (1).

Once the citizen observed the public goods provided, she decided whether to reduce the politician's salary ("punish") to 80 EP[24] by paying 30 EP, or to "approve" the politician and allow her to receive her full salary. The interaction ended once the citizen made her choice, and no feedback was provided within the round.

3.1.1 Additional Details

This interaction repeated over five decision rounds.[25] In each round, participants reported their first-order beliefs about their partner's behavior: politicians predicted the citizen's decision, while citizens predicted the politician's policy choice. These beliefs were incentive-compatible, as participants earned 5 EP each time they correctly predicted their partner's decision. To ensure expectations were recorded before observing others' behavior, each participant followed a separate experimental timeline.[26]

3.1.2 Treatments

We exploit a *within-subjects* design across all five rounds. Given that the cost of public goods (π_r) is randomly selected by the computer at the beginning of the round and was defined at the pair level, we denote the following two states of the economy:

- **High-cost environment -or bad economic times- (Control)**: Under this setting, the construction cost is set to **20EP**. Therefore, the maximum amount of public goods that can be produced is three, and the lowest amount of public goods that can be produced is one.
- **Low-cost environment -or good economic times- (Treatment)**: Under this setting, the construction cost is set to **10EP**. Therefore, the maximum amount of public goods that can be produced is six, and the lowest amount of public goods that can be produced is two.

[24] Given the structure of the experiment, the utility functions under the PCG are:

$$U_p = 120 + (1-h)20 - \delta(0.33)120 + h\phi_p$$

$$U_c = 100(1-0.4) + 30\boldsymbol{\Theta} + \delta\left((\alpha_c - \boldsymbol{\Theta})\lambda - 30\right)$$

[25] Participants knew that only one randomly selected round would count toward their final experimental payment.

[26] The experimental timeline for each type of participant is displayed in the Online Appendix C1

3.2 Coordination Game (Social Norms)

Finally, participants engaged in a coordination game based on Krupka and Weber (2013), used to elicit incentive-compatible normative expectations from citizens and politicians during the PCG.[27] Each participant rated the social appropriateness of the politician's behavior under each possible cost scenario. Specifically, subjects indicated whether allocating the entire budget was very socially inappropriate, socially inappropriate, socially appropriate, or very socially appropriate. Participants earned 5 EP each time their normative expectation aligned with the majority view among participants of their own type. At the end of the activity, they completed a non-incentivized socio-demographic questionnaire and provided payment details. Table D2 in the Appendix presents descriptive statistics.

4 Results

4.1 Political Corruption

Are politicians corrupt? Pooling participants across all five rounds, we find that they are corrupt 55% of the time. Notably, 22% of the politicians were always corrupt, while 12% of them were always honest.[28] Their willingness to engage in corruption is not independent of the economic cycle. Figure 6, which illustrates the percentage of corrupt politicians segmented by cost type, shows that the unconditional average corruption rate is 51% when the cost is high, rising to nearly 59% when the construction cost is low—a statistically significant increase at the 1% level. This increase represents a 14% rise in the probability of engaging in corruption compared to the high-cost environment.

To measure the causal treatment effects on political corruption, we estimate the following model:

$$\text{Corrupt}_{p,r,s} = \alpha_0 + \alpha_1 \text{Cost}_{p,r} + x'_p\gamma + \Lambda_r + \eta_s + \varepsilon_{p,r,s}, \tag{4}$$

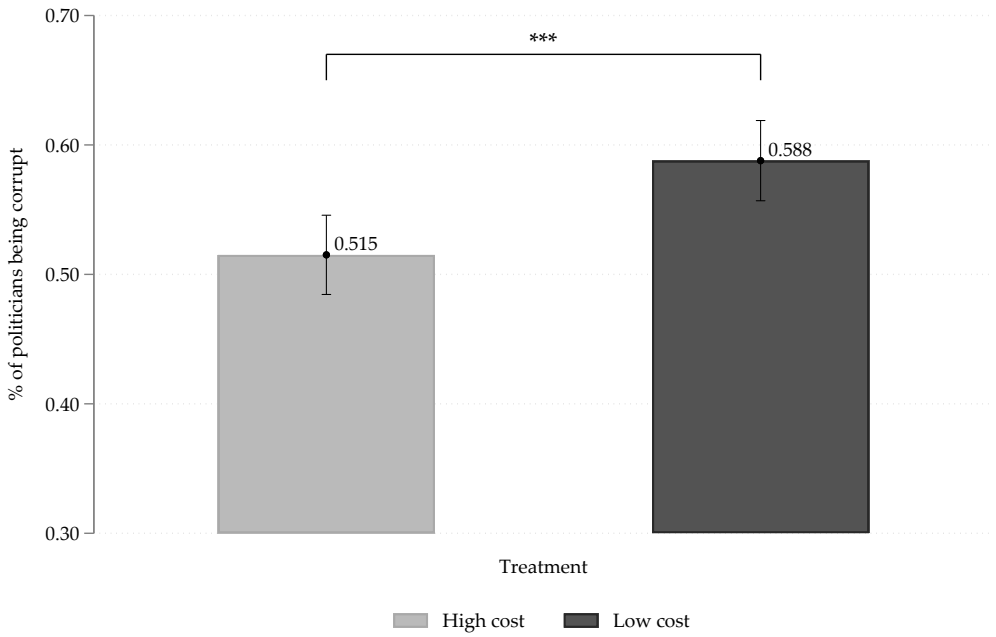
where $\text{Corrupt}_{i,r,s}$ is a dummy variable that takes the value of 1 if the politician was corrupt and 0 otherwise. $\text{Cost}_{p,r}$, the treatment variable, is a dummy variable that takes the value of 1 if the construction cost is low and 0 if the construction cost is high. Additionally, we include a set of controls (x'_p) that consider sociodemographic variables, a series of round

[27] The coordination game was implemented at the end of the experiment, with participant identities remaining anonymous.

[28] The distribution of the number of times politicians were corrupt is presented in Figure D5 in the Online Appendix.

fixed effects (Λ_r) and session fixed effects (η_s). Considering that there is no feedback on the experimental outcomes during the interaction, there is no reason to expect autocorrelation within the experimental group or session. Furthermore, our participants make multiple choices across rounds, so standard errors $(\varepsilon_{p,r,s})$ are clustered at participant level (Abadie et al., 2023).

Figure 6: Politician Corruption Segmented by Costs (π).



Notes: The figure depicts the average corruption rate by the politician across all rounds, segmented by production costs. **High cost** = 20 EP. **Low cost** = 10 EP. Dark lines show 95% clustered robust confidence intervals. No controls included. $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.1$, n.s $p > 0.1$. Reported significance levels come from hypothesis testing from a fully saturated linear model where the dependent variable is the politician's choice to be honest or corrupt, and regressors are dummy variables associated with treatments.

Table 1 presents the OLS estimates for equation (4).[29] In Column (1), we observe that a low cost of producing public goods leads to a 7.3*pp* increase in the probability of being corrupt. We add several controls, including participant age, sex, stratum, trust, risk preference, patience, and prior experiment participation. Column (2) shows that the effect of good economic times is robust to the inclusion of controls, indicating a 12% increase in corruption compared to the control group. Lastly, Column (3) includes round and session fixed effects to account for heterogeneity across rounds and sessions, revealing a robust average treatment effect of 6.5*pp* increase in corruption likelihood. Overall, the regression results confirm that good economic times, such as exogenous reductions in public goods costs, lead to an increase in local corruption of nearly 12%.

[29]The results are robust to other estimation methods. We present logit and probit estimates in Table D3 in the Online Appendix.

Table 1: Politicians **Corruption**: Main Results

Dep Var: **Politician Corruption**	OLS		
	(1)	(2)	(3)
Low cost (Treatment)	0.073***	0.066***	0.065***
	(0.02)	(0.02)	(0.02)
Constant	0.515***	0.389**	0.389**
	(0.02)	(0.15)	(0.17)
Mean Dep. Variable	0.52	0.52	0.52
Effect size (%)	14.1	12.8	12.6
Controls		✓	✓
Round FE			✓
Session FE			✓
Observations	2000	2000	2000
Clusters	400	400	400
Adjusted R-squared	0.01	0.07	0.10

Notes: The set of controls included in ***Controls*** are: age, sex, stratum, trust measure, risk measure, patience measure, and previous experience participating in experiments. The *Mean Dep. Variable* expresses the unconditional control mean of the dependent variable. The *Effect size (%)* denotes the size of the treatment effect as a percentage compared with the unconditional mean of the control group. Standard errors clustered at the individual level are presented in parentheses. $^{***}p < 0.01,^{**}p < 0.05,^{*}p < 0.1$

4.2 Citizens' Willingness to Punish Politicians

Do citizens punish corruption? Citizens disapprove of politicians 15% of the time, with 51% of citizens never punishing politicians and fewer than 1% always punishing.[30] Figure 7 incorporates the state of the economy and shows the percentage of citizens who disapprove of politicians, segmented by public good cost level. In high-cost environments (bad economic times), the disapproval rate is 16%; following a good economic period, it decreases slightly to 15%, although this difference is not statistically significant.

To measure the causal treatment effect of economic booms on citizen's willingness to punish politicians, we estimate the following model:

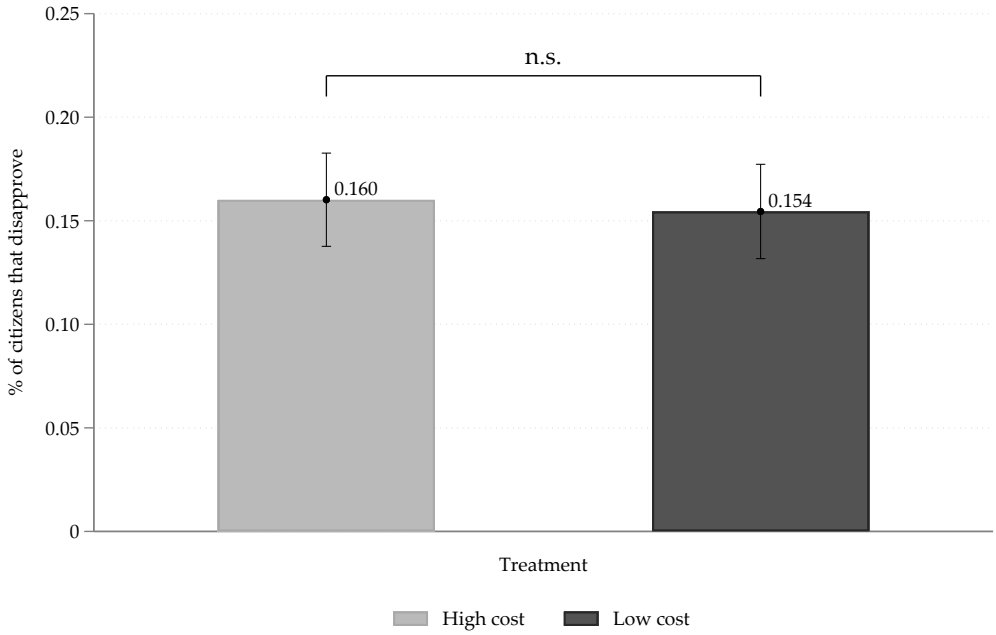
$$\text{Punish}_{c,r,s} = \alpha_0 + \alpha_1 \text{Cost}_{c,r} + x_c'\gamma + \Lambda_r + \eta_s + \varepsilon_{c,r,s}, \tag{5}$$

where $\text{Punish}_{c,r,s}$ is a dummy variable that takes the value of 1 if the citizen disapproves of the politician and 0 otherwise. The variable $\text{Cost}_{c,r}$, the treatment variable, is a dummy variable that takes the value of 1 if the construction cost is low and 0 if the construction

[30]This aligns with the notion of rational punishment, as citizens should not have incentives to punish honest politicians. The distribution of citizen punishment decisions is shown in Figure D6 in the Online Appendix.

cost is high. We include a set of controls (x'_c) that consider sociodemographic variables and a series of round (Λ_r) and session fixed effects (η_s). Similarly to our previous specification, standard errors $(\varepsilon_{c,r,s})$ are clustered at the participant level.

Figure 7: Citizen Punishment Segmented by Costs (π) .



Notes: The figure depicts the average disapproval rate by the citizen across all rounds, segmented by production costs. **High cost** = 20 EP. **Low cost** = 10 EP. Dark lines show 95% clustered robust confidence intervals. No controls included. $^{***}p < 0.01$,** $p < 0.05$,* $p < 0.1$, n.s $p > 0.1$. Reported significance levels are derived from hypothesis testing of a fully saturated linear model, where the dependent variable is the citizen's choice to disapprove of the incumbent, and the regressors are dummy variables associated with the treatments.

Table 2 present the OLS estimates for Equation (5). Column (1) shows that the unconditional disapproval rate in high-cost environments is 16%. While the point estimates suggest a reduced likelihood of punishment in low-cost environments, this difference is not statistically significant, indicating that citizens are not more willing to punish when the costs of producing public goods are lower.

Figure 8 shows the average disapproval rate, disaggregated by the cost of producing the public goods, and the public goods allocation observed by the citizens. As expected, when citizens observe a **low allocation**, the disapproval rate is 36% in high and low-cost cases. For an **ambiguous allocation**, the disapproval rate is 10% in high-cost and 9% in low-cost settings. Finally, when the public goods allocation is high, the disapproval rate is approximately 6% in high-cost environments and 3% in low-cost environments. We do not observe statistically significant differences across costs within public goods allocations.[31]

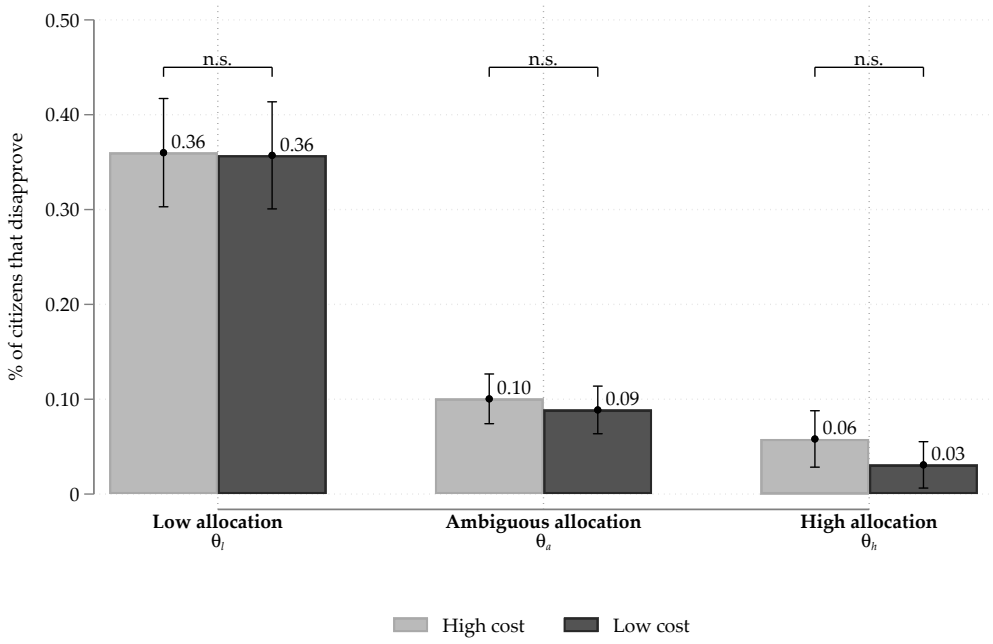
[31]OLS estimates for Equation (5), disaggregated by observed public goods allocation $(\boldsymbol{\Theta})$ are presented in

Table 2: Citizens **Disapproval Rate**: Main Results

Dep Var: **Citizen Punishment**	OLS		
	(1)	(2)	(3)
Low cost (Treatment)	-0.006	-0.003	-0.001
	(0.015)	(0.015)	(0.015)
Constant	0.160***	0.466***	0.336*
	(0.013)	(0.176)	(0.177)
Mean Dep. Variable	0.16	0.16	0.16
Effect size (%)	-3.5	-1.7	-0.5
Controls		✓	✓
Round FE			✓
Session FE			✓
Observations	1995	1995	1995
Clusters	399	399	399
Adjusted R-squared	0.00	0.02	0.04

Notes: The set of controls included in ***Controls*** are: age, sex, stratum, trust measure, risk measure, patience measure, previous experience participating in experiments, and the willingness-to-protest measure from a 6-item questionnaire. The *Mean Dep. Variable* expresses the unconditional control mean of the dependent variable. The *Effect size (%)* denotes the size of the treatment effect as a percentage compared with the unconditional mean of the control group. Standard errors clustered at individual level are presented in parentheses. $^{***}p < 0.01,^{**}p < 0.05,^{*}p < 0.1$

Figure 8: Citizen Punishment Segmented by Costs (π) across allocations $(\boldsymbol{\Theta})$.



Notes: The figure depicts the average disapproval rate by the citizen across all rounds, segmented by production costs. **High cost** = 20 EP. **Low cost** = 10 EP. Dark lines show 95% clustered robust confidence intervals. No controls included. $^{***}p < 0.01,^{**}p < 0.05,^{*}p < 0.1$, n.s $p > 0.1$. Reported significance levels come from hypothesis testing from a fully saturated linear model where the dependent variable is the citizen's choice to disapprove of the incumbent, and regressors are dummy variables associated with treatments.

Table D4 in the Online Appendix. Consistent with Table 2, these results do not reject the null hypothesis of no difference in punishment across cost environments.

4.3 The Signal of Politician's Quality: Public Goods (Θ)

Having established that the economic shock does not significantly impact the willingness to punish politicians, we evaluate the relevance of public goods' provision in the decision to disapprove them. To this end, we estimate the following model:

$$\text{Punish}_{c,r,s} = \alpha_0 + \alpha_1 \boldsymbol{\Theta_c^a} + \alpha_2 \boldsymbol{\Theta_c^l} + x_c'\gamma + \Lambda_r + \eta_s + \varepsilon_{c,r,s}, \tag{6}$$

where $\boldsymbol{\Theta_c^a}$ denotes a dummy variable that takes the value of 1 if the citizen observes an **ambiguous-allocation** signal, 0 otherwise. Similarly, the variable $\boldsymbol{\Theta_c^l}$ is a dummy variable that takes the value of 1 if the citizen observes a **low-allocation** signal, 0 otherwise. We add several controls, including the participant's age, sex, stratum, trust measure, preference for risk, patience measure, experience in previous experiments, and a metric for willingness to protest.[32]

Table 3: Citizens Disapproval Rate across Allocations ($\boldsymbol{\Theta}$), segmented by cost (π)

Dep Var: **Citizen Punishment**	Public goods cost (π)								
	Full sample			High cost ($\overline{\pi}$)			Low cost ($\underline{\pi}$)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Ambiguous allocation (θ^a)	0.049***	0.049***	0.048***	0.042**	0.041**	0.044**	0.058***	0.056***	0.057***
	(0.01)	(0.01)	(0.01)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Low allocation (θ^l)	0.313***	0.317***	0.319***	0.302***	0.299***	0.310***	0.326***	0.333***	0.333***
	(0.03)	(0.03)	(0.03)	(0.04)	(0.04)	(0.04)	(0.03)	(0.03)	(0.03)
Constant	0.046***	0.437**	0.300	0.058***	0.288*	0.064	0.031**	0.609***	0.581***
	(0.01)	(0.20)	(0.19)	(0.01)	(0.17)	(0.18)	(0.01)	(0.18)	(0.20)
Mean Dep. Variable	0.05	0.05	0.05	0.06	0.06	0.06	0.03	0.03	0.03
Controls		✓	✓		✓	✓		✓	✓
Round FE			✓			✓			✓
Session FE			✓			✓			✓
Observations	1995	1995	1995	1024	1024	1024	971	971	971
Clusters	399	399	399	389	389	389	379	379	379
Adjusted R-squared	0.12	0.14	0.16	0.11	0.13	0.16	0.13	0.17	0.20
p-value H_0: $\theta^a = \theta^l$	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Notes: The set of controls included in ***Controls*** are: age, sex, stratum, trust measure, risk measure, patience measure, previous experience participating in experiments, and a willingness-to-protest measure from a 6-item questionnaire. The *Mean Dep. Variable* expresses the unconditional control mean of the dependent variable. The p-value $H_0 : \theta^a = \theta^l$ denotes the statistical test of differences across allocations. Standard errors clustered at individual level are presented in parentheses. $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.1$

Table 3 presents the OLS estimates for Equation (6). Consistent with our theoretical model, citizens disapprove of politicians more as public goods allocations decrease. Columns (1) to (3) show the average treatment effect for the entire sample. The regression coefficients indicate that while the disapproval rate is close to 5% with a **high-allocation** signal, it

[32] We constructed a six-item index from LAPOP questions that measure the willingness to protest on various topics, including inequality, corruption, democracy, better salaries, and a better health system. The distribution of these variables is depicted in Figure D10 in the Online Appendix.

increases by 5*pp* for an **ambiguous-allocation** signal, and by 31*pp* for a **low-allocation** signal. Columns (4) to (6) show estimates under a high-cost environment, and Columns (7) to (9) present estimates under low-cost environments. The punishment pattern is consistent across both levels of costs, with the disapproval rate rising as public goods allocation decreases. Overall, these results suggest that citizens are willing to punish corrupt politicians, even at a relatively high personal cost.

5 Mechanisms

Having demonstrated that economic booms increase local political corruption, we now turn to assessing the mechanisms behind this result.

5.1 Empirical Expectations of Punishment

The theoretical framework outlined in Section 2 suggests that local politicians should expect a lower likelihood of punishment following a positive economic shock. This anticipated decline in disapproval increases the perceived benefits of corruption, potentially leading to higher rates of corruption. Since politicians first make policy decisions and then respond to empirical expectations, they may engage in strategic behavior, knowing that their choices influence the final allocation of public goods. Specifically, corrupt politicians may anticipate that during bad economic times, only low or ambiguous allocations are feasible, leading to a higher expected disapproval rate.

To assess this potential mechanism, we examine the unconditional expected punishment rate across policy decisions segmented by costs. Figure 9 illustrates this behavior pooled across rounds. The left panel shows that honest politicians in high-cost environments have an expected disapproval rate of 11% compared to 6% in low-cost environments (p-value = 0.007). Similarly, the right panel indicates that corrupt politicians in high-cost environments expect to be punished about 35% of the time, while in low-cost environments, they expect a 28% punishment rate (p-value = 0.034).

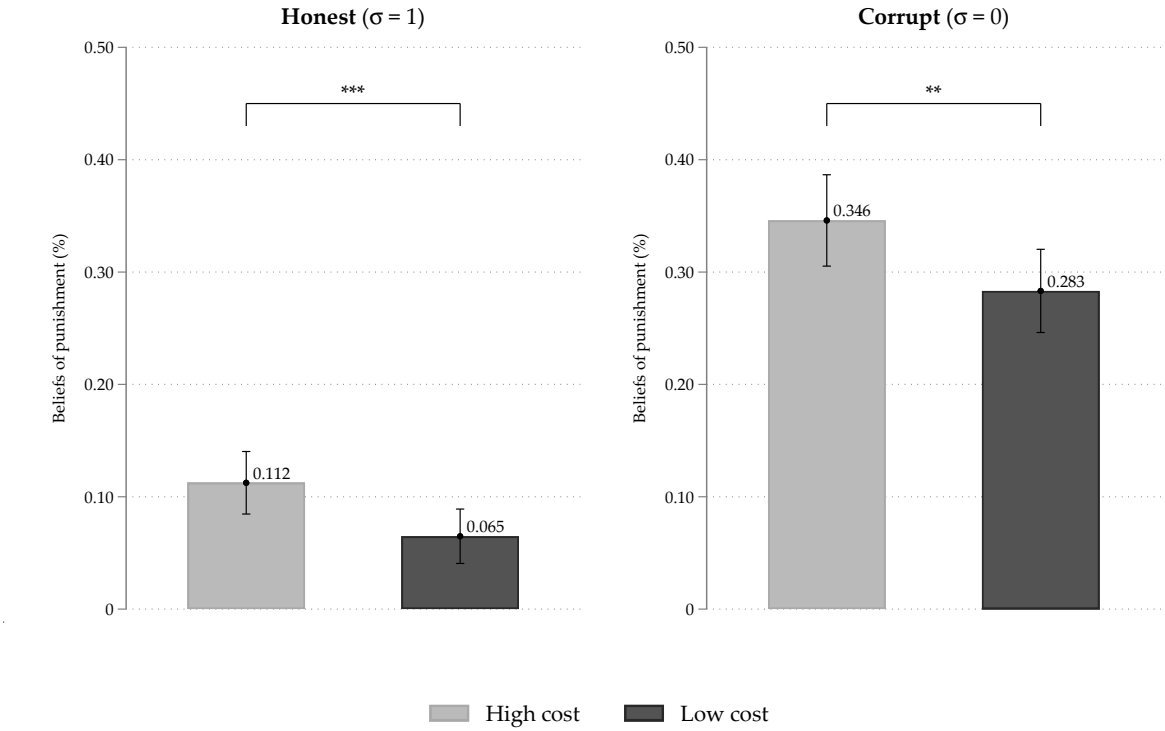
To validate this channel, we estimate the following model:

$$\text{ExpPunish}^{h}_{p,r,s} = \beta_0 + \beta_1 \text{Cost}_{p,r,s} + x'_p\gamma + \Lambda_r + \eta_s + \varepsilon_{p,r,s}, \tag{7}$$

where $\text{ExpPunish}^{h}_{p,r,s}$ denotes the politicians' expectations of disapproval. Once again, $\text{Cost}_{p,r,s}$ denotes the public goods construction cost and captures our main treatment. We

include the same set of sociodemographic controls presented in the estimation of equation (4). Finally, we estimate each model considering if the politician was honest $(\sigma = 1)$ or corrupt $(\sigma = 0)$.

Figure 9: Politician Expected Punishment Segmented by Costs (π) and Policy Choice



Notes: The figure depicts the average expected disapproval rate by the politician across all rounds, segmented by production costs and policy choices. **High cost** = 20 EP. **Low cost** = 10 EP. Dark lines show 95% clustered robust confidence intervals. No controls included. $^{***}p < 0.01$,** $p < 0.05$,* $p < 0.1$, n.s $p > 0.1$. Reported significance levels are derived from hypothesis testing of a fully saturated linear model, where the dependent variable is the citizen's choice to disapprove of the incumbent, and the regressors are dummy variables associated with the treatments.

The OLS estimates are shown in Table 4. Columns (1) to (3) present estimates for punishment expectations when the politician was honest. Honest politicians expect to be punished 11% of the time, consistent with the uncertainty surrounding the final public goods allocation, which may generate an ambiguous allocation signal. The coefficients further indicate that in good economic times, politicians expect a 4 to 5 percentage point reduction in the likelihood of punishment.

Similarly, Columns (4) to (6) present estimates for cases where the politician was corrupt. The expected disapproval rate is approximately 35%, significantly higher than that of honest politicians (p-value = 0.000). In good economic times, corrupt politicians anticipate a 6 to 7 percentage point reduction in the likelihood of punishment. The point estimates also reveal significant economic implications, with economic booms leading to a reduction

Table 4: Politicians Expected Punishment across cost (π), Segmented by Policy Choice

Dep Var: **Belief of punishment**	*Politician's decision (σ)*					
	Honest			Corrupt		
	(1)	(2)	(3)	(4)	(5)	(6)
Low cost (Treatment)	-0.048***	-0.044**	-0.041**	-0.063**	-0.068**	-0.064**
	(0.02)	(0.02)	(0.02)	(0.03)	(0.03)	(0.03)
Constant	0.112***	0.380***	0.411***	0.346***	0.574***	0.560***
	(0.02)	(0.13)	(0.13)	(0.02)	(0.17)	(0.19)
Mean Dep. Variable	0.11	0.11	0.11	0.35	0.35	0.35
Effect size (%)	-42.3	-39.0	-36.7	-18.1	-19.7	-18.5
Controls		✓	✓		✓	✓
Round FE			✓			✓
Session FE			✓			✓
Observations	899	899	899	1101	1101	1101
Clusters	309	309	309	351	351	351
Adjusted R-squared	0.01	0.09	0.16	0.00	0.02	0.08

Notes: The set of controls included in ***Controls*** are: age, sex, stratum, trust measure, risk measure, patience measure, and previous experience participating in experiments. The *Mean Dep. Variable* expresses the unconditional control mean of the dependent variable. The *Effect size (%)* denotes the size of the treatment effect as a percentage compared with the unconditional mean of the control group. Standard errors clustered at the individual level are presented in parentheses. $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.1$

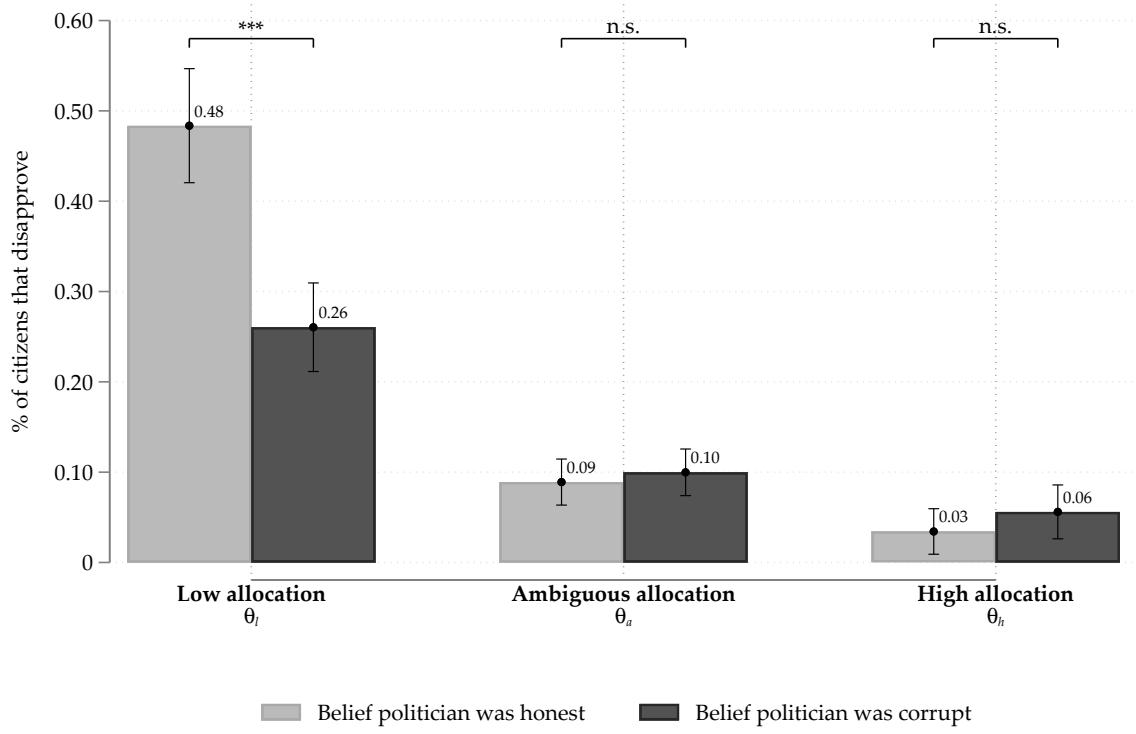
in the expected disapproval rate ranging from -42% to -20% compared to the control group.

5.2 Heterogeneous Punishment Behavior across Corruption Beliefs

Regarding political accountability behavior, we leverage citizens' incentive-compatible empirical expectations to estimate how prior beliefs about corruption influence punishment. Figure 10 displays the unconditional disapproval rate across the final public goods allocation of public goods segmented by prior beliefs of corruption.

From Figure 10, when citizens observe a low public goods allocation and expect the politician to be honest, the disapproval rate is 48%; however, for those expecting the politician to be corrupt, the rate drops to 26%, reflecting a 22*pp* reduction. Additionally, when citizens observe an ambiguous allocation, prior beliefs have minimal effect, with disapproval rates consistently around 9%. Lastly, for high allocations of public goods, disapproval is 3% among citizens expecting honesty and 6% among those expecting dishonesty, though this difference is not statistically significant.

Figure 10: Citizen Punishment Segmented by Prior beliefs (β) across Allocations ($\boldsymbol{\Theta}$).



Notes: The figure depicts the average disapproval rate of the citizen in all rounds, segmented by prior beliefs about corruption and the final allocation of public goods. **High cost** = 20 EP. **Low cost** = 10 EP. The dark lines show robust confidence intervals clustered with 95%. No controls included. $^{***}p < 0.01$,** $p < 0.05$,* $p < 0.1$, n.s $p > 0.1$. Reported significance levels come from hypothesis testing from a fully saturated linear model where the dependent variable is the citizen's choice to disapprove of the incumbent, and regressors are dummy variables associated with treatments.

Given that punishment seems to be heterogeneous across prior beliefs of corruption, and in order to formally test this mechanism, we estimate the following model:

$$\text{Punish}_{c,r,s} = \alpha_0 + \alpha_1 \boldsymbol{\Theta}_{\boldsymbol{c}}^{\boldsymbol{a}} + \alpha_2 \boldsymbol{\Theta}_{\boldsymbol{c}}^{\boldsymbol{l}} + \alpha_3 \text{BC} + \beta_1(\boldsymbol{\Theta}_{\boldsymbol{c}}^{\boldsymbol{a}} \times \text{BC}) + \beta_2(\boldsymbol{\Theta}_{\boldsymbol{c}}^{\boldsymbol{l}} \times \text{BC}) + \Lambda_r + \eta_s + \varepsilon_{c,r,s}, \quad (8)$$

where BC denotes the citizen's prior beliefs of corruption. Table 5 presents the results on the heterogeneous effect of prior beliefs across observed allocations.

Column (1) indicates that prior beliefs of corruption do not enhance political accountability; instead, they reduce the disapproval rate. While citizens are more likely to punish politicians when they observe a low allocation of public goods, the disapproval rate is significantly higher if they expect the politician to be honest (48%) compared to when they expect the politician to be corrupt (26%). This behavior suggests that when citizens anticipate poor performance, their motivation to punish decreases, aligning with the findings of De la Cuesta et al. (2022).

Table 5: Citizens Disapproval Rate across Allocations, by Prior Beliefs of Corruption (β)

Dep Var: **Citizen punishment**	OLS			
	(1)	(2)	(3)	(4)
Ambiguous allocation (θ^a)	0.048***	0.055***	0.047**	0.045**
	(0.01)	(0.02)	(0.02)	(0.02)
Low allocation (θ^l)	0.314***	0.449***	0.445***	0.442***
	(0.03)	(0.04)	(0.04)	(0.04)
Belief Corrupt	-0.051***	0.022	0.016	0.017
	(0.02)	(0.02)	(0.02)	(0.02)
Ambiguous allocation (θ^a) $\times$ Belief Corrupt		-0.011	-0.001	0.003
		(0.03)	(0.03)	(0.03)
Low allocation (θ^l) $\times$ Belief Corrupt		-0.245***	-0.236***	-0.227***
		(0.05)	(0.05)	(0.05)
Constant	0.073***	0.034***	0.356*	0.227
	(0.01)	(0.01)	(0.20)	(0.20)
Mean Dep. Variable	0.05	0.05	0.05	0.05
Effect size (%)	104.8	119.3	102.8	97.1
Controls			✓	✓
Round FE				✓
Session FE				✓
Observations	1995	1995	1995	1995
Clusters	399	399	399	399
Adjusted R-squared	0.13	0.15	0.17	0.18
p-value H_0: $\theta^a = \theta^l$	0.000	0.000	0.000	0.000

Notes: The set of controls included in ***Controls*** are: age, sex, stratum, trust measure, risk measure, patience measure, previous experience participating in experiments, and the willingness-to-protest measure from a 6-item questionnaire. The *Mean Dep. Variable* expresses the unconditional control mean of the dependent variable. The p-value $H_0 : \theta^a = \theta^l$ denotes the statistical test of differences across allocations. Standard errors clustered at the individual level are presented in parentheses. $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.1$

5.3 Does Cost Affect Prior Beliefs of Corruption?

Given that prior beliefs of corruption influence citizen's willingness to disapprove, we now estimate the effect of good economic times on those prior beliefs. Table 6 presents OLS estimates of the average treatment effect of low costs on prior beliefs of corruption. In low-cost environments, citizens expect politicians to be 8*pp* less corrupt than in high-cost environments. Furthermore, the point estimates indicate substantial economic impact, with reductions in incentive-compatible empirical expectations nearing 14%. These findings suggest that instead of anticipating increased corruption in good economic times, citizens' trust in politicians actually rises——a result consistent with observed behavior in the Latin America region (Figure 1).

The full punishment mechanism unfolds as follows: favorable economic conditions increase the incentives for corruption, leading to more frequent instances of inadequate public goods provision. Initially, citizens are more inclined to trust politicians, as good

Table 6: Citizens Prior Beliefs of Corruption across Cost (π)

Dep Var: **Belief of corruption**	*Prior beliefs (β)*		
	(1)	(2)	(3)
Low cost (Treatment)	-0.084***	-0.077***	-0.073***
	(0.03)	(0.03)	(0.03)
Constant	0.574***	0.356**	0.309**
	(0.02)	(0.14)	(0.14)
Mean Dep. Variable	0.57	0.57	0.57
Effect size (%)	-14.6	-13.5	-12.8
Controls		✓	✓
Round FE			✓
Session FE			✓
Observations	1995	1995	1995
Clusters	399	399	399
Adjusted R-squared	0.01	0.04	0.07

Notes: The set of controls included in ***Controls*** are: age, sex, stratum, trust measure, risk measure, patience measure, previous experience participating in experiments, and the willingness-to-protest measure from a 6-item questionnaire. The *Mean Dep. Variable* expresses the unconditional control mean of the dependent variable. The *Effect size (%)* denotes the size of the treatment effect as a percentage compared with the unconditional mean of the control group. Standard errors clustered at the individual level are presented in parentheses. $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.1$

times reduce their prior beliefs about the likelihood of corruption. However, when voters observe a low allocation of public goods and subsequently learn that the incumbent engaged in corrupt behavior, they experience a sense of betrayal, which leads to a sharp increase in political disapproval.

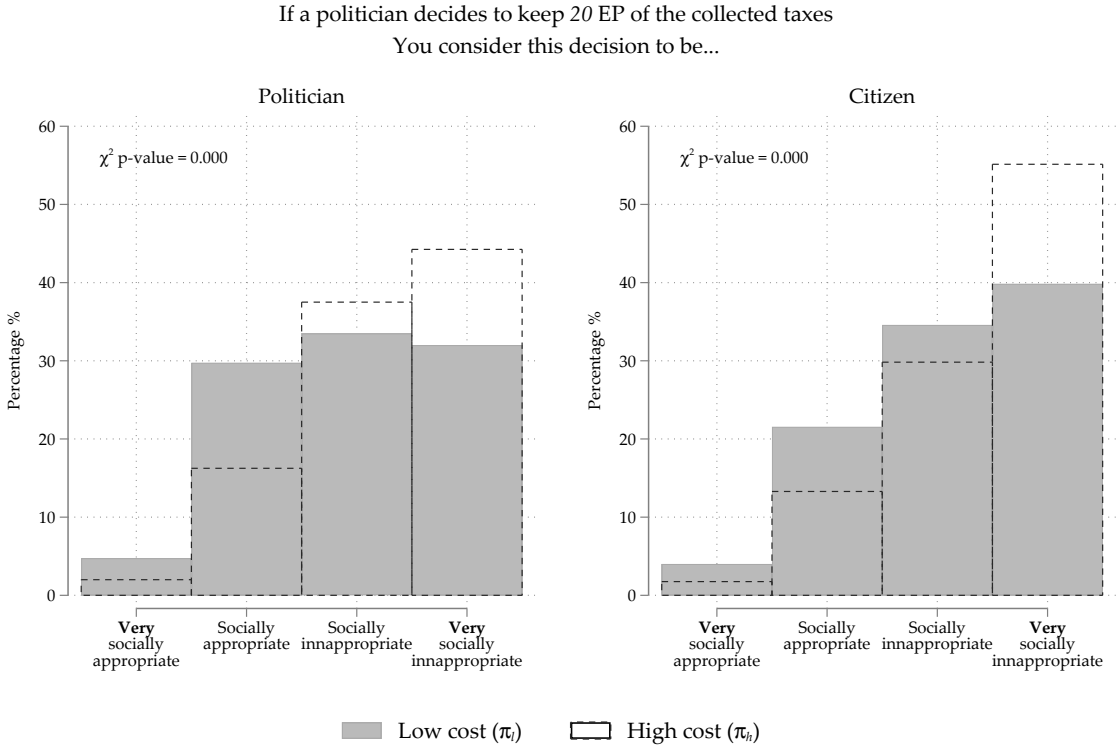
5.4 Social Norms

Lastly, we conduct a comparative analysis of responses from the coordination game questionnaire (Krupka and Weber, 2013), based on the following question: "*In the situation where the cost was (**High**, **Low**). If a politician decides to keep **20** EP of the collected taxes, you consider this decision to be...*?" In this task, participants attempted to match the modal appropriateness rating of others within their participant type.

As shown in the left panel of Figure 11, in high-cost scenarios, a consensus emerges among politicians that personally retaining a portion of collected taxes is very socially inappropriate. However, this behavior shifts dramatically in a low-cost environment where a significant portion of politicians view keeping some of the taxes for personal use

as more acceptable, illustrating a relaxation of social norms (descriptive norm) against corruption. Citizens' beliefs behave similarly. High-cost settings reinforce the strong social inappropriateness of corruption. Yet, in low-cost settings, fewer citizens view corrupt behavior as highly inappropriate, suggesting some degree of normalization or tolerance.

Figure 11: Social Norms Histogram across Costs and Participants Types



Notes: The figure illustrates the distribution of responses to the question: *"In the situation where the cost was (**High**, **Low**), if a politician decides to keep **20** EP of the collected taxes, do you consider this decision to be...?"* The histogram shows responses for low cost (in gray) and high cost (in dashed black). The left panel presents the distribution of responses for politicians, while the right panel shows those for citizens. (Left panel: χ^2 test p-value = 0.000; Right panel: χ^2 test p-value = 0.000.)

6 Discussion

In this study, we investigate the relationship between economic conditions and citizens' willingness to hold politicians accountable, as well as the impact of economic conditions on political corruption. While the literature extensively documents the adverse effects of corruption on economic performance, less attention has been given to how economic fluctuations shape corruption levels and accountability. By addressing this gap, our research offers new insights into political behavior during economic fluctuations and lays the groundwork for more effective governance strategies. We achieve this by developing a theoretical model and empirically testing its predictions through a laboratory experiment.

Our theoretical model provides key insights into the relationship between economic performance and political behavior. We predict that favorable economic conditions, characterized by low costs of providing public goods, increase politicians' inclination to engage in corrupt activities. This occurs largely because politicians anticipate a lower likelihood of punishment, as citizens may be more willing to overlook corruption in times of economic prosperity. Additionally, our model predicts that good economic times weaken political accountability, reflected in citizens' reduced willingness to disapprove of corrupt politicians. This highlights a broader trend of diminished punishment and weaker enforcement.

Empirically, our experiments support these theoretical findings. Corruption rates are significantly higher in low-cost environments, with an increase of approximately 14% compared to the baseline. Additionally, in these low-cost settings, politicians anticipated a nearly 7-percentage-point reduction in the disapproval rate, which further contributed to the rise in corruption. In contrast, citizens' willingness to disapprove of politicians remained relatively stable across cost levels, as they prioritized punishment based on the observed allocation of public goods (Ferejohn, 1986). Specifically, citizens are willing to punish corrupt politicians nearly 36% of the time when the allocation is low and between 3 and 6 percent of the time when the allocation is high.

Politicians recognize that citizens can identify corruption to some extent, believing they are three times more likely to be punished when corrupt than when honest. However, politicians do not expect to be punished every time they engage in corruption, and anticipate occasional punishment even when acting honestly. Consequently, incentives for corruption increase when politicians perceive citizens as less sophisticated. Moreover, citizens are less likely to punish corruption if they already suspect politicians are corrupt, suggesting that mistrust fosters conditions that enable further corruption.

While our findings offer novel insights, we acknowledge important limitations regarding external validity. First, as with any laboratory experiment, our setting abstracts from some features of real-world politics, most notably the stakes involved. In practice, corruption can carry legal consequences such as imprisonment and may deter reelection, whereas our experimental context imposes no such penalties. That said, although the magnitude of the incentives differs, we see no compelling reason to expect that the direction of behavioral responses or the underlying logic of the accountability mechanism would reverse in higher-stakes environments.

A second concern relates to the composition of our experimental sample, which consists primarily of university students. This may be viewed as a limitation, particularly in a

setting where other-regarding preferences play a central role. While earlier evidence suggests that students may exhibit higher levels of selfish behavior in economic games relative to non-students (Belot et al., 2015), more recent studies find broadly similar distributions of social preferences, especially with respect to altruistic and maximin (i.e., concern for improving the welfare of the worst-off) types across student and non-student populations, particularly among individuals with higher levels of educational attainment (Epper et al., 2023). This is particularly relevant given that both our sample and the population of interest (e.g., public officials) are highly educated. Even if one were to assume that selfish behavior exhibit by students may be driving the effects on political corruption, other studies using similar experimental samples in Colombia report that approximately 70% of college students behave in a non-selfish manner (Barinas-Forero and Scartascini, 2025), suggesting ample heterogeneity in other-regarding motives during our experimental interaction.[33]

Based on these findings, policymakers should remain vigilant about the risk of increased corruption during periods of economic growth. Strengthening transparency and accountability mechanisms, particularly during periods of prosperity, is crucial for maintaining high governance standards. Accountability becomes more challenging when citizens struggle to identify corruption, a difficulty amplified by economic cycles. When multiple levels of government share responsibility for providing public goods, holding politicians accountable becomes even more complex. Thus, while decentralization may offer a solution, it is effective only if it clarifies rather than obscures responsibilities.

Future research could expand on our findings by examining the long-term effects of good and bad economic times on corruption across different contexts and exploring how information dissemination might mitigate the negative impact of economic booms on political accountability. Additionally, analyzing the interaction between various types of economic shocks and political systems could provide valuable insights for combating corruption under diverse economic conditions.

[33]Our experiment was conducted at a leading Colombian university that has historically produced a significant share of the country's political elite, including multiple former presidents. Moreover, nearly 40% of participants are enrolled in degree programs directly related to public administration and governance, making this sample particularly relevant for studying the prospective behavior of future public officials.

References

Abadie, Alberto, Susan Athey, Guido W. Imbens, and Jeffrey M. Wooldridge, "When Should You Adjust Standard Errors for Clustering?," *The Quarterly Journal of Economics*, 2023, *138* (1), 1–35.

Abbink, Klaus and Danila Serra, "Anti-corruption Policies: Lessons from the Lab," in "New Advances in Experimental Research on Corruption," Emerald Group Publishing Limited, 2012.

_ **and Heike Hennig-Schmidt**, "Neutral versus loaded instructions in a bribery experiment," *Experimental Economics*, 2006, *9*, 103–121.

_ **, Bernd Irlenbusch, and Elke Renner**, "An Experimental Bribery Game," *Journal of Law, Economics and Organization*, 2002, *18* (2), 428–454.

Ades, Alberto and Rafael Di Tella, "Rents, competition, and corruption," *American Economic Review*, 1999, *89* (4), 982–993.

Adserà, Alícia, Carles Boix, and Mark Payne, "Are You Being Served? Political Accountability and Quality of Government," *The Journal of Law, Economics and Organization*, 2003, *19* (2), 445 –490.

Alekseev, Aleksandr, Gary Charness, and Uri Gneezy, "Experimental methods: When and why contextual instructions are important," *Journal of Economic Behavior and Organization*, 2017, *134*, 48–59.

Alesina, Alberto, Alberto Carrasquilla, and Juan Echevarria, "Decentralization in Colombia," in Alberto Alesina, ed., *Institutional Reforms in Colombia.*, MIT Press, 2005.

Almer, Christian, Jeremy Laurent-Lucchetti, and Manuel Oechslin, "Water scarcity and rioting: Disaggregated evidence from Sub-Saharan Africa," *Journal of Environmental Economics and Management*, 2017, *86* (1), 193–209.

Altunbaş, Yener and John Thornton, "Does financial development reduce corruption?," *Economics Letters*, 2012, *114* (2), 221–223.

Ardanaz, Martín and Alejandro Izquierdo, "Current expenditure upswings in good times and public investment downswings in bad times? New evidence from developing countries," *Journal of Comparative Economics*, 2022, *50* (1), 118–134.

Armand, Alex, Alexander Coutts, Pedro Vicente, and Ines Vilela, "Measuring corruption in the field using behavioral games," *Journal of Public Economics*, 2023, *218* (10), 47–99.

Armantier, Olivier and Amadou Boly, "A controlled field experiment on corruption," *European Economic Review*, 2011, *55* (8), 1072–1082.

Asher, Sam and Paul Novosad, "Rent-seeking and criminal politicians: Evidence from mining booms," *Review of Economics and Statistics*, 2023, *105* (1), 20–39.

Bai, Jie, Seema Jayachandran, Edmund J. Malesky, and Benjamin Olken, "Firm Growth and Corruption: Empirical Evidence from Vietnam," *The Economic Journal*, 2017, *129* (618), 651–677.

Banerjee, Abhijit, Sendhil Mullainathan, and Rema Hanna, "Corruption," Technical Report, National Bureau of Economic Research 2012.

Banerjee, Rtiwik, "On the interpretation of bribery in a laboratory corruption game: moral frames and social norms," *Experimental Economics*, 2016, *19*, 240–267.

Bardhan, Pranab, "Corruption and Development: A Review of Issues," *Journal of Economic Literature*, 1997, *35* (3), 1320–1346.

Barinas-Forero, Andres, "Why should my group trust yours? Collective trust and trustworthiness under Economic Shocks," *Documento CEDE*, 2024, (29), 1–48.

_ **and Carlos Scartascini**, "Wage cuts, Social Distributional Preferences and Political Corruption," *IDB Working Paper*, 2025, pp. 1–42.

Barr, Abigail and Danila Serra, "The effects of externalities and framing on bribery in a petty corruption experiment," *Experimental Economics*, 2009, *12*, 488–503.

Barro, Robert J, "The control of politicians: an economic model," *Public Choice*, 1973, *14*, 19–42.

Belot, Michele, Raymond Duch, and Luis Miller, "A comprehensive comparison of students and non-students in classic experimental games," *Journal of Economic Behavior & Organization*, 2015, *113*, 26–33.

Benton, Allyson Lucinda, "Dissatisfied Democrats or Retrospective Voters?: Economic Hardship, Political Institutions, and Voting Behavior in Latin America," *Comparative Political Studies*, 2005, *38* (4), 417–442.

Berninghaus, Siegfried, Sven Haller, Tyll Krüger, Thomas Neumann, Stephan Schosser, and Bodo Vogt, "Risk attitude, beliefs, and information in a Corruption Game–An experimental analysis," *Journal of Economic Psychology*, 2013, *34*, 46–60.

Bhattacharyya, Sambit and Roland Hodler, "Natural resources, democracy and corruption," *European Economic Review*, 2010, *54* (4), 608–621.

Bó, Ernesto Dal and Martín Rossi, "Corruption and inefficiency: Theory and evidence from electric utilities," *Journal of Public Economics*, 2007, *91* (5-6), 939–962.

Bonvecchi, Alejandro and Germán Lodola, "The dual logic of intergovernmental transfers: Presidents, governors, and the politics of coalition-building in Argentina," *Publius: The Journal of Federalism*, 2011, *41* (2), 179–206.

Burman, Leonard E. and Joel Slemrod, *Taxes in AmericaWhat Everyone Needs to Know®: What Everyone Needs to Know®*, New York, NY: Oxford University Press, 2020.

Cagala, Tobias, Ulrich Glogowsky, Veronika Grimm, and Johannes Rincke, "Public goods provision with rent-extracting administrators," *The Economic Journal*, 2019, *129* (620), 1593–1617.

Cameron, Lisa, Ananish Chaudhuri, Nisvan Erkal, and Lata Gangadharan, "Propensities to engage in and punish corrupt behavior: Experimental evidence from Australia, India, Indonesia and Singapore," *Journal of Public Economics*, 2009, *93* (7), 843–851.

Chen, Daniel, Martin Schonger, and Chris Wickens, "oTree – An open-source platform for laboratory, online, and field experiments," *Journal of Behavioral and Experimental Finance*, 2016, *9*, 88–97.

Chenoweth, Erica, Barton Hamilton, Hedwig Lee, Nicholas Papageorge, Stephen Roll, and Matthew Zahn, "Who protests, what do they protest, and why?," Technical Report, National Bureau of Economic Research 2022.

Chong, Alberto, Ana L. De La O, Dean Karlan, and Leonard Wantchekon, "Does Corruption Information Inspire the Fight or Quash the Hope? A Field Experiment in Mexico on Voter Turnout, Choice, and Party Identification," *The Journal of Politics*, 2015, *77* (1), 55–71.

Cieślik, Andrzej and Łukasz Goczek, "Control of corruption, international investment, and economic growth–Evidence from panel data," *World Development*, 2018, *103*, 323–335.

Corbacho, Ana, Daniel Gingerich, Virginia Oliveros, and Mauricio Ruiz-Vega, "Corruption as a self-fulfilling prophecy: Evidence from a survey experiment in Costa Rica," *American Journal of Political Science*, 2016, *60* (4), 1077–1092.

Cubel, Maria, Anastasia Papadopoulou, and Santiago Sánchez-Pagés, "Identity and political corruption: a laboratory experiment," *Economic Theory*, 2024, pp. 1–24.

Dube, Oeindrila and Juan F Vargas, "Commodity price shocks and civil conflict: Evidence from Colombia," *Review of Economic studies*, 2013, *80* (4), 1384–1421.

Epper, Thomas, Ernst Fehr, and Julien Senn, "Social preferences across subject pools: students vs. general population," *URPP Equality of Opportunity Discussion Paper Series*, 2023, (46).

Ferejohn, John, "Incumbent performance and electoral control," *Public Choice*, 1986, *50*, 5–25.

Ferraz, Claudio and Frederico Finan, "Electoral Accountability and Corruption: Evidence from the Audits of Local Governments," *American Economic Review*, 2011, *101* (4), 1274–1311.

Fisman, Raymond and Jakob Svensson, "Are corruption and taxation really harmful to growth? Firm level evidence," *Journal of Development Economics*, 2007, *83* (1), 63–75.

Galbraith, John Kenneth, *The great crash 1929*, Houghton Mifflin Harcourt, 1997.

Gokcekus, Omer and Yui Suzuki, "Business cycle and corruption," *Economics Letters*, 2011, *111* (2), 138–140.

Greiner, Ben, "Subject pool recruitment procedures: organizing experiments with ORSEE," *Journal of the Economics Science Association*, 2015, *1*, 114–125.

Grossman, Herschel, "A General Equilibrium Model of Insurrections," *American Economic Review*, 1991, *81* (4), 912–921.

Gründler, Klaus and Niklas Potrafke, "Corruption and economic growth: New empirical evidence," *European Journal of Political Economy*, 2019, *60*, 1–14.

Gupta, Sahil, Harold Davoodi, and Ricardo Alonso-Terme, "Does Corruption Affect Income Inequality and Poverty?," *IMF Working Paper*, 1998, (79), 1–43.

Hu, Lin and Mandar Oak, "Intermediated corruption under asymmetric punishment," *Journal of Economic Behavior Organization*, 2023, *215*, 490–499.

Huang, Chiung-Ju, "Is corruption bad for economic growth? Evidence from Asia-Pacific countries," *The North American Journal of Economics and Finance*, 2016, *35*, 247–256.

Ivlevs, Artjoms and Timothy Hinks, "Global economic crisis and corruption," *Public Choice*, 2015, *162*, 425–445.

Izquierdo, Alejandro, Carola Pessino, Guillermo Vuletin et al., *Better spending for better lives: how Latin America and the Caribbean can do more with less*, Vol. 10, Inter-American Development Bank, 2018.

Jha, Chandan Kumar, "Condoning corruption: Who votes for corrupt political parties?," *Journal of Economic Behavior Organization*, 2023, *215*, 74–88.

Kayser, Mark A. and Michael Peress, "Benchmarking across Borders: Electoral Accountability and the Necessity of Comparison," *American Political Science Review*, 2012, *106* (3), 661–684.

Keefer, Philip and Carlos Scartascini, *Trust: the key to social cohesion and growth in Latin America and the Caribbean*, [Washington, DC]: Inter-American Development Bank, 2022.

Kéïta, Kouramoudou and Hannu Laurila, "Do Business Booms Trigger Corruption?," *European Integration Studies*, 2021, *15*, 1–13.

Khlif, Hichem and Ines Amara, "Political connections, corruption and tax evasion: a cross-country investigation," *Journal of Financial Crime*, 2019, *26*, 401.

Klašnja, Marko and Joshua A. Tucker, "The economy, corruption, and the vote: Evidence from experiments in Sweden and Moldova," *Electoral Studies*, 2013, *32* (3), 536–543.

_ , Noam Lupu, and Joshua A. Tucker, "When do voters sanction corrupt politicians?," *Journal of Experimental Political Science*, 2021, *8* (2), 161–171.

Krupka, Erin L. and Roberto A. Weber, "Identifying Social Norms Using Coordination Games: Why Does Dictator Game Sharing Vary?," *Journal of the European Economic Association*, 2013, *11* (3), 495–524.

la Cuesta, Brandon De, Lucy Martin, Helen Milner, and Daniel Nielson, "Owning it: Accountability and citizens' ownership over oil, aid, and taxes," *The Journal of Politics*, 2022, *84* (1), 304–320.

LAPOP, "Latin American Public Opinion Project AmericasBarometer," 2024.

Leight, Jessica, Dana Foarta, Rohini Pande, and Laura Ralston, "Value for money? Vote-buying and politician accountability," *Journal of Public Economics*, 2020, *190*, 104227.

Leite, Carlos and Jens Weidmann, *Does mother nature corrupt? Natural resources, corruption, and economic growth*, International Monetary Fund, 1999.

Lewis-Beck, Michael S. and Mary Stegmaier, "Economic Determinants of Electoral Outcomes," *Annual Review of Political Science*, 2000, *3* (Volume 3, 2000), 183–219.

Lynge, Halfdan and Ferran Martinez i Coma, "The effect of economic downturns on voter turnout in Africa," *Electoral Studies*, 2022, *76*, 102456.

López-Laborda, Julio, Fernando Rodrigo, and Eduardo Sanz-Arcega, "Do you know to whom you pay your taxes? The case of decentralised Spain," *Social Policy and Administration*, 2023, *57* (5), 770–788.

Mani, Anandi and Sharun Mukand, "Democracy, visibility and public good provision," *Journal of Development Economics*, 2007, *83* (2), 506–529.

Martin, Lucy, "Taxation, loss aversion, and accountability: theory and experimental evidence for taxation's effect on citizen behavior," *Unpublished paper, Yale University, New Haven, CT*, 2014.

Martinelli, Cesar, "Accountability and Grand Corruption," *American Economic Journal: Microeconomics*, 2022, *14* (4).

Martinelli, César and Naila Sebastián Esandi, "Electoral accountability in the lab," *Games and Economic Behavior*, 2025.

Martínez, Luis R., "Natural resource rents, local taxes, and government performance: Evidence from Colombia," *Review of Economics and Statistics*, 2023, pp. 1–28.

Mauro, Paolo, "Corruption and Growth," *The Quarterly Journal of Economics*, 1995, *110* (3), 681–712.

Méon, Pierre-Guillaume and Khalid Sekkat, "Does corruption grease or sand the wheels of growth?," *Public Choice*, 2005, *122*, 69–97.

Mocan, Naci, "What determines corruption? International evidence from microdata," *Economic Inquiry*, 2008, *46* (4), 493–510.

Nannicini, Tommaso, Andrea Stella, Guido Tabellini, and Ugo Troiano, "Social capital and political accountability," *American Economic Journal: Economic Policy*, 2013, *5* (2), 222–250.

Olken, Benjamin, "Corruption and the costs of redistribution: Micro evidence from Indonesia," *Journal of Public Economics*, 2006, *90* (4-5), 853–870.

Ortiz, Isabel, Sara Burke, Mohamed Berrada, and Hernán Saenz Cortés, *World protests: A study of key protest issues in the 21st century*, Springer Nature, 2022.

Pavão, Nara, "Corruption as the only option: The limits to electoral accountability," *The Journal of Politics*, 2018, *80* (3), 996–1010.

Persson, Anna, Bo Rothstein, and Jan Teorell, "Why anticorruption reforms fail—systemic corruption as a collective action problem," *Governance*, 2013, *26* (3), 449–471.

Persson, Torsten and Guido Tabellini, *Political Economics: Explaining Economic Policy*, Vol. 2, The MIT Press, 2002.

_ , Gerard Roland, and Guido Tabellini, "Separation of powers and political accountability," *The Quarterly Journal of Economics*, 1997, *112* (4), 1163–1202.

Przeworski, Adam, Susan Stokes, and Bernand Manin, *Democracy, Accountability, and Representation*, Cambridge University Press, 1999.

Rijckeghem, Caroline Van and Beatrice Weder, "Bureaucratic corruption and the rate of temptation: do wages in the civil service affect corruption, and by how much?," *Journal of Development Economics*, 2001, *65* (2), 307–331.

Ristei, Mihaiela and Cristian Gugiu, "Economic Crisis and Corruption in the Eurepean Union," *Journal of Methods and Measurement in the Social Sciences*, 2016, *7*, 1–23.

Rivera, Eduardo, Enrique Seira, and Saumitra Jha, "Democracy Corrupted: Apex Corruption and the Erosion of Democratic Values," *Available at SSRN: https://ssrn.com/abstract=4828243*, 2024.

Romo, Cristian Márquez and Xavier Romero-Vidal, "In the Eyes of the Beholder? Understanding Policymakers' Perceptions of Corruption," *Legislative Studies Quarterly*, 2023, *48* (3), 535–559.

Saha, Shrabani and Kunal Sen, "Do economic and political crises lead to corruption? The role of institutions," *Economic Modelling*, 2023, *124*, 106307.

Schulze, Günther and Björn Frank, "Deterrence versus intrinsic motivation: Experimental evidence on the determinants of corruptibility," *Economics of Governance*, 2003, *4*, 143–160.

Serra, Danila, "Empirical determinants of corruption: A sensitivity analysis," *Public Choice*, 2006, *126* (1), 225–256.

Shleifer, Andrei and Robert W. Vishny, "Corruption," *The Quarterly Journal of Economics*, 1993, *108* (3), 599–617.

Sjursen, Ingrid Hoem, "Accountability and taxation: Experimental evidence," *Journal of Economic Behavior and Organization*, 2023, *216*, 386–432.

Solaz, Hector, Catherine E. De Vries, and Roosmarijn A. De Geus, "In-group loyalty and the punishment of corruption," *Comparative Political Studies*, 2019, *52* (6), 896–926.

Tanner, Carmen, Stefan Linder, and Matthias Sohn, "Does moral commitment predict resistance to corruption? experimental evidence from a bribery game," *Plos One*, 2022, *17* (1), 1–22.

Tavares, Jose, "Does foreign aid corrupt?," *Economics Letters*, 2003, *79* (1), 99–106.

Torsello, Davide, "Corruption and the economic crisis: empirical indications from Eastern Europe," *International Issues and Slovak Foreign Policy Affairs*, 2010, *19* (02), 65–75.

Treisman, Daniel, "The causes of corruption: a cross-national study," *Journal of Public Economics*, 2000, *76* (3), 399–457.

Voors, Maarten J, Erwin H Bulte, and Richard Damania, "Income shocks and corruption in Africa: does a virtuous cycle exist?," *Journal of African Economies*, 2011, *20* (3), 395–418.

Weitz-Shapiro, Rebecca and Matthew S. Winters, "Lacking Information or Condoning Corruption: When Do Voters Support Corrupt Politicians?," *Comparative Politics*, 2013, *45* (4), 418–436.

_ **and** _ , "Who's in Charge Here? Direct and Indirect Accusations and Voter Punishment of Corruption," *Political Research Quarterly*, 2016, *69* (2), 207–219.

_ **and** _ , "Can Citizens Discern? Information Credibility, Political Sophistication, and the Punishment of Corruption in Brazil," *Comparative Politics*, 2017, *79* (1), 60–74.

Zechmeister, Elizabeth J. and Daniel Zizumbo-Colunga, "The Varying Political Toll of Concerns About Corruption in Good Versus Bad Economic Times," *Comparative Political Studies*, 2013, *46* (10), 1190–1218.

Online Appendix

Corruption and Political Accountability in Good and Bad Economic Times

Andres Barinas-Forero Carlos Scartascini

Contents

A Extensive Representation Payoffs **2**

B Proofs **2**

B.1 Proof Lemma 1 2

B.2 Proof Lemma 2 3

B.3 Proof Lemma 3 4

B.4 Proof Lemma 4 5

B.5 Proof Theorem 1 7

B.6 Proof Corollary 1 8

B.7 Proof Theorem 2 9

C Experimental Design **9**

C.1 Participants' Timelines 9

D Additional Figures and Tables **10**

E Experimental Screen Instructions **20**

E.1 *Original Instructions in Spanish* 20

A Extensive Representation Payoffs

Regarding our experimental setting, we have the following vector of public goods allocations: *Hig-cost* environment: $\left(\boldsymbol{\Theta}^h, \boldsymbol{\Theta}^a, \boldsymbol{\Theta}^l | \overline{\pi} = 20\text{EP}\right) = (3, 2, 1)$ and *Low-cost* environment: $\left(\boldsymbol{\Theta}^h, \boldsymbol{\Theta}^a, \boldsymbol{\Theta}^l | \underline{\pi} = 10\text{EP}\right) = (6, 4, 2)$.

Politician's Payoffs:

$$\begin{pmatrix} U^1_{\text{pol.}} \\ U^2_{\text{pol.}} \\ U^3_{\text{pol.}} \\ U^4_{\text{pol.}} \end{pmatrix} = \begin{pmatrix} w(1-\eta) + \phi_p \\ w + \phi_p \\ w(1-\eta) + r \\ w + r \end{pmatrix}$$

Citizen's Payoffs:

$$\begin{pmatrix} U^1_{\text{citi.}} \\ U^2_{\text{citi.}} \\ U^3_{\text{citi.}} \\ U^4_{\text{citi.}} \\ U^5_{\text{citi.}} \\ U^6_{\text{citi.}} \end{pmatrix} = \begin{pmatrix} y(1-\tau) + \kappa(\boldsymbol{\Theta}^h) - \kappa \\ y(1-\tau) + \kappa(\boldsymbol{\Theta}^h) \\ y(1-\tau) + \kappa(\boldsymbol{\Theta}^a) + ((\alpha_c - \boldsymbol{\Theta}^a)\lambda - \kappa) \\ y(1-\tau) + \kappa(\boldsymbol{\Theta}^a) \\ y(1-\tau) + \kappa(\boldsymbol{\Theta}^l) + ((\alpha_c - \boldsymbol{\Theta}^l) - \kappa) \\ y(1-\tau) + \kappa(\boldsymbol{\Theta}^l) \end{pmatrix}$$

B Proofs

B.1 Proof Lemma 1

Proof. Under a high-allocation $\boldsymbol{\Theta}^h$, the citizen updates her posterior beliefs of corruption following,

$$\lambda \equiv \Pr\left(\sigma = 0 | \boldsymbol{\Theta}(\pi) = \boldsymbol{\Theta}^h(\pi)\right) = \frac{\Pr\left(\boldsymbol{\Theta}(\pi) = \boldsymbol{\Theta}^h(\pi) | \sigma = 0\right) \times \Pr\left(\sigma = 0\right)}{\Pr\left(\boldsymbol{\Theta}(\pi) = \boldsymbol{\Theta}^h(\pi)\right)} \tag{B1}$$

From Figure 2, observe that a higher allocation of public goods is only possible if the politician is honest. Therefore, $\Pr(\boldsymbol{\Theta}(\pi) = \boldsymbol{\Theta}^h(\pi) | \sigma = 0) = 0$. Updating (λ) following equation (B1), we find that $\lambda = 0$. Given this signal, the optimization payoff presented in Equation (3) can be expressed as:

$$\max_{\delta \in \{0,1\}} \quad y(1-\tau) + \kappa \boldsymbol{\Theta}^h(\pi) + \delta(-\kappa).$$

Finally, note that any strategy $\nu > 0$ results in a lower material payoff. Intuitively, this occurs because, under a high allocation signal, the citizen has no reason to protest, as the politician was honest. Therefore, under this signal, and regardless of the cost (π), the citizen plays:

$$\nu(\boldsymbol{\Theta}^h(\pi)) = 0 \quad \forall\, \alpha_c \in [0, \overline{\alpha}]$$

□

B.2 Proof Lemma 2

Proof. Under a low-allocation $\boldsymbol{\Theta}^l$, the citizen updates her posterior beliefs of corruption following Equation (B1). Once again, from Figure 2 observe that a low-allocation of public goods is only possible if the politician was corrupt. Moreover, the belief-updating process (λ) results in the following: $\lambda = \frac{(1-\mu)\times\beta}{(1-\mu)(\beta)} = 1$. Given this signal and the updated beliefs, the optimization problem for the citizen can be written as:

$$\max_{\delta\in\{0,1\}} \quad y(1-\tau) + \kappa\boldsymbol{\Theta}^l(\pi) + \delta\left((\alpha_c - \boldsymbol{\Theta}^l(\pi)) - \kappa\right)$$

Intuitively, the citizen optimizes by choosing a mixed strategy ν that maximizes her payoff, which depends entirely on: $\left((\alpha_c - \boldsymbol{\Theta}^l(\pi)) - \kappa\right)$. From Section 2, recall that $\boldsymbol{\Theta}^l(\pi) = \left(\frac{\tau y - r}{\pi}\right)$. Rearranging terms, we find that the optimal strategy under a low-allocation signal is:

$$\nu(\alpha_c, \boldsymbol{\Theta}^l(\pi)) = \begin{cases} 1 \quad \text{if} \quad \alpha_c \geq \kappa + \left(\frac{\tau y - r}{\pi}\right) \\ 0 \quad \text{if} \quad \alpha_c < \kappa + \left(\frac{\tau y - r}{\pi}\right) \end{cases} \tag{B2}$$

Moreover, regarding the optimal cutoff given a low-allocation signal, we find that:

$$\alpha^*(\boldsymbol{\Theta}^l(\pi)) = \left\{\kappa + \left(\frac{\tau y - r}{\pi}\right)\right\} \in (0, \overline{\alpha}]$$

Following the same logic, given an ambiguous-allocation $\boldsymbol{\Theta}^a$, the belief-updating process results in: $\lambda = \frac{\mu\beta}{\mu\beta + (1-\beta)(1-\mu)}$. Given this signal, the optimization problem for the citizen can be expressed as:

$$\max_{\delta\in\{0,1\}} \quad y(1-\tau) + \kappa\boldsymbol{\Theta}^a(\pi) + \delta\left((\alpha_c - \boldsymbol{\Theta}^a(\pi))\left(\frac{\mu\beta}{\mu\beta + (1-\beta)(1-\mu)}\right) - \kappa\right)$$

Similarly, the citizen chooses a mixed strategy ν that maximizes her payoff, which depends on: $\left((\alpha_c - \boldsymbol{\Theta}^a(\pi))\left(\frac{\mu\beta}{\mu\beta+(1-\beta)(1-\mu)}\right) - \kappa\right)$. Given that $\boldsymbol{\Theta}^a(\pi) = \left(\frac{\tau y}{\pi}\right)$, and rearranging terms, the optimal strategy under an ambiguous-allocation signal is:

$$\nu(\alpha_c, \boldsymbol{\Theta}^a(\pi)) = \begin{cases} 1 \quad \text{if} \quad \alpha_c \geq \kappa\left(\dfrac{\mu\beta + (1-\beta)(1-\mu)}{\mu\beta}\right) + \left(\dfrac{\tau y}{\pi}\right) \\ 0 \quad \text{if} \quad \alpha_c < \kappa\left(\dfrac{\mu\beta + (1-\beta)(1-\mu)}{\mu\beta}\right) + \left(\dfrac{\tau y}{\pi}\right) \end{cases} \tag{B3}$$

Moreover, regarding the optimal cutoff under an ambiguous allocation signal, we find that:

$$\alpha^*(\boldsymbol{\Theta}^a(\pi)) = \left\{\kappa\left(\frac{\mu\beta + (1-\beta)(1-\mu)}{\mu\beta}\right) + \left(\frac{\tau y}{\pi}\right)\right\} \in (0, \overline{\alpha}]$$

Given that cutoff points determine the citizen's optimal strategy under both signals and regardless of the cost, this proves the lemma. □

B.3 Proof Lemma 3

Proof. Employing Lemma 2, and focusing on a particular $\pi = \{\overline{\pi}, \underline{\pi}\}$ without loss of generality, we can define the citizen's set of optimal cutoffs

$$\alpha^*(\boldsymbol{\Theta}^l, \boldsymbol{\Theta}^a) = \left(\kappa + \left(\frac{\tau y - r}{\pi}\right), \kappa\left(\frac{\mu\beta + (1-\beta)(1-\mu)}{\mu\beta}\right) + \left(\frac{\tau y}{\pi}\right)\right)$$

Therefore, we compared these cutoffs following:

$$\underbrace{\kappa + \left(\frac{\tau y - r}{\pi}\right)}_{\alpha^*(\boldsymbol{\Theta}^l)} < \underbrace{\kappa\left(\frac{\mu\beta + (1-\beta)(1-\mu)}{\mu\beta}\right) + \left(\frac{\tau y}{\pi}\right)}_{\alpha^*(\boldsymbol{\Theta}^a)}$$

Rearranging terms,

$$\kappa - \left(\frac{r}{\pi}\right) < \kappa\left(\frac{\mu\beta + (1-\beta)(1-\mu)}{\mu\beta}\right)$$

Given that $\kappa \in \Re^+$, it is possible to rewrite this last expression as:

$$1 - \left(\frac{r}{\kappa\pi}\right) < \left(\frac{\mu\beta + (1-\beta)(1-\mu)}{\mu\beta}\right)$$

Simplifying, we obtain

$$1-\left(\frac{r}{\kappa\pi}\right)<1+\left(\frac{(1-\beta)(1-\mu)}{\mu\beta}\right)$$

Given that $r\in(0,\tau y)$, $\kappa\in\Re^{+}$, and $\pi\in\{\overline{\pi},\underline{\pi}\}\subset\Re^{+}$, the expression $\left(\frac{r}{\kappa\pi}\right)$ is always positive. On the other hand, given that $\mu\in(0,1)$, and $\beta\in(0,1)$, the expression $\left(\frac{\mu\beta+(1-\beta)(1-\mu)}{\mu\beta}\right)$ is always positive.

Therefore, the expression

$$1-\underbrace{\left(\frac{r}{\kappa\pi}\right)}_{\text{Positive}}<1+\underbrace{\left(\frac{(1-\beta)(1-\mu)}{\mu\beta}\right)}_{\text{Positive}},$$

is always true. This completes the proof. □

B.4 Proof Lemma 4

Proof. The politician may consider a strategy (σ) that aligns with the citizen's strategy ν and the citizen's updated beliefs λ, forming an equilibrium. Hence, once the public good costs are determined, the politician compares the expected payoff of each possible action, considering the citizens' behavior, and chooses the highest.

Strategy 1: Expected payoff of being Corrupt

If the politician chooses to be corrupt, there is a probability (μ) that the national authority fully covers her decision $(\gamma=1)$, resulting in the citizen observing an ambiguous allocation signal $\boldsymbol{\Theta}^{a}$. In this case, the citizen will punish the politician if equation (B3) holds. Conversely, with a probability of $(1-\mu)$, the national authority does not provide resources $(\gamma=0)$, leading the citizen to observe a low allocation signal $\boldsymbol{\Theta}^{l}$, and the citizen will punish if equation (B2) holds. Therefore, the expected payoff for being corrupt $(h=0)$ is given by

$$\mathbb{E}\left[U_p|h=0,\pi\right]=w+r-\mu\left[P\left(\alpha_c>\alpha^{*}(\boldsymbol{\Theta}^{a}(\pi))\right)(\eta w)\right]-(1-\mu)\left[P\left(\alpha_c>\alpha^{*}(\boldsymbol{\Theta}^{l}(\pi))\right)(\eta w)\right]$$

Given that $\alpha_c\sim f[0,\overline{\alpha}]$, we can rewrite this expression as

$$w+r-\mu\left(1-\int_{0}^{\alpha^{*}(\boldsymbol{\Theta}^{a}(\pi))}f(\alpha)\,d\alpha\right)(\eta w)-(1-\mu)\left(1-\int_{0}^{\alpha^{*}(\boldsymbol{\Theta}^{l}(\pi))}f(\alpha)\,d\alpha\right)(\eta w)\qquad\text{(B4)}$$

Strategy 2: Expected payoff of being honest

If the politician decides to be honest, she knows that with probability (μ), the national authority will provide additional resources $(\gamma = 1)$, and the citizen observes a high allocation signal $\boldsymbol{\Theta}^h$. From Lemma 1, we know that under this signal the citizen will never punish the politician. Conversely, with probability $(1-\mu)$, the national authority does not provide resources $(\gamma = 0)$, and the citizen observes an **ambiguous** signal $\boldsymbol{\Theta}^a$ of public goods. Again, under an ambiguous signal, the citizen will disapprove if equation (B3) holds. Therefore, the expected payoff of being honest $(h = 1)$ is given by

$$\mathbb{E}[U_p|h=1,\pi] = w + \phi_p - (1-\mu)\left[P\left(\alpha_c > \alpha^*(\boldsymbol{\Theta}^a(\pi))\right)(\eta w)\right]$$

Rewriting this expression, we obtain

$$\mathbb{E}[U_p|h=1,\pi] = w + \phi_p - (1-\mu)\left(1 - \int_0^{\alpha^*(\boldsymbol{\Theta}^l(\pi))} f(\alpha)\,d\alpha\right)(\eta w) \tag{B5}$$

To find an optimal strategy (σ), the politician considers both expected payoffs and chooses the one that maximizes her utility. Under both scenarios, the politician receives her salary (w) and also faces the possibility of punishment under an ambiguous signal. Therefore, she compares the expected utility of being honest over being corrupt, given a particular public goods cost (π). This comparison can be expressed as

$$\underbrace{\phi_p - (1-\mu)\left(1 - \int_0^{\alpha^*(\boldsymbol{\Theta}^a(\pi))} f(\alpha)\,d\alpha\right)(\eta w)}_{\text{Being honest}} \geq \underbrace{r - \mu\left(1 - \int_0^{\alpha^*(\boldsymbol{\Theta}^a(\pi))} f(\alpha)\,d\alpha\right)(\eta w) - (1-\mu)\left(1 - \int_0^{\alpha^*(\boldsymbol{\Theta}^l(\pi))} f(\alpha)\,d\alpha\right)(\eta w)}_{\text{Being corrupt}} \tag{B6}$$

Factorizing, we obtain

$$\phi_p \geq r + (1-2\mu)\left(1 - \int_0^{\alpha^*(\boldsymbol{\Theta}^a(\pi))} f(\alpha)\,d\alpha\right) - (1-\mu)\left(1 - \int_0^{\alpha^*(\boldsymbol{\Theta}^l(\pi))} f(\alpha)\,d\alpha\right)$$

Therefore,

$$\phi^*(\pi) = r + (1-2\mu)\left(1 - \int_0^{\alpha^*(\boldsymbol{\Theta}^a(\pi))} f(\alpha)\,d\alpha\right) - (1-\mu)\left(1 - \int_0^{\alpha^*(\boldsymbol{\Theta}^l(\pi))} f(\alpha)\,d\alpha\right) \tag{B7}$$

Finally, only rests to prove that $\phi^*(\pi) < \widehat{\phi}$. First, given that $\mu \in (0,1)$ behaves monotonically, we can focus on the extreme value cases of μ.

Case 1: $\lim_{\mu \to 0}$

$$\phi^*(\pi) = r + \left(1 - \int_0^{\alpha^*(\boldsymbol{\Theta}^a(\pi))} f(\alpha)\, d\alpha \right) - \left(1 - \int_0^{\alpha^*(\boldsymbol{\Theta}^l(\pi))} f(\alpha)\, d\alpha \right)$$

From Lemma 3, it follows that $\int_0^{\alpha^*(\boldsymbol{\Theta}^l(\pi))} f(\alpha)\, d\alpha \leq \int_0^{\alpha^*(\boldsymbol{\Theta}^a(\pi))} f(\alpha)\, d\alpha$, and given that $r < \widehat{\phi}$, we obtain

$$\phi^*(\pi) = r + \left(1 - \int_0^{\alpha^*(\boldsymbol{\Theta}^a(\pi))} f(\alpha)\, d\alpha \right) - \left(1 - \int_0^{\alpha^*(\boldsymbol{\Theta}^l(\pi))} f(\alpha)\, d\alpha \right) < r < \widehat{\phi}$$

So, when $\lim_{\mu \to 0}$,

$$\phi^*(\pi) < \widehat{\phi}$$

Case 2: $\lim_{\mu \to 1}$

$$\phi^*(\pi) = r - \left(1 - \int_0^{\alpha^*(\boldsymbol{\Theta}^a(\pi))} f(\alpha)\, d\alpha \right)$$

Given that $\alpha^*(\boldsymbol{\Theta}^a(\pi)) > 0$, we obtain

$$\phi^*(\pi) = r - \left(1 - \int_0^{\alpha^*(\boldsymbol{\Theta}^a(\pi))} f(\alpha)\, d\alpha \right) < r < \widehat{\phi}$$

So, when $\lim_{\mu \to 1}$,

$$\phi^*(\pi) < \widehat{\phi}$$

This completes the proof. □

B.5 Proof Theorem 1

Proof. Case 1: (Low allocation):

$$\alpha^*(\boldsymbol{\Theta}^l(\pi)) = \kappa + \left(\frac{\tau y - r}{\pi}\right)$$

Derivative with respect to π:

$$\frac{d}{d\pi}\alpha^*(\boldsymbol{\Theta}^l(\pi)) = \frac{d}{d\pi}\left(\kappa + \frac{\tau y - r}{\pi}\right) = -\frac{\tau y - r}{\pi^2}$$

Since $\tau y - r$ is a positive constant and π^2 is always positive, $-\frac{\tau y - r}{\pi^2}$ is negative. Therefore, $\alpha^*(\boldsymbol{\Theta}^l(\pi))$ is a decreasing function of π. Since $\alpha^*(\boldsymbol{\Theta}^l(\pi))$ decreases as π increases, it follows that:

$$\alpha^*(\boldsymbol{\Theta}^l(\overline{\pi})) \leq \alpha^*(\boldsymbol{\Theta}^l(\underline{\pi}))$$

whenever $\overline{\pi} \geq \underline{\pi}$.

Case 2: (Ambiguous allocation):

$$\alpha^*(\boldsymbol{\Theta}^a(\pi)) = \kappa \left(\frac{\mu\beta + (1-\beta)(1-\mu)}{\mu\beta} \right) + \left(\frac{\tau y}{\pi} \right)$$

Derivative with respect to π:

$$\frac{d}{d\pi}\alpha^*(\boldsymbol{\Theta}^a(\pi)) = \frac{d}{d\pi}\left(\kappa \left(\frac{\mu\beta + (1-\beta)(1-\mu)}{\mu\beta} \right) + \frac{\tau y}{\pi} \right) = -\frac{\tau y}{\pi^2}$$

Since τy is a positive constant and π^2 is always positive, $-\frac{\tau y}{\pi^2}$ is negative. Therefore, $\alpha^*(\boldsymbol{\Theta}^a(\pi))$ is a decreasing function of π. Since $\alpha^*(\boldsymbol{\Theta}^a(\pi))$ decreases as π increases, it follows that:

$$\alpha^*(\boldsymbol{\Theta}^a(\overline{\pi})) \leq \alpha^*(\boldsymbol{\Theta}^a(\underline{\pi}))$$

whenever $\overline{\pi} \geq \underline{\pi}$. This completes the proof of Theorem 1. □

B.6 Proof Corollary 1

Proof. From Theorem 1, $\alpha^*(\boldsymbol{\Theta}^m(\overline{\pi})) \geq \alpha^*(\boldsymbol{\Theta}^m(\underline{\pi}))$, then rearranging terms,

$$\int_0^{\alpha^*(\boldsymbol{\Theta}^m(\underline{\pi}))} f(\alpha)\, d\alpha + \int_{\alpha^*(\boldsymbol{\Theta})^m(\underline{\pi})}^{\alpha^*(\boldsymbol{\Theta}^m(\overline{\pi}))} f(\alpha)\, d\alpha \geq \int_0^{\alpha^*(\boldsymbol{\Theta}^m(\underline{\pi}))} f(\alpha)\, d\alpha$$

$$\int_{\alpha^*(\boldsymbol{\Theta})^m(\underline{\pi})}^{\alpha^*(\boldsymbol{\Theta}^m(\overline{\pi}))} f(\alpha)\, d\alpha \geq 0$$

This completes the proof of Corollary 1. □

B.7 Proof Theorem 2

Proof. From Equation (B7), we have that

$$\phi^*(\pi) = r + (1-2\mu)\left(1 - \int_0^{\alpha^*(\boldsymbol{\Theta}^a(\pi))} f(\alpha)\, d\alpha\right) - (1-\mu)\left(1 - \int_0^{\alpha^*(\boldsymbol{\Theta}^l(\pi))} f(\alpha)\, d\alpha\right)$$

From Corollary 1, we can readily see that

$$\left(1 - \int_0^{\alpha^*(\boldsymbol{\Theta}^m(\overline{\pi}))} f(\alpha)\, d\alpha\right) \leq \left(1 - \int_0^{\alpha^*(\boldsymbol{\Theta}^m(\underline{\pi}))} f(\alpha)\, d\alpha\right) \;\; \forall m = \{a, l\}$$

Then, it follows that

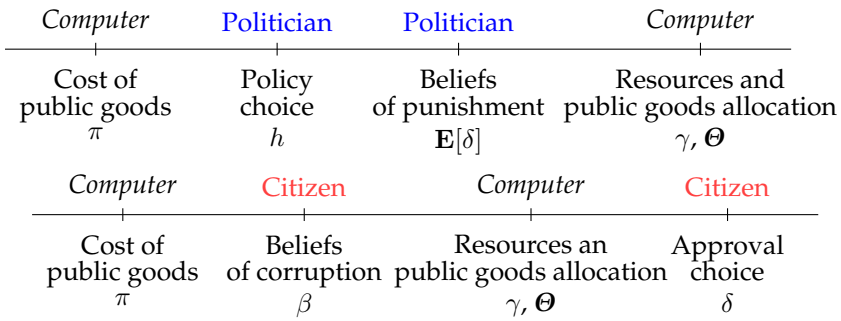
$$\phi^*(\overline{\pi}) \leq \phi^*(\underline{\pi})$$

□

C Experimental Design

C.1 Participants' Timelines

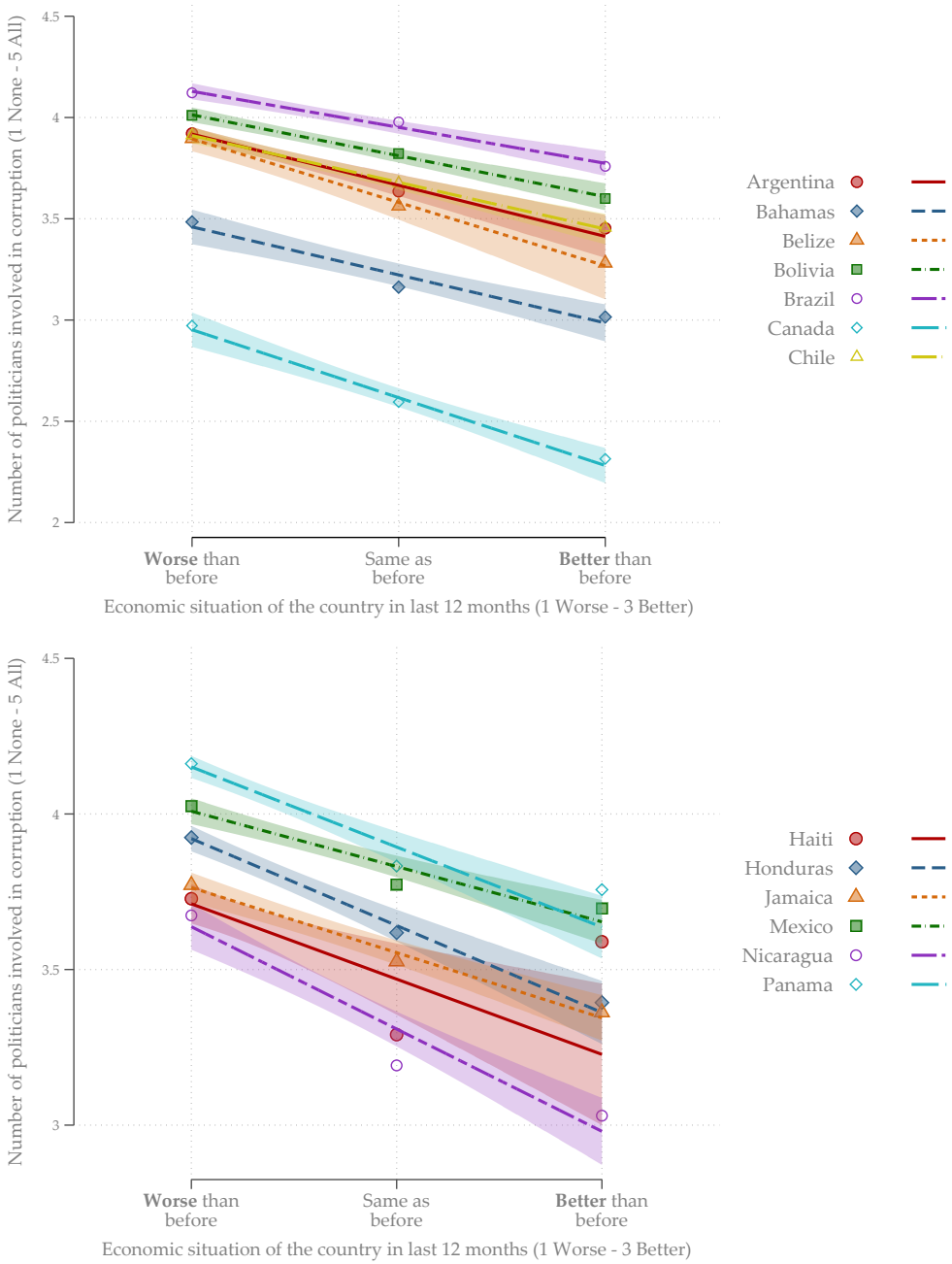
Figure C1: Experimental Timeline in PCG across Participants Types



Notes: The figure illustrates the experimental timeline across different participant types. **Beliefs of punishment** refers to the politician's response to the question: "Do you think the citizen would disapprove of you?" **Beliefs of corruption** refers to the citizen's response to the question: "How many resources do you think the politician would allocate?"

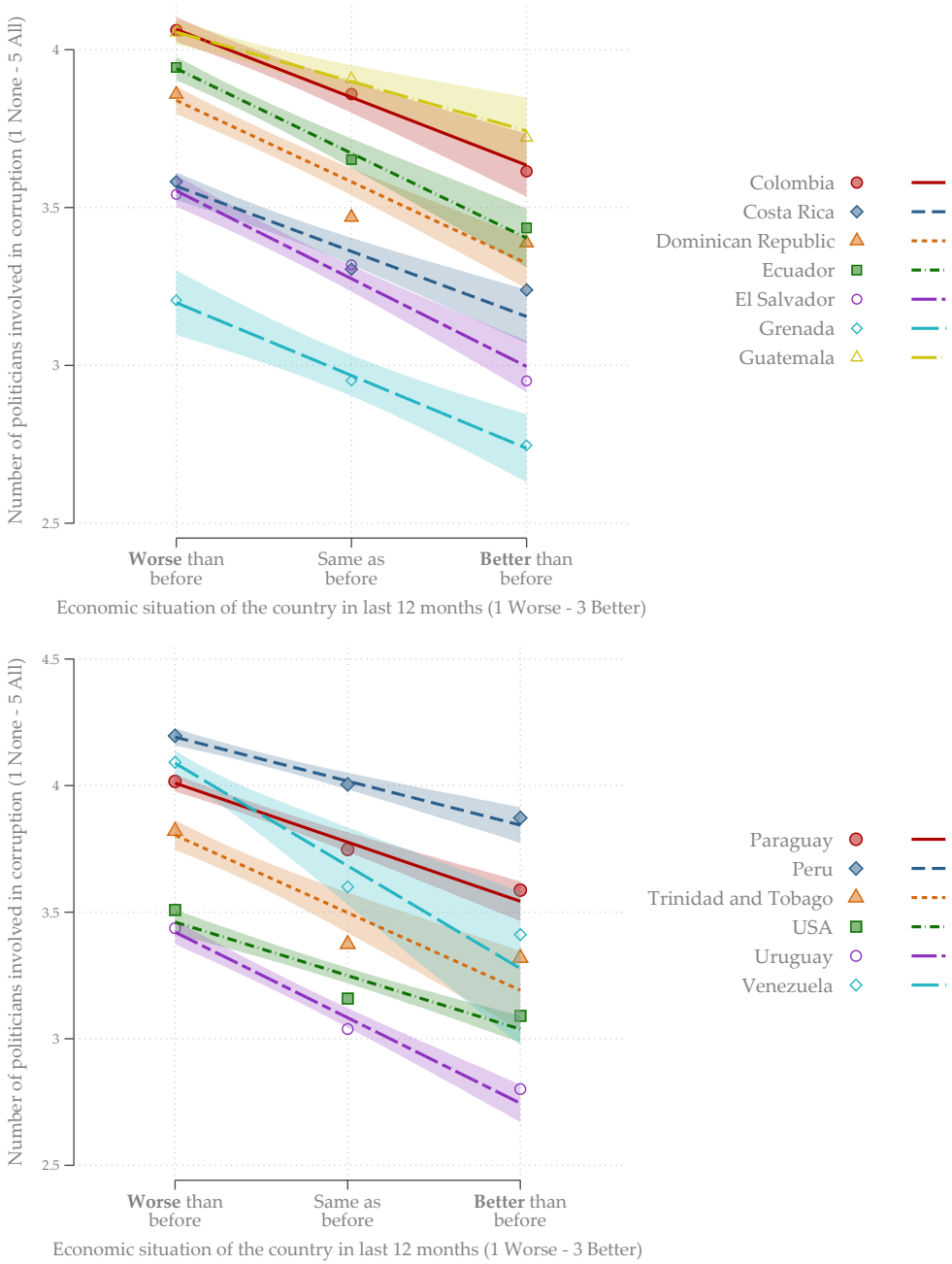
D Additional Figures and Tables

Figure D1: Perception of Number of Politicians Involved in Corruption across National Economic Situation Perception.



Notes: The figure depicts the scatterplot binned between the perception of the number of politicians involved in corruption and the perception of the country's economic situation over the past 12 months. Scatter binned estimation for all respondents to the LAPOP question in 26 countries between 2016 and 2023. The dark line indicates the linear fit. Gray dashed lines represent the 95% robust confidence intervals. The analysis includes country and year fixed effects. Statistical significance levels: $^{***}p < 0.01$,** $p < 0.05$,* $p < 0.1$, n.s $p > 0.1$.

Figure D2: Perception of Number of Politicians Involved in Corruption across National Economic Situation Perception



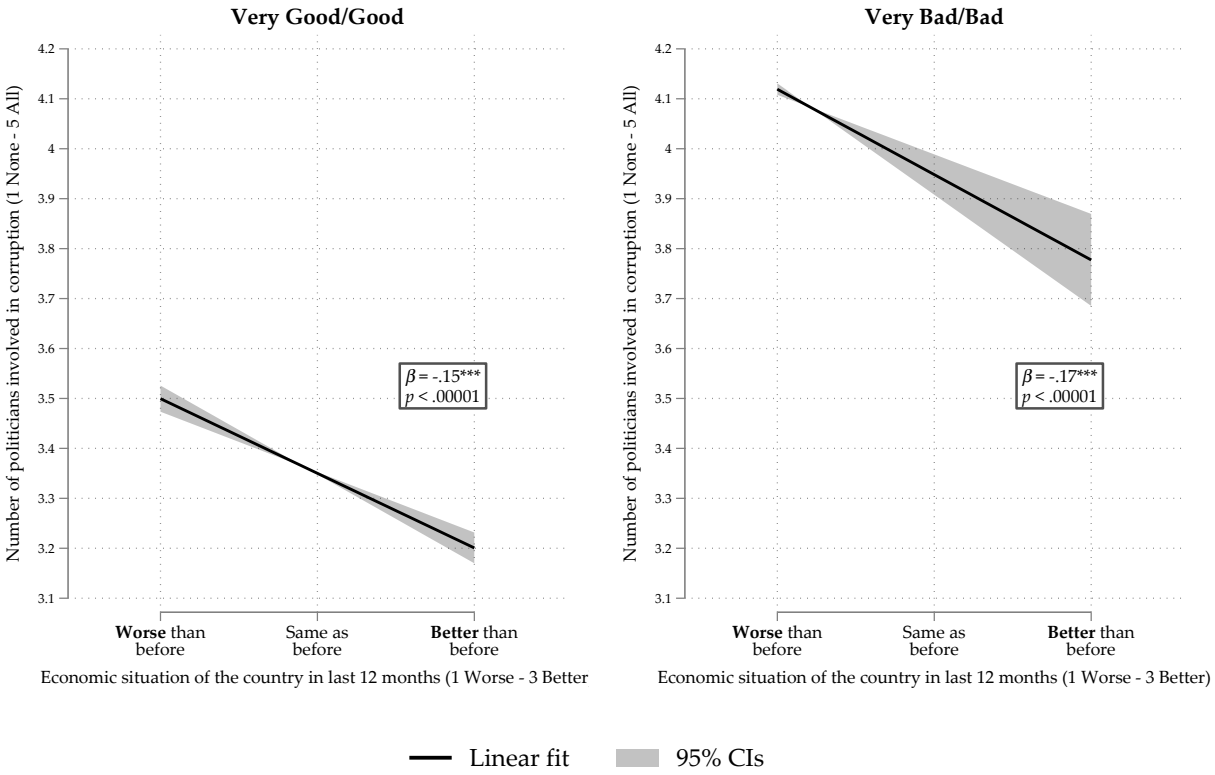
Notes: The figure depicts the scatterplot binned between the perception of the number of politicians involved in corruption and the perception of the country's economic situation over the past 12 months. Scatter binned estimation for all respondents to the LAPOP question in 26 countries between 2016 and 2023. The dark line indicates the linear fit. Gray dashed lines represent the 95% robust confidence intervals. The analysis includes country and year fixed effects. Statistical significance levels: $^{***}p < 0.01$,** $p < 0.05$,* $p < 0.1$, n.s $p > 0.1$.

Table D1: Politician Corruption (LAPOP measure) and Economic Perceptions

Dep Var: **Politicians Corrupt - LAPOP**	OLS			
	(1)	(2)	(3)	(4)
Economic situation of the country in last 12 months	-0.310***	-0.255***		
	(0.03)	(0.01)		
Personal economic situation in last 12 months			-0.203***	-0.154***
			(0.03)	(0.01)
Constant	4.198***	4.113***	4.048***	3.964***
	(0.04)	(0.02)	(0.05)	(0.01)
Country and Year FE		✓		✓
Weight	✓	✓	✓	✓
Observations	79787	79787	108813	108813
Clusters	26	26	26	26
Adjusted R-squared	0.04	0.11	0.02	0.11

Notes: **Politicians corrupt - LAPOP**: Measures the perception of the number of politicians involved in corruption. **Economic situation of the country in the last 12 months** and **Personal economic situation in the last 12 months** are measured on a scale from 1 to 3 (1 = Worse, 3 = Better). Data were collected from respondents to the LAPOP survey across 26 countries between 2016 and 2023. Country and year fixed effects are included in the analysis. Statistical significance levels: $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.1$, n.s $p > 0.1$.

Figure D3: Perception of Number of Politicians Involved in Corruption across National Economic Situation Perception by Presidential Job Approval



Notes: The figure depicts the scatterplot binned between the perception of the number of politicians involved in corruption and the perception of the country's economic situation over the past 12 months disaggregated by presidential job approval. Scatter binned estimation for all respondents to the LAPOP question in 26 countries between 2016 and 2023. The dark line indicates the linear fit from a regression model. Gray area represent the 95% clustered confidence intervals at the country level. The analysis includes country and year fixed effects. Statistical significance levels: $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.1$, n.s $p > 0.1$.

Table D2: Descriptive Statistics

Variable	N	Mean	Std	Min	Max
Panel A: Sociodemographic variables					
Age	799	22.08	4.69	0	63
Gender	799	0.54	0.51	0	2
Semester	799	5.05	3.26	0	13
Stratum	799	3.10	1.04	0	6
Mother education	799	3.79	0.95	0	5
Father education	799	3.55	1.30	0	5
Self-perception of wealth (income stair)	799	5.70	1.62	1	10
Panel B: Experimental variables					
Generalized Trust Measure (0: Low - 10: High)	799	4.57	2.19	0	10
Trust in his neighbors	799	2.97	0.64	1	4
Generalized Risk Measure (0: Low - 10: High)	799	6.82	1.92	0	10
Patience measure (1: Low - 5:High)	799	4.05	0.79	1	5
Guess 2/3 of the average	799	36.69	22.32	0	100
Have you participated in experiments before?	799	0.66	0.47	0	1
Panel C: Political opinions					
Expect negative consequences of expressing political opinion	799	0.66	0.47	0	1
Willing to protest for better salaries	799	7.98	2.31	0	10
Willing to protest for better health	799	8.65	2.04	0	10
Willing to protest against climate change	799	7.29	2.86	0	10
Willing to protest for democracy rights	799	7.47	2.46	0	10
Willing to protest against corruption	799	7.78	2.34	0	10
Willing to protest against inequality	799	7.52	2.73	0	10
How likely is for the president to be involved in corruption?	799	6.62	2.67	0	10
How likely is for the congress to be involved in corruption?	799	8.22	1.82	0	10
How likely is for the mayor to be involved in corruption?	799	7.70	1.93	0	10
How likely is for the police to be involved in corruption?	799	8.03	1.97	0	10
How likely is for the judges to be involved in corruption?	799	6.57	2.36	0	10
Political preference (1: Right - 5: Left)	799	3.54	1.46	1	6

Notes: **Self-perception of wealth**: is a categorical variable that denotes the participant's answer to the question: "Imagine a staircase with 10 steps. People with the lowest incomes are placed on step 1 and people with the highest incomes on step 10. Where would you place your family group?". **Trust in neighbors:** denotes the participant's answer to the question: On a scale from 1 to 4, how much do you trust your neighbors?**Patience measure:** denotes the participant's answer to the question: On a scale from 1 (low) to 5 (high), how patient are you? **Guess 2/3 of the average**: denotes the participant's answer to the beauty contest of *Guess 2/3 of the average*. **Political Preference:** On a scale from 1 (right) to 5 (left), where would you place your political preference?.**Expectation of Negative Consequences:** How likely is it that expressing your political opinion will result in negative consequences?. **Willingness to protest for ... :** where 0 denotes "Not willing to", and 10 "Very willing to". **Perception of Corruption in the ...** : How likely do you think it is that ... is involved in corruption? (Scale: 1-10).

Table D3: Politicians **Corruption**: Logit and Probit Estimates

Dep Var: **Politician Corruption**	Logit				Probit			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
Low cost (Treatment)	0.295***	0.284***	0.292***	0.447***	0.184***	0.172***	0.175***	0.266***
	(0.10)	(0.10)	(0.10)	(0.12)	(0.06)	(0.06)	(0.06)	(0.07)
Corrupt - round before				0.704***				0.433***
				(0.15)				(0.09)
Constant	0.060	-0.457	-1.054	-0.763	0.038	-0.290	-0.656	-0.489
	(0.08)	(0.66)	(0.85)	(0.75)	(0.05)	(0.40)	(0.51)	(0.46)
Controls		✓	✓	✓		✓	✓	✓
Round FE			✓	✓			✓	✓
Session FE			✓	✓			✓	✓
Observations	2000	2000	2000	1600	2000	2000	2000	1600
Clusters	400	400	400	400	400	400	400	400
Pseudo R-squared	0.00	0.05	0.07	0.09	0.00	0.05	0.07	0.09

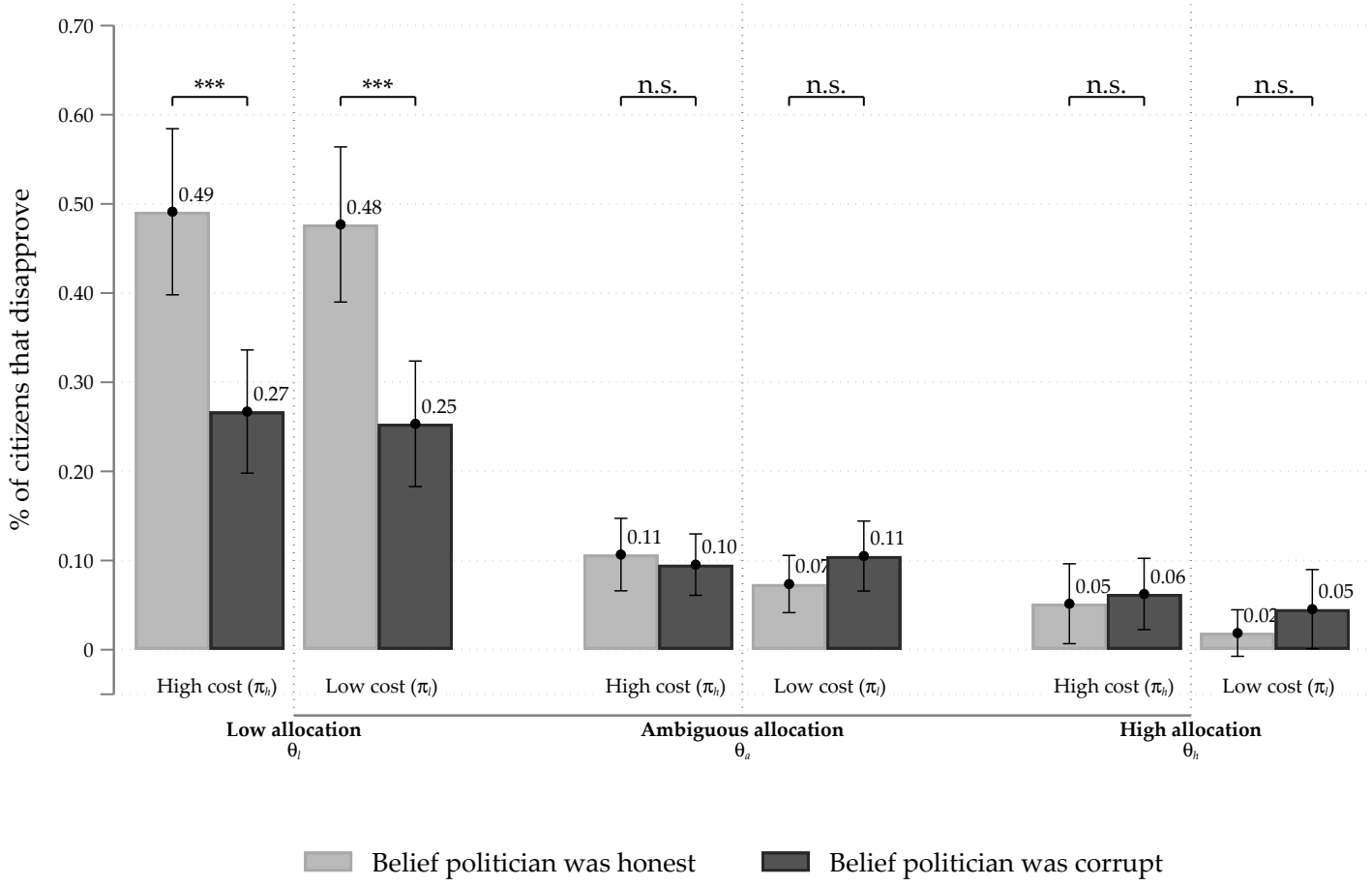
Notes: The set of controls included in ***Controls*** are: age, sex, stratum, trust measure, risk measure, patience measure and previous experience participating in experiments. The *Mean Dep. Variable* express the unconditional control mean of the dependent variable. The *Effect size (%)* denotes the size of the treatment effect as a percentage compared with the unconditional mean of the control group. Standard errors clustered at individual level are presented in parenthesis. $^{***}p < 0.01,^{**}p < 0.05,^{*}p < 0.1$

Table D4: Citizens **Disapproval Rate** across Allocations

Dep Var: **Citizen Punishment**	Public goods cost (π)								
	Low allocation (θ^l)			Ambiguous allocation (θ^a)			High allocation (θ^h)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Low cost (Treatment)	-0.003	-0.000	-0.003	-0.012	-0.008	-0.006	-0.027	-0.026	-0.030*
	(0.039)	(0.039)	(0.039)	(0.018)	(0.017)	(0.017)	(0.018)	(0.018)	(0.018)
Constant	0.360***	0.854***	0.742***	0.100***	0.529	0.466	0.058***	0.182	0.206
	(0.032)	(0.234)	(0.244)	(0.015)	(0.335)	(0.305)	(0.015)	(0.184)	(0.165)
Mean Dep. Variable	0.36	0.36	0.36	0.10	0.10	0.10	0.06	0.06	0.06
Effect size (%)	-0.8	0.0	-0.9	-11.6	-8.1	-6.1	-47.0	-45.0	-52.0
Controls		✓	✓		✓	✓		✓	✓
Round FE			✓			✓			✓
Session FE			✓			✓			✓
Observations	555	555	555	1004	1004	1004	436	436	436
Clusters	317	317	317	389	389	389	278	278	278
Adjusted R-squared	0.00	0.04	0.10	0.00	0.05	0.08	0.00	0.05	0.11

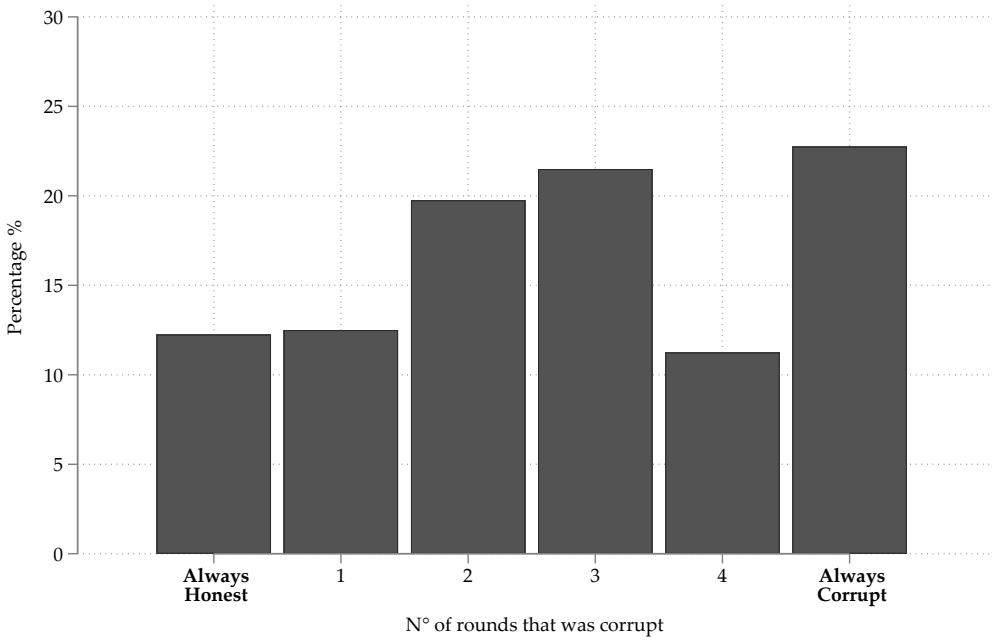
Notes: The set of controls included in ***Controls*** are: age, sex, stratum, trust measure, risk measure, patience measure and previous experience participating in experiments. The *Mean Dep. Variable* expresses the unconditional control mean of the dependent variable. The *Effect size (%)* denotes the size of the treatment effect as a percentage compared with the unconditional mean of the control group. Standard errors clustered at individual level are presented in parenthesis. $^{***}p < 0.01,^{**}p < 0.05,^{*}p < 0.1$

Figure D4: Citizen Punishment in the Corruption Game



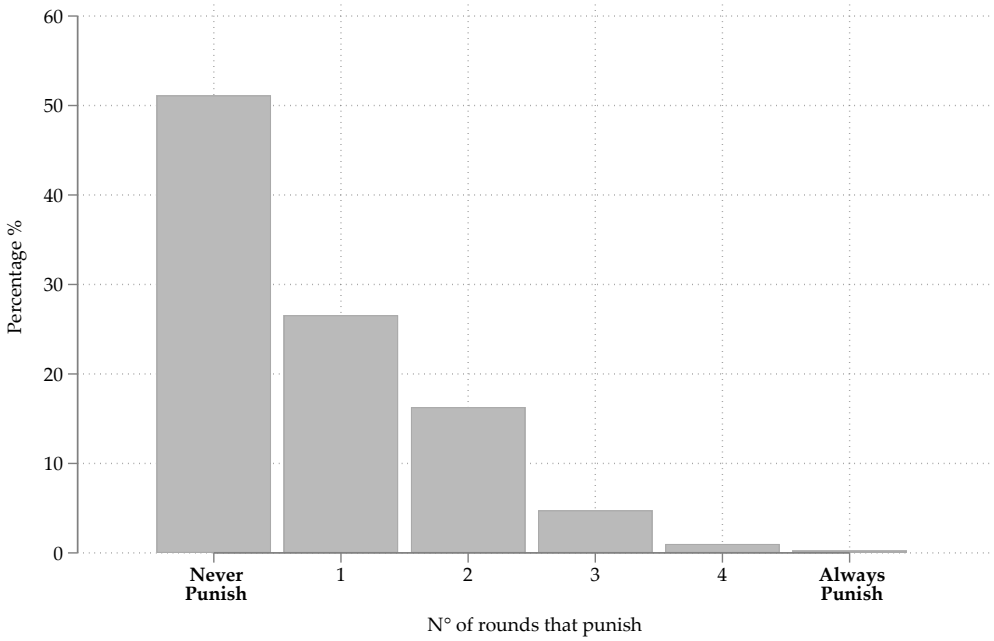
Notes: The figure depicts the average disapproval rate by the citizen across all rounds, segmented by prior beliefs of corruption and the final allocation of public goods. **High cost** = 20 EP. **Low cost** = 10 EP. Dark lines show 95% clustered robust confidence intervals. No controls included. $^{***}p < 0.01, ^{**}p < 0.05, ^{*}p < 0.1$, n.s $p > 0.1$. Reported significance levels come from hypothesis testing from a fully saturated linear model where the dependent variable is the citizen's choice to disapprove of the incumbent, and regressors are dummy variables associated with treatments.

Figure D5: Distribution of Total Corruption in the Political Corruption Game (PCG)



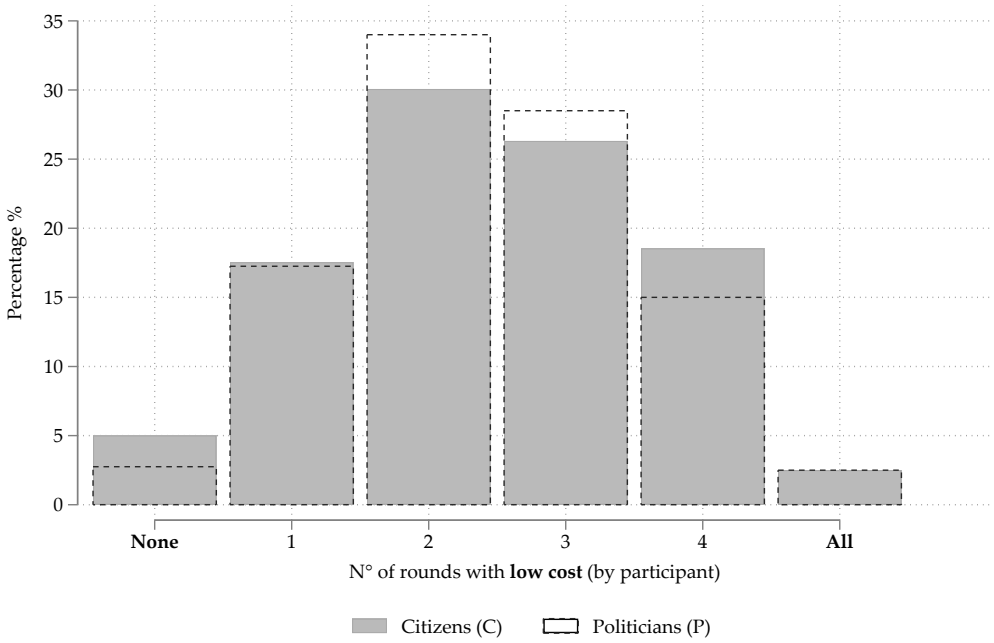
Notes: The figure illustrates the percentage distribution of the number of rounds in which politicians were corrupt during the game. The histogram represents different categories of corruption behavior, ranging from "Always Honest" (0 rounds of corruption) to "Always Corrupt" (5 rounds of corruption).

Figure D6: Distribution of Total Punishment in the Political Corruption Game (PCG)



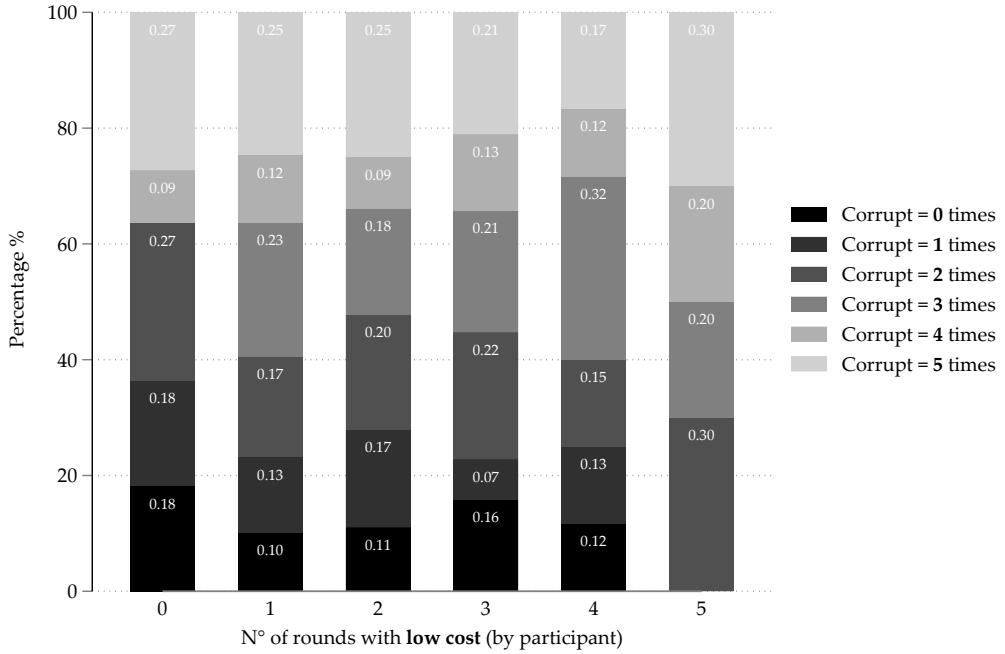
Notes: The figure illustrates the percentage distribution of the number of rounds in which citizens disapprove of politicians during the game. The histogram represents different categories of political accountability behavior, ranging from "Never punish" (0 rounds of disapproval) to "Always punish" (5 rounds of political disapproval).

Figure D7: Distribution of **Treated Rounds** by Participant Type (PCG)



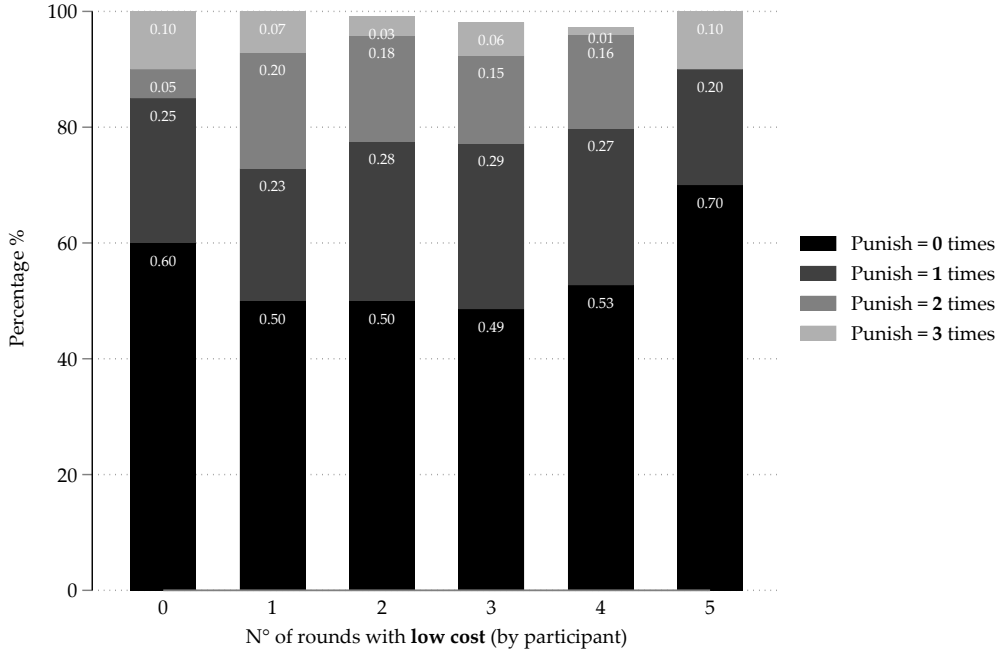
Notes: The figure illustrates the percentage distribution of the number of rounds in which citizens disapprove of politicians during the game. The histogram represents different categories of political accountability behavior, ranging from "Never punish" (0 rounds of disapproval) to "Always punish" (5 rounds of political disapproval).

Figure D8: Distribution of Total Corruption in the Political Corruption Game (PCG)



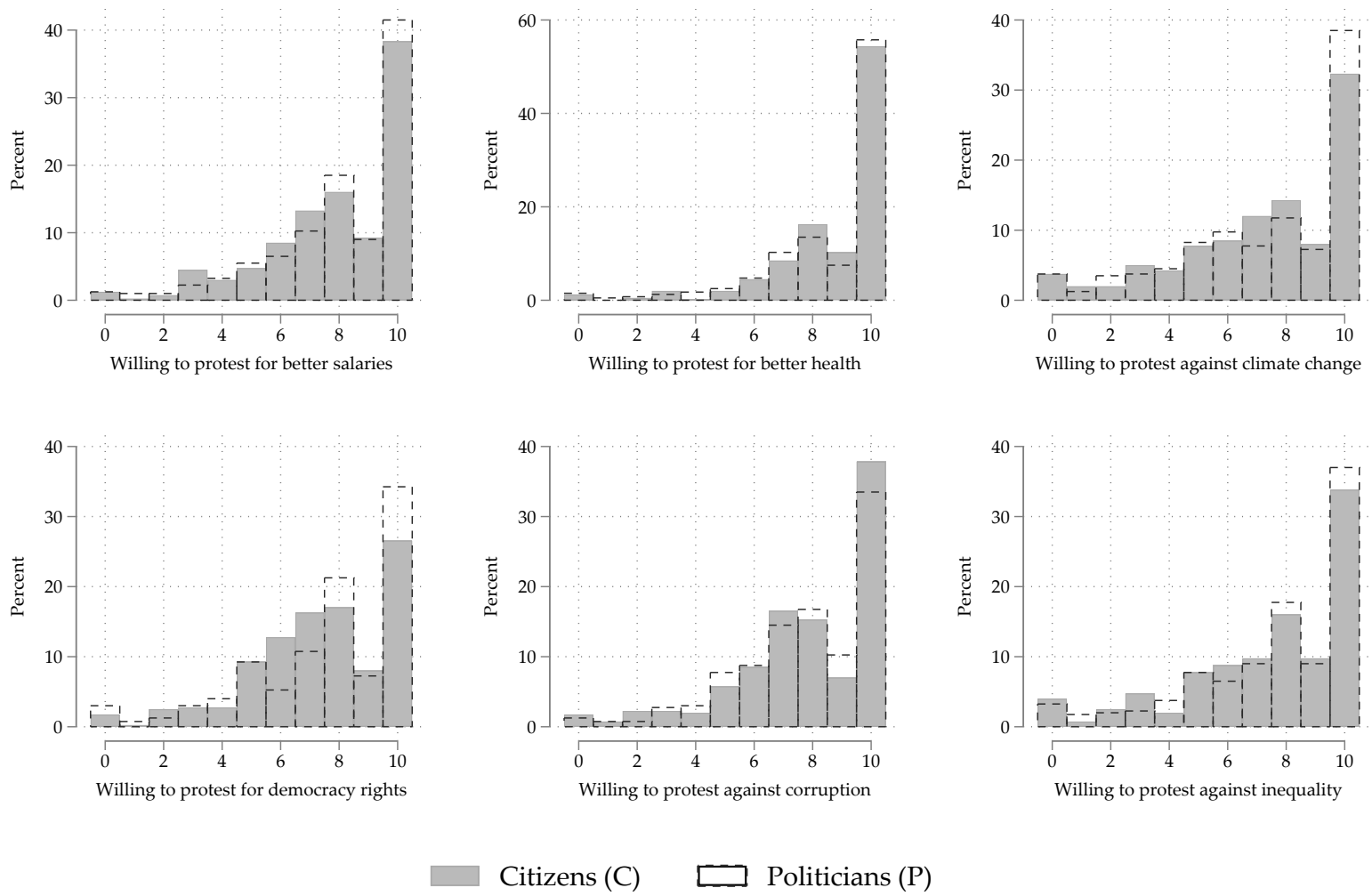
Notes: We present the distribution of total number of times being corrupt across the total number of rounds that the politician face a low cost. Notice that "Always honest" politicians face several rounds with low cost settings.

Figure D9: Distribution of Total Punishment in the Political Corruption Game (PCG)



Notes: We present the distribution of total number of punishment rounds across the total number of rounds that the citizen faces a low cost. Notice that "Never punish" citizens face several rounds with low cost settings.

Figure D10: Willingness to Punish by LAPOP Question, by Participant Type

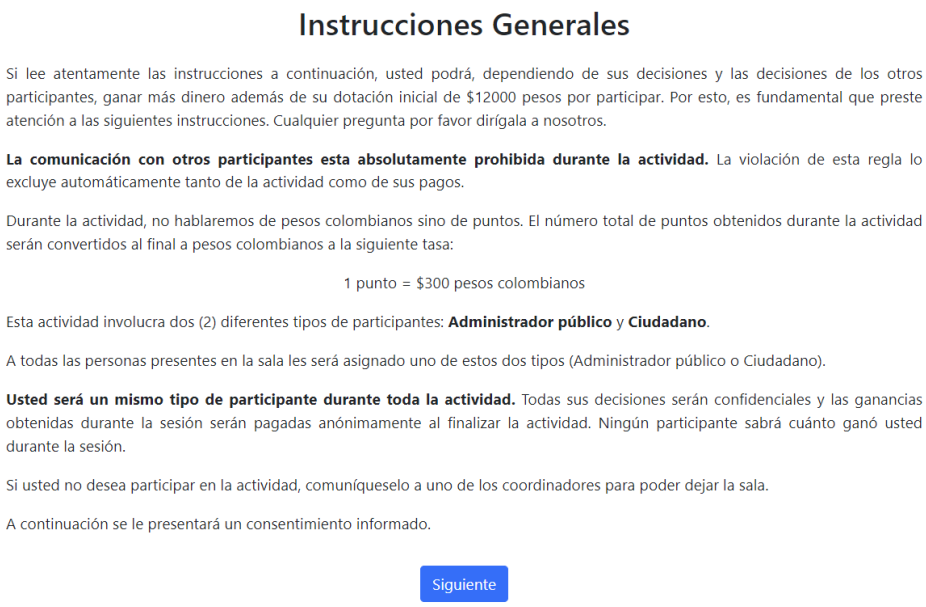


Notes: The figure illustrates the willingness of citizens (in gray) and politicians (in red) to protest for various causes, including better salaries, health, climate change, democracy rights, corruption, and inequality. The histogram represents the willingness to protest, scored from 0 to 10.

E Experimental Screen Instructions

E.1 *Original Instructions in Spanish*

Figure E1: General instructions



Instrucciones Generales

Si lee atentamente las instrucciones a continuación, usted podrá, dependiendo de sus decisiones y las decisiones de los otros participantes, ganar más dinero además de su dotación inicial de $12000 pesos por participar. Por esto, es fundamental que preste atención a las siguientes instrucciones. Cualquier pregunta por favor diríjala a nosotros.

La comunicación con otros participantes esta absolutamente prohibida durante la actividad. La violación de esta regla lo excluye automáticamente tanto de la actividad como de sus pagos.

Durante la actividad, no hablaremos de pesos colombianos sino de puntos. El número total de puntos obtenidos durante la actividad serán convertidos al final a pesos colombianos a la siguiente tasa:

1 punto = $300 pesos colombianos

Esta actividad involucra dos (2) diferentes tipos de participantes: **Administrador público** y **Ciudadano**.

A todas las personas presentes en la sala les será asignado uno de estos dos tipos (Administrador público o Ciudadano).

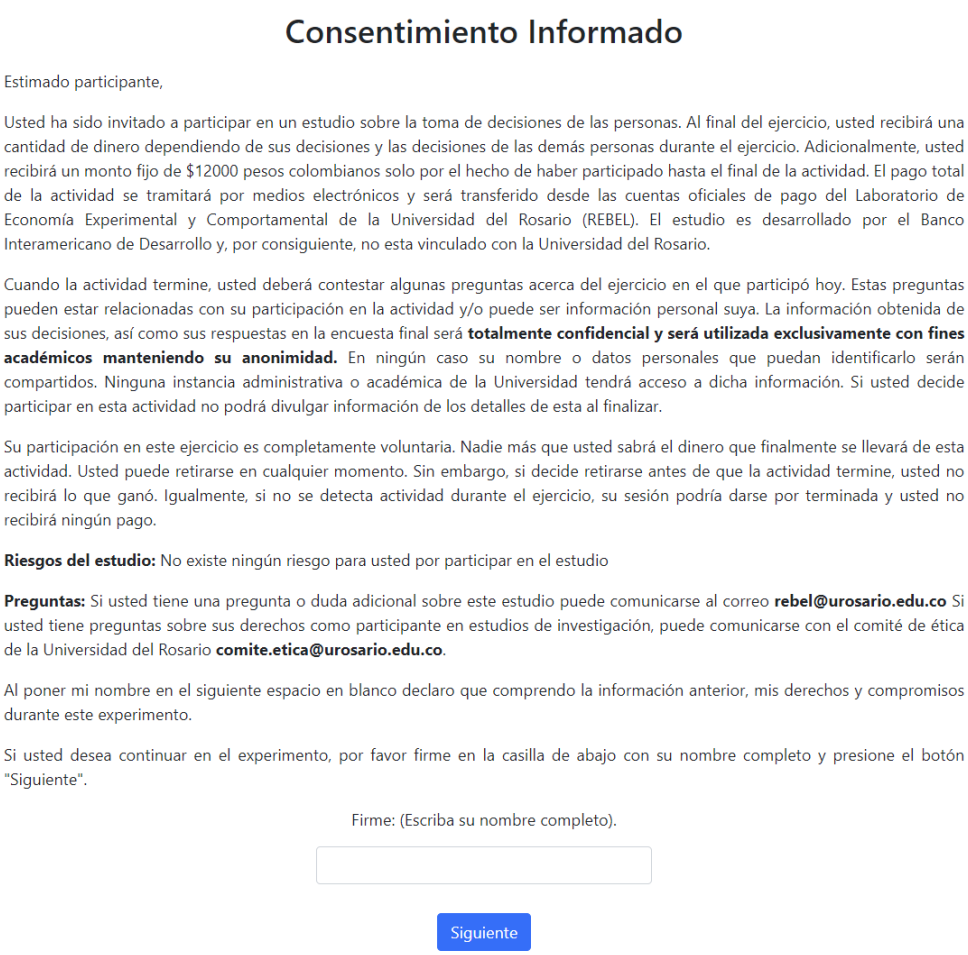
Usted será un mismo tipo de participante durante toda la actividad. Todas sus decisiones serán confidenciales y las ganancias obtenidas durante la sesión serán pagadas anónimamente al finalizar la actividad. Ningún participante sabrá cuánto ganó usted durante la sesión.

Si usted no desea participar en la actividad, comuníqueselo a uno de los coordinadores para poder dejar la sala.

A continuación se le presentará un consentimiento informado.

Siguiente

Figure E2: Informed Consent



Consentimiento Informado

Estimado participante,

Usted ha sido invitado a participar en un estudio sobre la toma de decisiones de las personas. Al final del ejercicio, usted recibirá una cantidad de dinero dependiendo de sus decisiones y las decisiones de las demás personas durante el ejercicio. Adicionalmente, usted recibirá un monto fijo de $12000 pesos colombianos solo por el hecho de haber participado hasta el final de la actividad. El pago total de la actividad se tramitará por medios electrónicos y será transferido desde las cuentas oficiales de pago del Laboratorio de Economía Experimental y Comportamental de la Universidad del Rosario (REBEL). El estudio es desarrollado por el Banco Interamericano de Desarrollo y, por consiguiente, no esta vinculado con la Universidad del Rosario.

Cuando la actividad termine, usted deberá contestar algunas preguntas acerca del ejercicio en el que participó hoy. Estas preguntas pueden estar relacionadas con su participación en la actividad y/o puede ser información personal suya. La información obtenida de sus decisiones, así como sus respuestas en la encuesta final será **totalmente confidencial y será utilizada exclusivamente con fines académicos manteniendo su anonimidad.** En ningún caso su nombre o datos personales que puedan identificarlo serán compartidos. Ninguna instancia administrativa o académica de la Universidad tendrá acceso a dicha información. Si usted decide participar en esta actividad no podrá divulgar información de los detalles de esta al finalizar.

Su participación en este ejercicio es completamente voluntaria. Nadie más que usted sabrá el dinero que finalmente se llevará de esta actividad. Usted puede retirarse en cualquier momento. Sin embargo, si decide retirarse antes de que la actividad termine, usted no recibirá lo que ganó. Igualmente, si no se detecta actividad durante el ejercicio, su sesión podría darse por terminada y usted no recibirá ningún pago.

Riesgos del estudio: No existe ningún riesgo para usted por participar en el estudio

Preguntas: Si usted tiene una pregunta o duda adicional sobre este estudio puede comunicarse al correo **rebel@urosario.edu.co** Si usted tiene preguntas sobre sus derechos como participante en estudios de investigación, puede comunicarse con el comité de ética de la Universidad del Rosario **comite.etica@urosario.edu.co**.

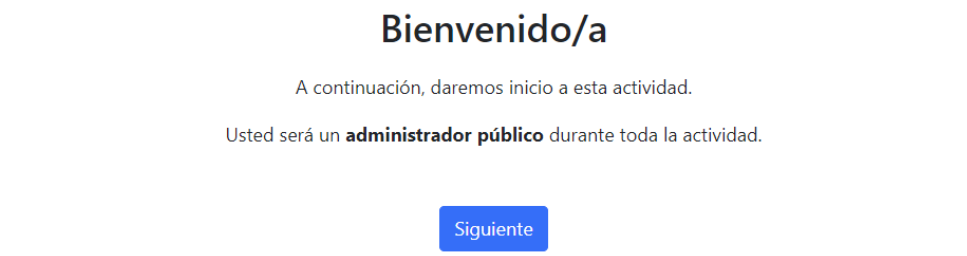
Al poner mi nombre en el siguiente espacio en blanco declaro que comprendo la información anterior, mis derechos y compromisos durante este experimento.

Si usted desea continuar en el experimento, por favor firme en la casilla de abajo con su nombre completo y presione el botón "Siguiente".

Firme: (Escriba su nombre completo).

Siguiente

Figure E3: Welcome



Bienvenido/a

A continuación, daremos inicio a esta actividad.

Usted será un **administrador público** durante toda la actividad.

Siguiente

Figure E4: Initial Instructions

Instrucciones iniciales

La siguiente actividad consta de cinco rondas de interacción.

A lo largo de la sesión usted producirá un pago para cada una de las cinco rondas. Sin embargo, usted **solo** recibirá el pago de **una** de las cinco rondas.

Al finalizar la última ronda, el computador seleccionará aleatoriamente una de las cinco rondas para ser pagada y el pago de dicha ronda será adicionado a su pago por participar de 12000 pesos.

Las cinco rondas tienen la misma probabilidad de salir, así que lo mejor que usted puede hacer es tomar decisiones en cada ronda pensando que dicha ronda será seleccionada para el pago.

A continuación le describimos las instrucciones de la actividad.

Siguiente

Figure E5: Instructions - 1

Instrucciones General

Recuerde que usted es un administrador público

Durante esta actividad, cada **ciudadano** tendrá un ingreso inicial de 80 puntos y cada **administrador público** tendrá un salario inicial de 120 puntos.

Una vez comienza la actividad:

1. El ciudadano pagará **40 puntos** de impuestos y conservará 40 puntos.
2. Usted recibirá los **40 puntos** de impuestos pagados por el ciudadano y su trabajo será decidir qué tantos impuestos utilizar en la construcción de bienes públicos, denotados con la siguiente imagen:



¿Por qué son importantes los bienes públicos en esta actividad?

Los bienes públicos son importantes dado que representan ingresos para el **ciudadano**.

Por cada bien público que usted construya, el ciudadano recibirá **30 puntos** adicionales.

Note que durante la actividad:

Solo el ciudadano recibe ingresos adicionales por la construcción de bienes públicos.

Solo el ciudadano paga impuestos para construir bienes públicos.

Figure E6: Instructions - 2



Figure E7: Round stages

Etapas de la ronda

Etapa 1 - Emparejamiento.

En la primera etapa de la ronda, cada administrador público será emparejado aleatoriamente con un ciudadano.

Usted siempre será emparejado con una persona **distinta**. Por consiguiente, usted no interactuará dos veces con la misma persona.

Usted nunca sabrá la identidad de la persona con la que fue emparejado/a.

Etapa 2 - Anuncio del costo de producción de los bienes públicos.

En la segunda etapa de la ronda, se definirá **aleatoriamente** el costo de producción de los bienes públicos. El costo de producción puede ser:

Costo Alto

20 puntos



Costo Bajo

10 puntos



Ambos costos son igual de probables.

Todos los participantes, sin importar su tipo, conocerán el costo del bien público.

Etapa 3 - Decisión política.

En la tercera etapa de la ronda, solo usted toma decisiónes.

En esta etapa, usted recibe los **40 puntos** de impuestos pagados por el ciudadano y deberá decidir si:

- Utiliza **20 puntos** en la construcción de bienes públicos y **conserva 20 puntos para usted.**
- Utiliza **40 puntos** en la construcción de bienes públicos y **no conserva nada para usted.**

Usted puede conservar 20 puntos que se adicionan a su salario de 120 puntos. Si usted decide conservar los 20 puntos, estos se convierten automáticamente en un ingreso adicional para usted.

Figure E8: Round stages - 2

Etapa 4 - Financiamiento adicional.

En la cuarta etapa, solo el computador toma decisiones.

En esta etapa, el computador podrá aleatoriamente dar un financiamiento adicional de **20 puntos** para construir más bienes públicos.

Con 50% de probabilidad, el computador destinará **20 puntos** adicionales y con 50% de probabilidad **no destinará** puntos adicionales.

Una vez el computador haya tomado su decisión, los recursos destinados por el administrador público y por el computador se sumarán y se destinarán en la construcción de bienes públicos.

Si usted destinó **40 puntos** y el computador destinó **20 puntos**, se emplearán **60 puntos** para construir bienes públicos.
Si usted destinó **40 puntos** y el computador destinó 0 puntos, se emplearán **40 puntos** para construir bienes públicos.
Si usted destinó **20 puntos** y el computador destinó **20 puntos**, se emplearán **40 puntos** para construir bienes públicos.
Si usted destinó **20 puntos** y el computador destinó 0 puntos, se emplearán **20 puntos** para construir bienes públicos.

Los recursos sumados se dividirán entre el **costo del bien público** y se construirán los respectivos bienes públicos.

Ningún participante conocerá la decisión del computador.

Al final de la interacción, el ciudadano **solo** observará los bienes públicos construidos.

Etapa 5 - Rendición de cuentas.

En la última etapa, solo el ciudadano toma decisiones.

Una vez el ciudadano observe los bienes públicos construidos, él recibirá **30 puntos** por cada bien público construido y, posteriormente, deberá decidir si:

- **Aprobar** su labor como administrador público, y que usted reciba el salario completo de **120 puntos.**
- **Desaprobar** su labor como administrador público, y que usted reciba un salario de **80 puntos.**

Por último:

Si el ciudadano decide desaprobar su labor como administrador público, él deberá pagar 30 puntos.

Figure E9: Payment

¿Cómo se calcula el pago de la ronda?

Así es como calculamos los pagos del administrador público y el ciudadano:

El pago del **administrador público** es:

Pago = **120 puntos** de salario + 20 puntos de impuestos *[si el administrador público decide conservarlos]* - 40 puntos por desaprobación *[si el ciudadano decide desaprobar al administrador público]*

El pago del **ciudadano** es:

Pago = **80 puntos** de ingreso - **40 puntos** de impuestos + (**30 puntos** x **número de bienes públicos**) - 30 puntos *[si el ciudadano decide desaprobar al administrador público]*

A continuación le presentamos un ejemplo de la actividad.

Siguiente

Figure E10: Control Question A - 1

Ejemplo 1

Información general

El ingreso del ciudadano, antes de pagar impuestos, es de **80 puntos**.

Recuerde que el total de impuestos pagados por el ciudadano es de **40 puntos**.

El salario del administrador público es de **120 puntos**.

Situación hipotética:

Suponga que durante esta interacción, el costo de producción de bienes públicos fue de **10 puntos**, es decir: un costo **bajo**

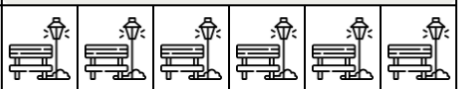
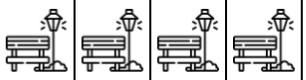


Recuerde que el administrador público pudo destinar los **40 puntos** de impuestos a la construcción de bienes públicos o destinar solo **20 puntos** de impuestos y conservar 20 puntos para él.

Finalmente, recuerde que el computador pudo dar un financiamiento adicional de **20 puntos** o no dar nada.

Figure E11: Control Question A - 2

A continuación, le presentamos una tabla que resume lo que el ciudadano puede observar en esta interacción

Recursos destinados por:		Recursos totales	Costo del bien público	Número de bienes públicos
Administrador público	Computador			
40 puntos	20 puntos	60 puntos	10 puntos	
40 puntos	0 puntos	40 puntos	10 puntos	
20 puntos	20 puntos			
20 puntos	0 puntos	20 puntos	10 puntos	

Suponga que el **ciudadano** observa que se construyeron seis (6) bienes públicos



¿Qué se puede inferir con la construcción de estos bienes públicos?

- Que el administrador público destinó solo 20 puntos de los impuestos, conservó la otra mitad y el computador SI dio un financiamiento adicional de 20 puntos.
- Que el administrador público destinó todos los 40 puntos de impuestos y el computador SI dio un financiamiento adicional de 20 puntos.
- Que el administrador público destinó todos los 40 puntos de impuestos y el computador NO dio un financiamiento adicional.

Siguiente

Figure E12: Control Question B - 1

Ejemplo 2

Información general

El ingreso del ciudadano, antes de pagar impuestos, es de **80 puntos**.

Recuerde que el total de impuestos pagados por el ciudadano es de **40 puntos**.

El salario del administrador público es de **120 puntos**.

Situación hipotética:

Suponga que durante esta interacción, el costo de producción de bienes públicos fue de **20 puntos**, es decir: un costo **alto**



A continuación, le presentamos una tabla que resume lo que el ciudadano pudo observar en esta interacción

Recursos destinados por:		Recursos totales	Costo del bien público	Número de bienes públicos		
Administrador público	Computador					
40 puntos	20 puntos	60 puntos	20 puntos			
40 puntos	0 puntos	40 puntos	20 puntos			
20 puntos	20 puntos					
20 puntos	0 puntos	20 puntos	20 puntos			

Figure E13: Control Question B - 2

Suponga que el **ciudadano** observa que se construyeron (3) bienes públicos



Recuerde que el ciudadano recibe **30 puntos** por cada bien público construido.

Por consiguiente, él ciudadano recibiría 90 puntos adicionales que se suman a sus **40 puntos** de ingreso después de impuestos.

Hasta este punto, el pago acumulado del ciudadano es de 130 puntos.

Recuerde que el administrador público inicia con un salario de 120 puntos.

Recuerde también que si el ciudadano desaprueba la labor del administrador público, el ciudadano deberá **pagar** 30 puntos y el administrador público recibirá un salario de 80 puntos.

Pregunta

Si el ciudadano decide desaprobar al administrador público, ¿cuál es el pago final de ambos jugadores?

Opciones	Pago del **administrador público**	Pago del **ciudadano**
Opción **A**	80 puntos	130 puntos
Opción **B**	120 puntos	100 puntos
Opción **C**	80 puntos	100 puntos

- ○ **Opción A**
- ○ **Opción B**
- ○ **Opción C**

Siguiente

Figure E14: General instructions

Inicio de la actividad

A continuación, daremos comienzo a la actividad.

Ahora lo vamos a emparejar **aleatoriamente con un ciudadano** de la sesión.

Siguiente

Figure E15: Politician's Decision
Cost (π) = *20 EP*

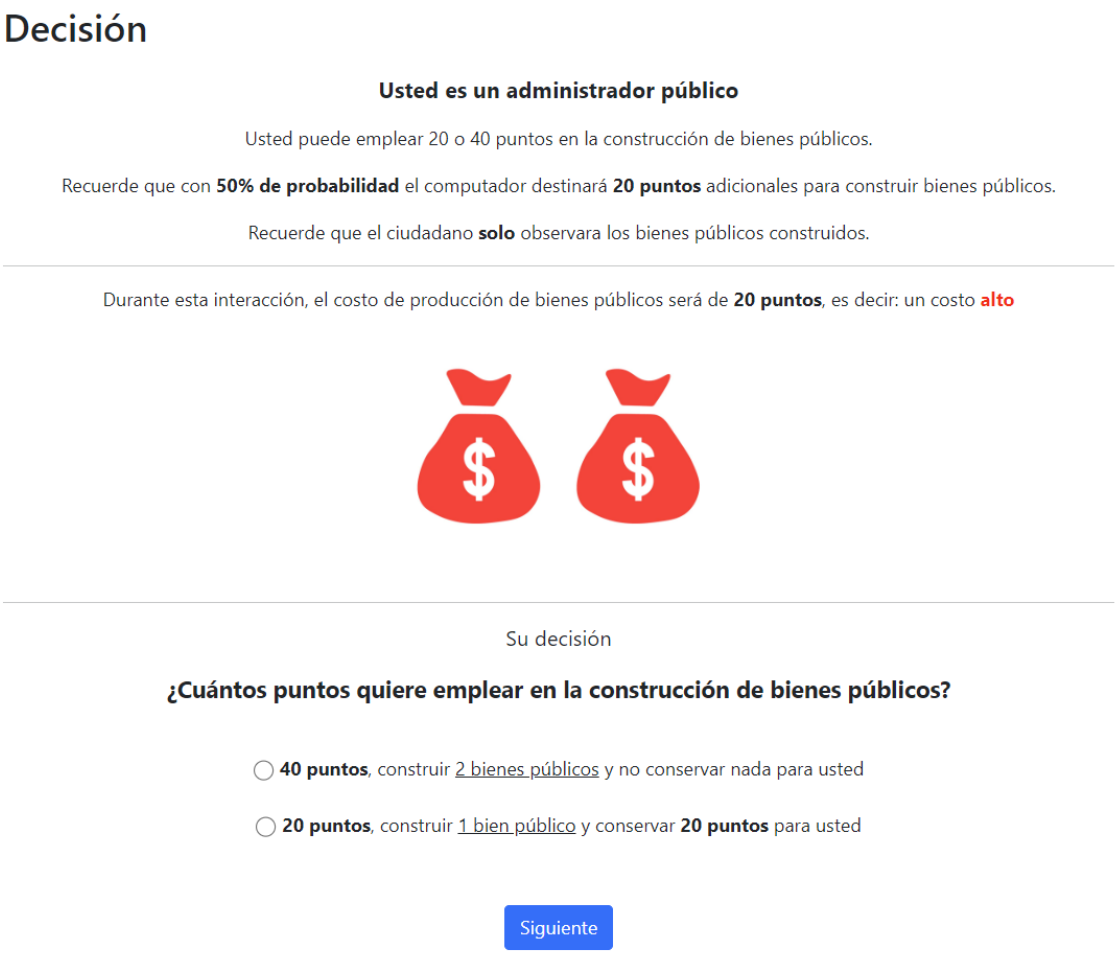


Figure E16: Politician's Belief of Punishment
Cost $(\pi) = 20$ *EP*

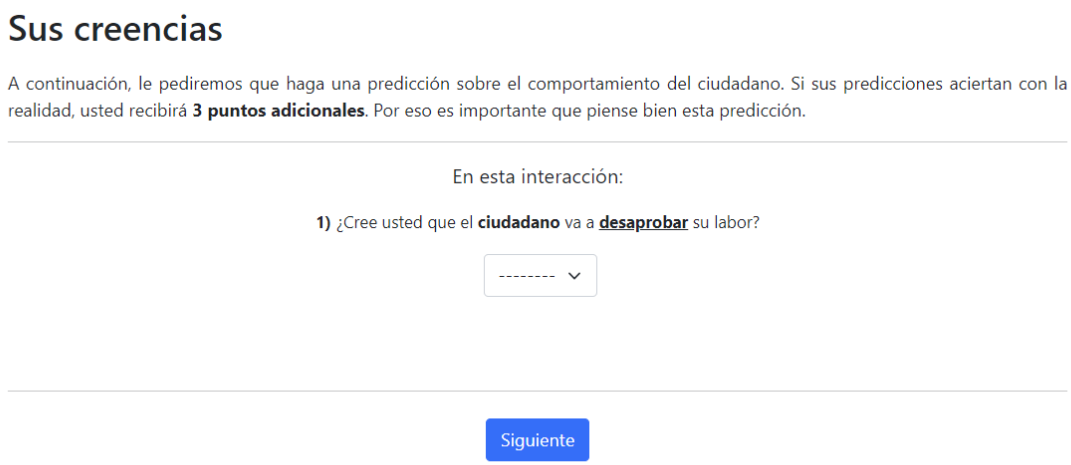


Figure E17: Citizen's Belief of Corruption
Cost $(\pi) = 20$ *EP*



Figure E18: Citizen's Decision
Cost (π) = *20 EP*

Rendición de cuentas

Durante esta interacción, el costo de producción de bienes públicos fue de **20 puntos**, es decir: un costo **alto**



Con un costo **alto**, usted podría observar:

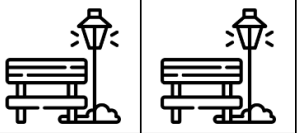
Si el administrador público destina	Si el computador destina	Recursos totales	Costo del bien público	Número de bienes públicos que usted observa:
40 puntos	20 puntos	60 puntos	20 puntos	
40 puntos	0 puntos	40 puntos	20 puntos	
20 puntos	20 puntos			
20 puntos	0 puntos	20 puntos	20 puntos	



Figure E19: End of the round

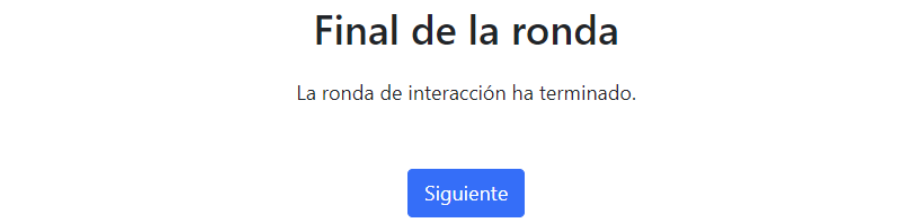


Figure E20: New round screen

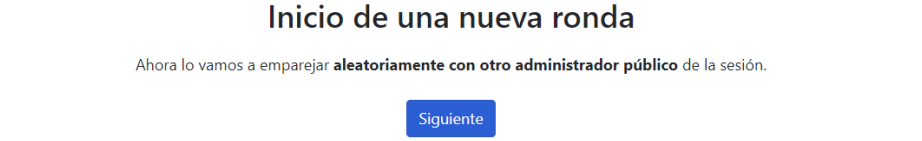


Figure E21: Citizen's Social Norms Screen

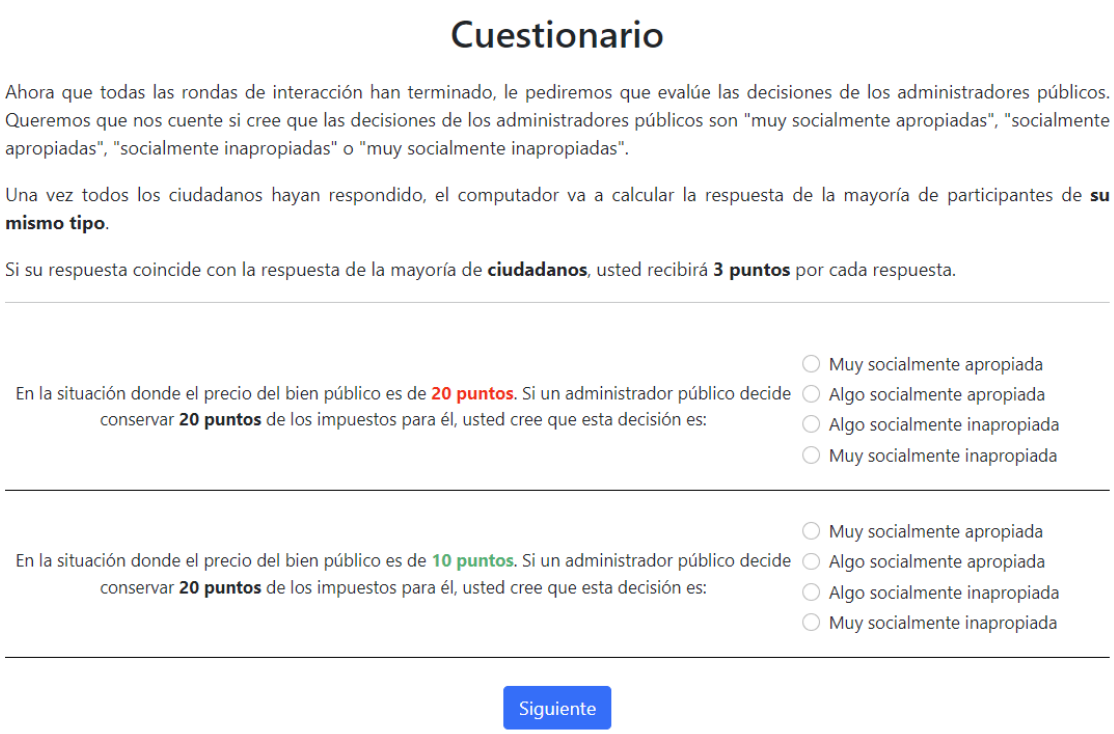


Figure E22: Politician's Social Norms Screen

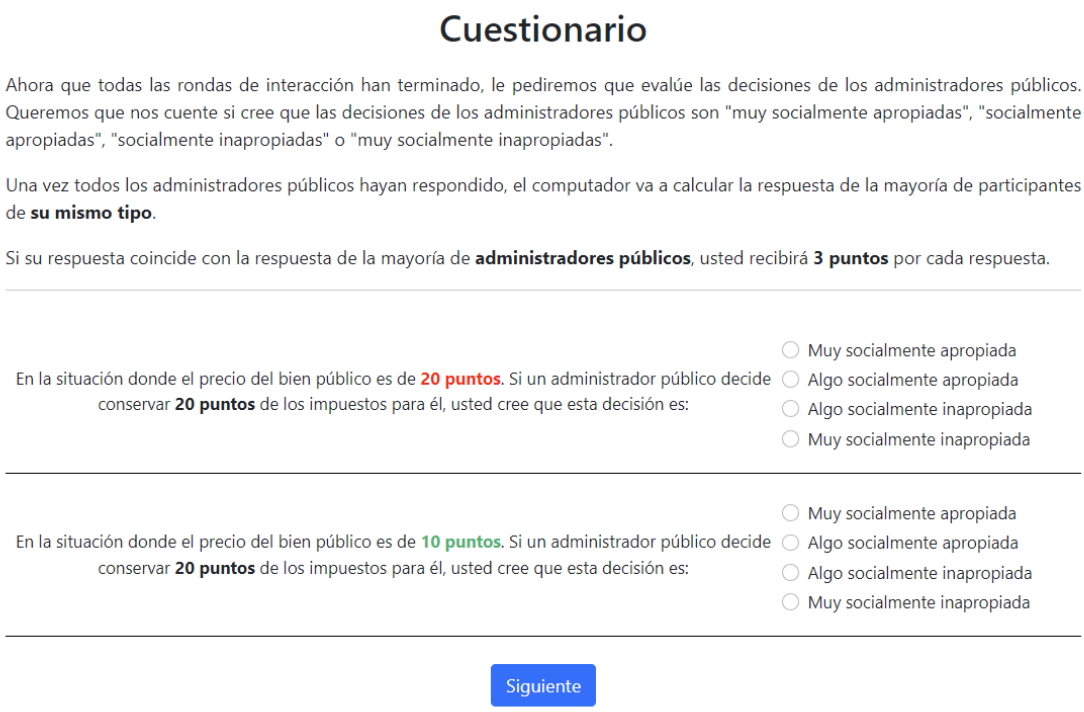


Figure E23: Final Results

Pago final del experimento

Usted era un **ciudadano**

A continuación le presentamos información relevante sobre sus pagos.

La ronda seleccionada aleatoriamente para el pago fue:	Ronda 3
En dicha ronda, el costo de los bienes públicos fue:	**10 puntos**

Decisiones

En dicha ronda, usted observó:	6 bienes públicos
Estos bienes públicos, sumado a su ingreso después de impuestos sumaba:	220 puntos
En dicha ronda, ¿Usted decidió **desaprobar** al administrador público?	**NO**

Pago de la ronda

Por consiguiente, su pago fue:	**220 puntos**

Creencias

Durante el desarrollo de la actividad le pedimos que nos contara sus creencias sobre el comportamiento del administrador público. Recuerde que usted recibirá 3 puntos en caso de que su predicción haya acertado.

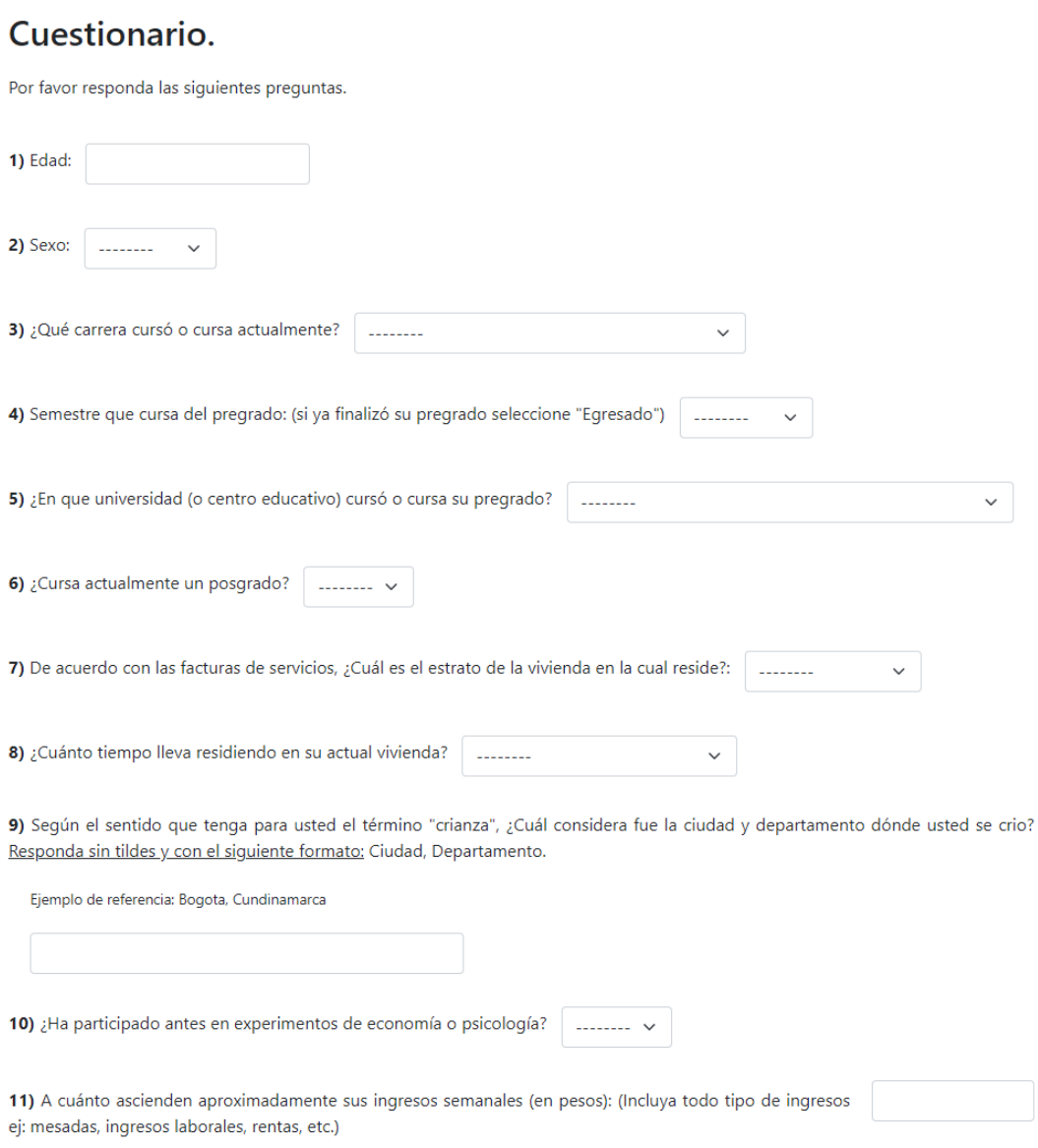
Tipo de creencia	Su respuesta	Recursos asignados por el administrador público	Su respuesta coincició con la realidad:
Recursos que asignará el administrador público para construir bienes públicos.	40 puntos	40 puntos	**SI**

Por sus creencias, usted recibe un pago adicional de: **3 puntos**

Figure E24: Final Results - 2



Figure E25: Questionnaire



Cuestionario.

Por favor responda las siguientes preguntas.

1) Edad:

2) Sexo: --------

3) ¿Qué carrera cursó o cursa actualmente? --------

4) Semestre que cursa del pregrado: (si ya finalizó su pregrado seleccione "Egresado") --------

5) ¿En que universidad (o centro educativo) cursó o cursa su pregrado? --------

6) ¿Cursa actualmente un posgrado? --------

7) De acuerdo con las facturas de servicios, ¿Cuál es el estrato de la vivienda en la cual reside?: --------

8) ¿Cuánto tiempo lleva residiendo en su actual vivienda? --------

9) Según el sentido que tenga para usted el término "crianza", ¿Cuál considera fue la ciudad y departamento dónde usted se crio? Responda sin tildes y con el siguiente formato: Ciudad, Departamento.

Ejemplo de referencia: Bogota, Cundinamarca

10) ¿Ha participado antes en experimentos de economía o psicología? --------

11) A cuánto ascienden aproximadamente sus ingresos semanales (en pesos): (Incluya todo tipo de ingresos ej: mesadas, ingresos laborales, rentas, etc.)

12) ¿Cuál es el máximo nivel de educación alcanzado por su madre? --------

13) ¿Cuál es el máximo nivel de educación alcanzado por su padre? --------

14) Imagine una escalera con 10 escalones. Las personas con los ingresos más **bajos** se sitúan en el escalón 1 y las personas con los ingresos más **altos** en el escalón 10. ¿Dónde situarías a tu grupo familiar?

1	2	3	4	5	6	7	8	9	10

15) En una escala del 0 al 10, donde **0 significa que "no está dispuesto(a) a tomar riesgos en lo absoluto"** y **10 significa que "está muy dispuesto(a) a tomar riesgos"**, por favor, dígame, en general ¿qué tan dispuesto (a) está o no está usted a tomar riesgos? Puede usar cualquier número entre 0 y 10 para indicar en qué lugar de la escala se encuentra:

0	1	2	3	4	5	6	7	8	9	10

16) En una escala del 0 al 10, donde **0 significa que "Por lo general, hay que tener mucho cuidado de las personas"** y **10 significa "Por lo general, se puede confiar en la gente"**, ¿usted diría que se puede confiar en la mayoría de la gente o que hay que tener mucho cuidado al tratar con las personas?

0	1	2	3	4	5	6	7	8	9	10

17) ¿Usted diría que la gente de su barrio es "muy confiable", "algo confiable", "poco confiable" o "nada confiable"? --------

18) ¿En su vida personal, ¿en qué medida está dispuesto a renunciar a algo que puede ser beneficioso para usted hoy a cambio de obtener mayores beneficios en el futuro? Para ello utilice la siguiente escala de 1 al 5, donde 1 significa que usted **no está dispuesto** a renunciar a obtener los beneficios hoy y 5 significa que usted está **totalmente dispuesto** a esperar a cambio de beneficios mayores en el futuro

1	2	3	4	5

19) ¿Si usted fuera víctima de un robo o asalto, ¿cuánto confiaría que el sistema judicial castigue al culpable? --------

20) De la siguiente lista de actividades políticas que las personas pueden hacer, le pedimos que indique si: Alguna vez la ha hecho, si alguna vez la haría o si nunca la haría

Firmar una petición --------

Participar en una manifestación autorizada --------

Participar en una manifestación no autorizada --------

Bloquear el tráfico --------

21) Pensando en los políticos, ¿cuántos cree que están involucrados en corrupción? --------

22) Si una persona se enferma, ¿quién debe pagar su tratamiento?
(Puede deslizar la barra indicando quien tiene más responsabilidad en el pago del tratamiento)

El estado debe pagar todo el tratamiento **El paciente debe pagar todo el tratamiento**

23) ¿Si usted expresa libre y abiertamente su opinión sobre los problemas nacionales, cree que esto podría tener consecuencias negativas para usted?

----

24) En una escala de 1 a 10, siendo 1 "no esta dispuesto en lo absoluto" y 10 "completamente dispuesto", ¿Qué tan dispuesto esta a protestar por...

Mejores salarios y mejores condiciones de trabajo --------

Mejor salud y educación --------

Cambio climático --------

Defender los derechos democráticos --------

Pelear en contra de la corrupción --------

Una sociedad más igualitaria --------

25) Imagine que vamos a otorgar un premio de 50.000 COP al ganador del siguiente juego. Usted debe escoger un número entre el 0 y el 100. El ganador será aquel participante de la sala cuyo número elegido sea el más cercano a 2/3 (dos terceras partes) del promedio de los números elegidos por todos los participantes.

¿Qué número escogería?

26) Pensando en lo que más se acerque a su punto de vista, ¿Cómo debería hacer uno para que usted y el país avancen?

----

27) Muchas de las calles de la ciudad están en mal estado. Supongamos que el gobierno va a asignar recursos para su mantenimiento a los barrios donde los vecinos logren reunir por lo menos 500 firmas en una petición. ¿Qué tan probable es que en el barrio donde usted vive se logre reunir las 500 firmas para la petición?

----

28) Si logran juntar en el barrio las 500 firmas necesarias para la petición ¿Qué tan probable es que las autoridades efectivamente solucionen el problema?

----

29) ¿Un oficial de policía le ha pedido un soborno en los últimos doce meses? --------

30) En los últimos doce meses, ¿algún empleado del gobierno le pidió un soborno? --------

31) Qué tanto confía que el gobierno nacional haga lo correcto? --------

32) Siendo 1 "Nada involucrado" y 10 "Totalmente involucrado", ¿Qué tanto cree que están involucrados en corrupción ...

El presidente y los trabajadores de su oficina --------

Los congresistas --------

Los gobernadores --------

Los alcaldes --------

Los policias --------

Los jueces y magistrados --------

La DIAN --------

33) Hoy en día cuando se habla de tendencias políticas, mucha gente habla de aquellos que simpatizan más con la izquierda o con la derecha. Según el sentido que tengan para usted los términos "izquierda" y "derecha" cuando piensa sobre **su punto de vista** político, ¿usted se describiría como de derecha, centro derecha, centro, centro izquierda, o izquierda?

----

34) Qué tan de acuerdo está con la siguiente afirmación: **Cada individuo es responsable de su propio bienestar**

----

35) Qué tan de acuerdo está con la siguiente afirmación: **El gobierno debe poner en práctica políticas firmes para reducir la desigualdad entre ricos y pobres**

----

36) ¿Votó usted en las últimas elecciones presidenciales?

Siguiente